

Department of Chemical and Biomolecular Engineering

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A Case Study of Low-Carbon Data Centre Deployment in Singapore

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Executive Summary

Singapore can expand its data centre industry while cutting the sector's electricity-related emissions by 90%, by combining cooling optimisation with rooftop solar photovoltaics (PV), low-carbon ammonia generation and, in the longer term, small modular reactors (SMRs). Data centres already use more than a tenth of national electricity on a grid that is about 94% gas-fired, so growth depends on efficiency and on cleaner supply. This report assesses a 100 MW IT reference facility in the planned Jurong Island low-carbon data centre park. At an 85% load factor and a power usage effectiveness (PUE) of 1.25, it draws 930.8 GWh a year, emits 374 kt CO_2e on today's gas-dominated grid and spends S$186 million on electricity.

The available low-carbon options were screened against the demands of a flat 24/7 load. Rooftop PV generates 1.6 GWh a year at S$85/MWh, below the grid price. A 60 MWe direct-ammonia combined-cycle plant generates 446.8 GWh a year at S$342/MWh and, with certified renewable fuel, 0.111 tCO_2e/MWh on a life-cycle basis, 72% below the grid. A 77 MWe SMR supplies 627.3 GWh a year of firm electricity at a central S$230/MWh. Efficiency is applied before supply is allocated, so that each source serves the reduced demand. Physics-informed predictive control of the chilled-water plant, calibrated at the NUS Sustainable Tropical Data Centre Testbed, and SS 697 temperature optimisation save 17.0 GWh a year and lower PUE to 1.227 at a net saving of S$2.9 to 3.4 million a year. Hybrid liquid cooling then reaches PUE 1.20, and its S$4.0 million annual saving repays up to S$36.8 million of capital.

Six gaps limit Singapore's current approach, namely AI demand that outpaces efficiency, diminishing returns from PUE, intermittent solar output, fuel and supply security, inconsistent low-carbon accounting, and a policy focus on efficiency rather than absolute consumption, and each becomes a design requirement. The measures are combined into six configurations. The integrated long-term portfolio uses rooftop PV, SMR generation and 264.6 GWh of electricity allocated from a shared ammonia plant, with the Singapore grid providing balancing and operational reliability. It reduces annual electricity-related emissions to 37.0 $ktCO_2e$, avoiding approximately 337 $ktCO_2e$ (90.1%) at an abatement cost of S$145/$tCO_2e$, and an illustrative hourly carbon-free electricity matching score of 97.1% shows how closely supply can follow a continuous load. Nuclear integration alone abates at S$43/$tCO_2e$, and an existing site starting from PUE 1.50 avoids 404 kt a year. The ammonia price and its certified life-cycle intensity dominate the cost result. Operators should deploy predictive cooling control and rooftop PV now, design liquid cooling into new AI halls, contract certified ammonia electricity from Jurong Island and prepare for SMR supply, while IMDA and EMA write plant metering and hourly reporting into licence conditions.

Chapter 1: Introduction and Scope

1.1 Background and Research Objective

Singapore is the major data centre hub of the Asia-Pacific region. By October 2026 it hosted about 70 data centres with a maximum IT load capacity of around 1.6 GW, including 200 MW provisionally allocated under the second Data Centre Call for Application (DC-CFA2) [1]. These facilities consumed about 6.5 TWh of electricity in 2024, accounting for more than a tenth of national consumption and up from about 7% in 2020 [2].

Computing capacity matters to Singapore's economy: the digital economy generated S$144.1 billion of value added in 2025, 19.3% of GDP [3], and workloads in finance, healthcare, digital identity and AI depend on domestic capacity for latency, security and resilience [1]. However, data centres operate within a power system in which natural gas supplied about 94% of generation and the average grid emission factor (GEF) was 0.402 kg CO_2/kWh in 2024 [4]. At the same time, Singapore has also committed to net-zero emissions by 2050, with an intermediate target of 45 to 50 $MtCO_2e$ by 2035 [5], so every additional megawatt of data centre load must be supplied by a decarbonising power system.

Against this background, this study investigates a central question: To what extent can a grid-connected data centre in Singapore reduce its annual electricity-related emissions by combining energy efficiency improvements with low-carbon electricity supply, and at what cost? The study aims to develop and evaluate an integrated strategy that considers Singapore's technical, resource and operational constraints. The assessment draws on measured operational evidence, calibrated simulations, published standards and stated design assumptions, informed by Mytton and Ashtine [6].

1.2 Singapore's Data Centre Landscape and Policy Framework

Operational supply is tight: colocation vacancy was 4.9% in the second half of 2025 [7] and 2% in early 2026, the lowest in the Asia-Pacific region [8], so new capacity is scarce and its allocation carries high value.

That allocation is now explicitly tied to environmental performance. Concerns over land, electricity and water led the Government to pause new data centre development in 2019 [9]. Capacity allocation resumed through the pilot DC-CFA, which sought proposals advancing energy efficiency, decarbonisation, connectivity and economic objectives [10], and about 280 MW has been allocated through two competitive tenders since 2022 [2]. Table 1.1 summarises the instruments that now govern efficiency and supply.

Table 1.1. Policy instruments and standards governing data centre efficiency and supply in Singapore

Instrument	Requirements
Green Data Centre Roadmap (May 2024)	At least 300 MW of further capacity for green data centres; aspiration for all data centres to reach PUE 1.3 or better at 100% IT load within about ten years; WUE of 2.0 m^3/MWh or better [11]
SS 697:2023 Tropical DC Standard	Roadmap to operate at 26 °C and above; each 1 °C rise saves 2% to 5% of cooling energy [12]
SS 715:2025 IT energy efficiency	Energy efficiency of data centre IT equipment, workload consolidation and higher utilisation [13]

Instrument	Requirements
DC-CFA2 (Dec 2025)	At least 200 MW; PUE 1.25 or better at 100% IT load; SS 715 equipment; Green Mark Platinum; at least 50% of new capacity powered by eligible green pathways (biomethane, low-carbon ammonia or hydrogen, fuel cells with CCS, building-integrated PV) [14]
Digital Infrastructure Bill (passed 7 Oct 2026)	Licence for every site with at least 3 MW of critical IT load; PUE-based efficiency standards set by IMDA; green energy pledges written into licences; fines up to S$1 million or 10% of local turnover [2], [15]
SS 726:2026 Liquid cooling standard (Aug 2026)	Design, deployment and operation of liquid cooling in tropical data centres; liquid cooling can cut energy use by more than 30% against air cooling [16]

Two features of this framework shape the proposal. Firstly, the new licensing framework is expected to introduce phased sustainability requirements for both existing and new data centres, so efficiency at existing sites carries regulatory as well as financial value [17]. Secondly, DC-CFA2 names low-carbon ammonia among its eligible green pathways [14], which the proposed Jurong Island ammonia power project can supply (Section 2.5). Growth will continue, but within climate and resource constraints and for value rather than volume [1].

Singapore is preparing both firm options assessed here: nuclear capability building ahead of the IAEA Integrated Nuclear Infrastructure Review (INIR) Phase 1 mission in 2027 [18], [19] (Section 2.4), and front-end design of a 55 to 65 MW Jurong Island combined-cycle plant fuelled by imported low- or zero-carbon ammonia [20] (Section 2.5).

1.3 Scope of the Assessment

Reference facility and geographical scope

The study assesses one hypothetical grid-connected data centre in the planned Jurong Island low-carbon data centre park (Section 5.1), with 100 MW of rated IT capacity, an 85% electrical load factor, an annual PUE of 1.25 and 8,760 operating hours, giving 744,600 MWh of IT energy and 930,750 MWh of facility electricity a year (Equation 1.1). The grid, at 0.402 tCO_2/MWh [3] and an all-in S$200/MWh, is the reference supply. Each configuration is assessed for annual electricity, comparative emissions emitted and avoided, annual and incremental cost, and abatement cost.

PUE and electrical load factor

DC-CFA2 requires new capacity to achieve a PUE of 1.25 or better at 100% IT load [14]. The report adopts this value as the representative annual PUE of a new facility operating at the assumed load. The 85% load factor is an independent assumption for average IT electrical demand relative to rated capacity (Table 1.3), and combining IT power with PUE is the bottom-up approach reviewed by Mytton and Ashtine [6].

Emission scopes and carbon boundary

Data centre emissions are classified under the Greenhouse Gas (GHG) Protocol into three scopes [21]. Scope 1 covers direct emissions from diesel backup generators and fugitive refrigerant leakage from chillers and cooling units. Scope 2 covers emissions from purchased electricity. Scope 3 covers indirect emissions across the value chain, including embodied emissions from construction materials, servers, GPUs, power and cooling equipment, upstream fuel production and transport, and waste generated in operations (Figure 1.1).

Scope 3 is significant and rising: Microsoft's Scope 3 emissions rose 30.9% against its 2020 baseline, driven partly by data centre construction and IT hardware [22]. Scope 1 is small in

normal operation: monthly one-hour testing of 140 MW of diesel generation at 30% load (0.27 L/kWh and 2.68 kg CO_2/L, design assumptions) emits about 0.4 kt CO_2 a year, roughly 0.1% of the reference facility's electricity-related emissions.

This study assesses electricity-related emissions rather than a full Scope 1–3 GHG inventory. Direct on-site emissions and other data centre value-chain emissions are excluded, while selected life-cycle emissions from electricity supply are included. Differences in emission-factor boundaries are discussed below.

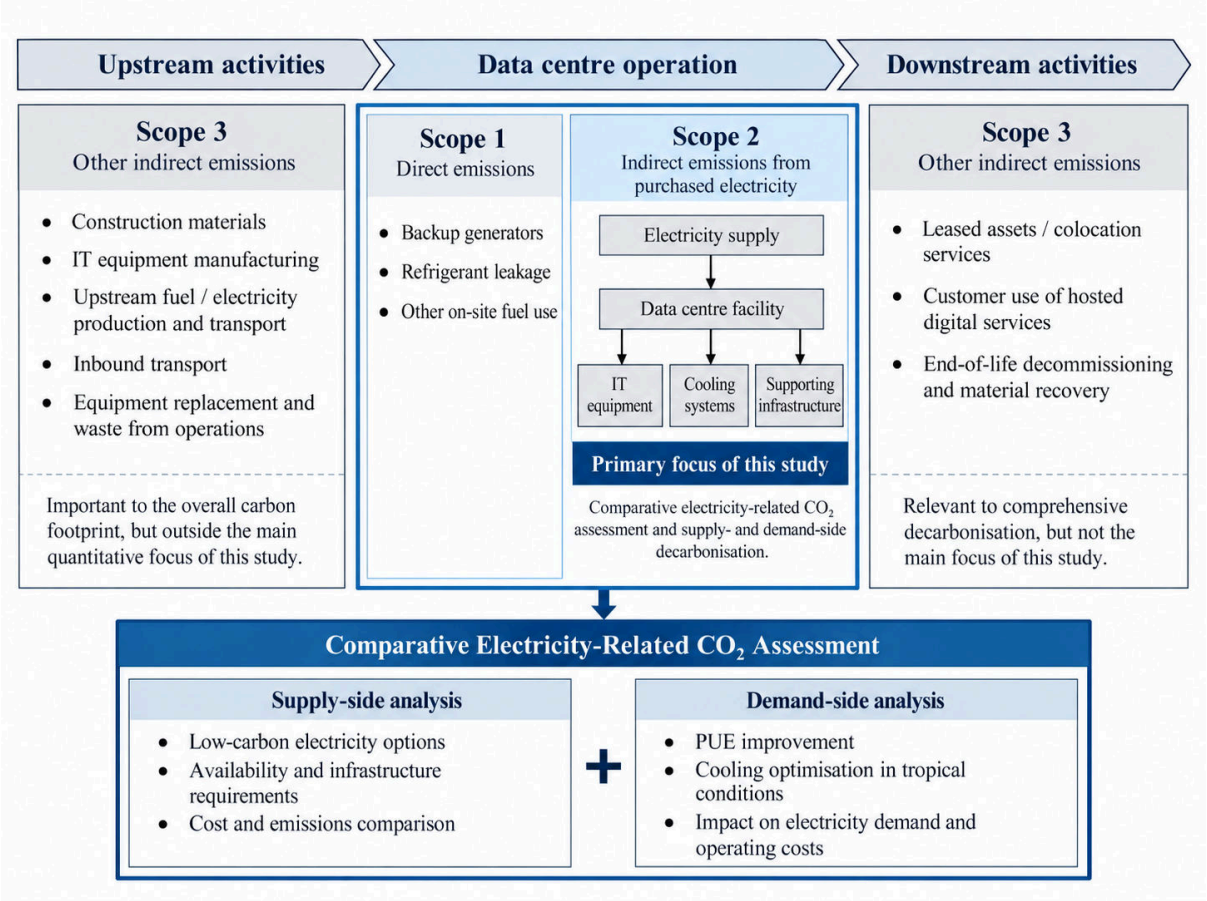


Figure 1.1. Data centre greenhouse gas emissions across the value chain and the assessment boundary of this study.

Shared generation and annual allocation

Three generation pathways are quantified (Table 1.4): rooftop PV (Scenario A), one 77 MWe SMR module (Scenario B) and a 60 MWe ammonia combined-cycle plant (Scenario C). The grid supplies residual demand, balancing and backup; regional imports and the other screened technologies remain policy context (Section 2.2). Each MWh is allocated to one source and one consumer, and allocations sum to facility demand (Equation 1.2), the annual volumetric basis reviewed by Khosravi et al. [23] and He et al. [24]. The shared ammonia plant supplies the facility's demand after PV and SMR output, 264,634 MWh a year at PUE 1.20, and other Jurong Island consumers take the remaining 182,126 MWh. Because annual matching does not show whether supply and demand coincide in time [25], hourly carbon-free electricity matching is the operational verification metric (Section 4.5). Hourly dispatch, transmission and market modelling lie outside the assessment.

Comparative emissions methodology

Grid electricity is charged at EMA's average grid emission factor, which counts combustion CO_2 [3]. The three generation pathways share a life-cycle boundary: rooftop PV and SMR output carry IPCC life-cycle medians of 0.041 and 0.012 tCO_2e/MWh [26], which include plant manufacture, and ammonia electricity carries fuel production, the embodied emissions of the renewable electricity used to make the fuel, maritime delivery and combustion N_2O, giving 0.111 tCO_2e/MWh (Section 2.5). Combined results are reported in tCO_2e, with the grid factor's CO_2 treated as CO_2e. The results are comparative electricity-related emissions indicators designed for consistent comparison of supply configurations; a life-cycle grid factor of 0.49 tCO_2e/MWh [26] would raise every reduction (Appendix C.11). Certificates serve to verify delivery, so each reduction is counted once.

Cooling optimisation

Cooling measures apply published Singapore testbed, calibrated simulation and standard results to the reference facility. IT electricity and non-cooling overhead are held constant, so each cooling saving becomes an equal reduction in facility electricity and PUE (Section 3.5), and the IT-side saving of liquid cooling is held in reserve as additional upside. Transferring the NUS testbed performance to a commercial plant is an engineering assumption supported by commercial evidence (Section 3.4); plant-specific calibration, equipment selection and commissioning are outside the quantitative assessment.

Economic assessment

All configurations share one economic basis (Table 1.4): an all-in grid price of S$200/MWh, levelised generation costs for PV, the SMR and the ammonia plant (Equation 2.4), capital recovery at 7%, a cooling implementation cost of S$1 to 5 million and break-even capital where costs are unpublished (Appendix C). Financing, procurement and investment approval are excluded.

Technology development and siting

The ammonia plant is an emerging pathway linked to the Jurong Island project, now at front-end engineering design [20]. SMRs are a longer-term firm option consistent with Singapore's current assessment [27], [18]; Singapore has not decided to deploy nuclear energy and considers siting parameters premature [28].

Sensitivity analysis

Sensitivity is tested one variable at a time for the principal price, operating and technology assumptions (Section 5.5); no multivariable uncertainty model is used.

1.4 Research Methodology and Integrated Assessment Model

This study adopts a scenario-based techno-economic assessment based on published research, testbed results, calibrated simulations, technical standards and stated design assumptions. A grid-supplied reference case is first established, followed by the assessment of demand-side efficiency measures, including predictive cooling control and hybrid liquid cooling, and three low-carbon electricity pathways: rooftop PV, ammonia-fired generation and SMR electricity. Efficiency savings are applied before allocating electricity supply to the reduced demand. These measures are then combined into six configurations (Chapter 5) and compared in terms of annual electricity consumption, emissions reductions, electricity costs and carbon abatement costs. Sensitivity analysis (Section 5.5) and illustrative hourly carbon-free electricity matching (Section 4.5) complement the assessment.

The reference data centre is defined with a rated IT capacity of 100 MW, an electrical load factor of 85% and an annual PUE of 1.25. The PUE follows the DC-CFA2 benchmark, applied as the representative annual value (Section 1.3). The load factor represents average IT electrical demand relative to rated capacity, so annual electricity is estimated independently of the maximum installed capacity. This is the bottom-up approach of Mytton and Ashtine [6], in which IT power characteristics and facility PUE are combined to estimate total consumption, rather than a top-down estimate from national or sector aggregates. Equation 1.1 gives the annual electricity, and Equations 1.2 and 1.3 convert it to CO_2 and cost; Tables 1.2 to 1.4 define the variables, the reference case and the technology assumptions:

$$E_{DC} = P_{IT} \times LF \times PUE \times H \quad (1.1)$$

$$CO_2 = \Sigma_i E_i EF_i, \quad \Sigma_i E_i = E_{DC} \quad (1.2)$$

$$C = \Sigma_i E_i p_i + C_{eff} \quad (1.3)$$

$$C_{eff} = CAPEX_{eff} \times CRF, \quad CRF = r(1 + r)^n / [(1 + r)^n - 1] \quad (1.4)$$

$$\Delta CO_2 = CO_{2,baseline} - CO_{2,proposed} \quad (1.5)$$

$$\Delta C = C_{proposed} - C_{baseline} \quad (1.6)$$

$$AC = \Delta C / \Delta CO_2 \quad (1.7)$$

Table 1.2. Nomenclature for the integrated model

Symbol	Definition	Unit
P_{IT}	Rated IT capacity	MW
LF	Electrical load factor	-
PUE	Power usage effectiveness, E_{DC}/E_{IT} (baseline PUE_0, proposal PUE_1)	-
H	Annual operating hours	h/yr
E_{IT}	Annual IT energy, $P_{IT} \times LF \times H$	MWh/yr
E_{DC}	Annual facility electricity demand	MWh/yr
E_i	Electricity allocated to the facility from source i (grid, PV, SMR or ammonia)	MWh/yr
EF_i	Emission factor of source i (Section 1.3)	tCO_2e/MWh
p_i	Delivered grid price, or levelised generation cost of PV, SMR or ammonia output	S$/MWh
P_{NH3}, CF_{NH3}	Ammonia plant net capacity and capacity factor	MW, -
η_{NH3}, LHV_{NH3}	Net efficiency of the ammonia plant; lower heating value of ammonia	-, GJ/t
SFC_{NH3}	Specific fuel consumption, $3.6/(\eta_{NH3} \times LHV_{NH3})$	tNH_3/MWh
I_{NH3}	Life-cycle intensity of delivered ammonia (production and shipping)	tCO_2e/tNH_3
E_{NH3}, $E_{NH3,DC}$	Ammonia plant output, and the share allocated to the facility	MWh/yr
C_{eff}	Annualised implementation cost of the efficiency measures	S$/yr
$CAPEX_{eff}$	Capital cost of the efficiency measures	S$
CRF, r, n	Capital recovery factor, discount rate and project life	-, -, yr
CO_2, C	Annual emissions and annual cost	tCO_2e/yr, S$/yr
ΔCO_2	Annual emissions avoided against the baseline	tCO_2e/yr
ΔC	Incremental annual cost against the baseline (negative values are savings)	S$/yr
AC	Carbon abatement cost	S/tCO_2e$

Table 1.3. Reference data centre: assumptions and baseline outputs

Parameter	Value
IT capacity, P_{IT}	100 MW (reference case)
Electrical load factor, LF	0.85: high-utilisation central design assumption for average electrical draw as a share of rated IT capacity; tested at 0.75 to 0.95 (Table 1.5)
Power usage effectiveness, PUE	1.25 (DC-CFA2 benchmark at 100% IT load [14], adopted as the annual value; Section 1.3)
Annual operating hours, H	8,760 h/yr (continuous operation)
Annual IT energy, E_{IT}	744,600 MWh/yr ($P_{IT} \times LF \times H$)
Annual facility electricity, E_{DC}	930,750 MWh/yr (Equation 1.1)
Grid emission factor, EF_{grid}	0.402 tCO_2/MWh (EMA, 2024 [3])
Reference electricity price, p_{grid}	S$200/MWh, all-in delivered (Table 1.7)
Baseline annual emissions	374,162 tCO_2/yr (Equation 1.2)
Baseline annual electricity cost	S$186.15 million/yr (Equation 1.3)

Table 1.4. Technology and economic assumptions of the three generation pathways

Parameter	A: Rooftop solar PV	B: SMR	C: Ammonia GTCC
Capacity	1.14 MWp: 13,600 m² roof × 0.6 × 0.14 kWp/m² (Section 2.3) [29], [30]	77 MWe, one module of a six-module, 462 MWe plant [31]	60 MWe, midpoint of the 55 to 65 MW project [20]
Yield or capacity factor	1.385 MWh/kWp/yr, CF 15.8% [30]	93% [32]	85% (design assumption; 50% to 93% in Table 5.7)
Annual generation	1,582 MWh (Equation 2.1)	627,304 MWh (Equation 2.1)	446,760 MWh (Equation 2.1)
Conversion	Net of system losses, within the SERIS specific yield [30]	Pressurised-water steam cycle (Figure 2.5) [33]	Direct liquid NH_3 GTCC, 47.7% net [34]; 0.401 tNH_3/MWh (Equation 2.2)
Emission factor	0.041 tCO_2e/MWh [26]	0.012 tCO_2e/MWh [26]	0.111 tCO_2e/MWh: 0.276 tCO_2e/tNH_3 (production 0.03 [35], embodied wind electricity 0.126 [26], [36], shipping 0.12 [36]) and N_2O [34]
Capital cost	S$1,000 to 1,400/kWp, central 1,200 [37]	Within generation cost; first-of-a-kind US$20,139/kW [31]	US$4,281/kW (S$5,462/kW) [34]
Operating and fuel cost	1.0% to 1.45% of capital a year, central 1.225% [37]	Within generation cost [31], [38]	Maintenance 2.5% and insurance 2% of capital a year; fuel US$480/t delivered [34]
Life and discount rate	25 years (design assumption), 7%	40-year cost recovery (design assumption; 60-year technical life [32]), 7%	25 years [34], 7%
Generation cost (S$/MWh)	85 (66 to 106); Equation 2.4, Appendix C.2	230 (114 to 281) [31], [38]	342 (Equation 2.4, Appendix C.6); 260 to 531 for fuel at US$320 to 850/t [39]

Grid: 0.402 tCO_2/MWh [3] and S$200/MWh (Table 1.3); S$1.276 per US$ [40]. The 7% discount rate is a design assumption for all pathways; Ahmad et al. use 10% (Appendix C.6).

Grid electricity is priced at its delivered price, including transmission and balancing, and PV, SMR and ammonia output at levelised generation cost (Equation 2.4) under cost-reflective purchase agreements. Capital is therefore counted once, and C_{eff} carries only the metering, control and commissioning cost of the efficiency measures. ΔC and AC are reported on an electricity-expenditure basis and then net of annualised implementation cost (Section 5.3); a negative AC means that the proposal saves money while it reduces emissions.

Singapore's National Electricity Market recorded 58.4 TWh of consumption in 2025 [41], so the reference facility alone equals about 1.6% of national demand. The load factor is distinct from server utilisation and rack power density. The same 100 MW of IT draws between 0.79 and 1.17 TWh a year across realistic load factors and PUE values (Table 1.5).

Table 1.5. Annual electricity demand (TWh) of 100 MW of IT capacity by load factor and PUE

Electrical load factor	PUE 1.20	PUE 1.25	PUE 1.40
75%	0.79	0.82	0.92
85%	0.89	**0.93**	1.04
95%	1.00	1.04	1.17

Reference case in bold.

At an 85% load factor, improving PUE from 1.40 to 1.25 saves 112 GWh a year, or 11%, which makes facility efficiency a valuable lever at existing sites; the PUE 1.25 baseline therefore understates the opportunity at the installed base, which Section 5.4 assesses at PUE 1.50. Scaled to the volumes of new capacity now under discussion, every additional 100 MW of IT adds about 0.93 TWh and 0.37 Mt CO_2 a year on today's grid (Table 1.6).

Table 1.6. Scale-up of incremental IT capacity at the reference assumptions

Incremental IT capacity	Annual electricity demand	Share of 2025 NEMS consumption	Grid-based emissions
100 MW	0.93 TWh	1.6%	0.37 Mt CO_2
300 MW	2.79 TWh	4.8%	1.12 Mt CO_2
500 MW	4.65 TWh	8.0%	1.87 Mt CO_2

Data centre loads are also large and continuous: at full load, 300 MW of IT capacity needs about 375 MW of facility power. Singapore's peak demand reached 8.6 GW in August 2026 and could rise to 9.9 to 13 GW by 2031, and EMA is responding with five new gas-fired plants that expand capacity by about 26% [42], including a 670 MW hydrogen-ready unit awarded to Tuas Power [43]. An efficient facility still adds to gas-fired capacity unless its residual demand is matched with low-carbon supply.

Electricity is also the dominant operating cost. The Uniform Singapore Energy Price (USEP) averaged S$116.6/MWh in 2025, but rose through 2026 to an average of S$196/MWh for January to August and S$332.5/MWh in August alone [44]. USEP is the wholesale energy component only, so this report uses an all-in price of S$200/MWh as the central value and treats higher prices as sensitivities (Table 1.7).

Table 1.7. Electricity cost sensitivity for the 100 MW reference case (0.93 TWh/yr)

All-in electricity price	Annual electricity cost
S$150/MWh	S$140 million
S$200/MWh (central)	S$186 million
S$250/MWh	S$233 million
S$300/MWh	S$279 million

Each S$10/MWh change in price moves annual expenditure by S$9.3 million, so the 112 GWh saved by moving from PUE 1.40 to 1.25 is worth S$22 million a year at the central price, and the 2026 price trend would raise the value of every efficiency result. The carbon tax is S$45/t$CO_2$e in 2026 and 2027, rising to S$50 to 80/t$CO_2$e by 2030 [45]; it is levied on generators, reaches data centres through the tariff, and serves in this report only as a benchmark for abatement cost.

1.5 Constraints on Sustainable Data Centre Growth

Singapore sets no fixed ceiling on data centre capacity. Growth instead takes place inside a declining economy-wide emissions envelope of 45 to 50 MtCO_2e by 2035 [5], and under the licence conditions in Table 1.1. Four constraints define what a low-carbon facility must achieve.

Firstly, low-carbon electricity must be secured at scale. Singapore had 2.3 GWp of solar in October 2026 and targets 3 GWp by 2030 and 4 GWp by 2035 [42], but solar can meet only about a tenth of projected 2050 needs [46]. Singapore therefore pursues a portfolio of regional imports, about 6 GW by 2035 [47] or a third of projected demand [2], and domestic low-carbon generation from ammonia, hydrogen, geothermal and, potentially, nuclear energy [48], [18].

Secondly, continuous data centre operation requires reliable electricity delivery, supported by firm generation, grid balancing and backup infrastructure. Singapore plans its system around a 27% reserve margin [49], and EMA plans to grow energy storage from about 350 MWh to just over 1 GWh by 2030 [42]. Because the facility load is flat over 24 hours, the hour-by-hour match between low-carbon delivery and load determines how much of the contracted volume can be verified as carbon-free (Section 4.5).

Thirdly, land and water favour integrated sites such as the planned low-carbon data centre park on Jurong Island [50], which Section 5.1 adopts as the development context of the reference facility. Median water use of about 2.1 m^3/MWh [51], or about 1.6 million m^3 a year on the reference facility's IT energy, is tracked on the performance dashboard of Chapter 4.

Fourthly, Singapore's heat and humidity raise the energy needed for cooling, which can account for up to 40% of a typical Singapore data centre's energy [12], while AI racks can draw up to 50 times the power of a conventional rack [16], making liquid and hybrid cooling a design requirement for new AI halls (Chapter 3). Together, these constraints define the conditions under which data centre growth is compatible with Singapore's climate commitments.

1.6 Decarbonisation Challenge and Proposed Strategy

The decarbonisation gap is the difference between the electricity that data centre growth requires and the low-carbon electricity that can be supplied within Singapore's climate,

resource and reliability constraints. The gap is material even under efficient operation: the reference facility at PUE 1.25 still needs 0.93 TWh and emits 374 kt CO_2 a year on today's grid.

Closing the gap requires both levers in Equations 1.1 and 1.2: efficiency shrinks the electricity that must be supplied and is the cheapest lever, and low-carbon generation lowers the carbon intensity of what remains. The strategy reduces demand first, adds rooftop PV, allocates shared ammonia electricity to the reduced load, prepares for SMR supply, keeps the grid for balancing and verifies the result hour by hour, cutting the reference facility's annual emissions by 90% in the long term (Chapter 5).

Chapter 2: Low-Carbon Electricity Supply Assessment

2.1 Introduction and Assessment Framework

Low-carbon supply is central to decarbonising Singapore's data centres, whose continuous demand makes the reliability, carbon intensity and cost of supply decisive. Limited land, scarce renewable resources, and imported fuels constrain domestic generation, and Khosravi et al. [23] distinguish on-site generation, limited by space, from off-site supply, which scales further but depends on third parties. Each option is screened against technological maturity, resource availability, scalability, reliability, economic feasibility, infrastructure requirements and deployment timeline (Table 2.1).

Three technologies are shortlisted: rooftop solar PV (Scenario A), SMRs (Scenario B) and low-carbon ammonia-fired generation (Scenario C). Each is applied to the reference data centre of Table 1.3, with the grid supplying residual demand, and evaluated for generation, emissions, expenditure and abatement cost. Generation follows from capacity and capacity factor:

$$E_i = P_i \times CF_i \times H \tag{2.1}$$

where P_i is the installed capacity (MW) and CF_i the capacity factor; for PV, $E_{PV} = P_{PV} \times Y_{PV}$, with Y_{PV} the specific yield (MWh/kWp/yr). Each MWh is allocated to one source (Equation 1.2), and the results feed the integrated proposal of Chapter 5.

Table 2.1. Low-carbon electricity options for Singapore and their status

Category	Option	Status in Singapore
Domestic renewable or low-carbon generation	Solar PV	Deployed at scale; limited by land; Scenario A
	Wind	No viable resource
	Hydropower	No suitable sites
	Geothermal	Under evaluation
	Nuclear (SMR)	National assessment under way; Scenario B
Imported fuels and electricity	Low-carbon ammonia	Jurong Island 55 to 65 MW project at front-end engineering design; Scenario C
	Hydrogen	Future fuel; new gas units built hydrogen-ready
	Gas with carbon capture (CCS)	Limited by geology and energy penalty
	Regional low-carbon electricity	Import links being developed; national policy context

2.2 Feasibility of Renewable and Low-Carbon Energy Sources

Singapore's land scarcity, low wind speeds and lack of fast-flowing rivers rule out most domestic renewable options [52]. The five generation technologies in Figure 2.1, together with low-carbon ammonia, are rated against the screening criteria in Figure 2.2.

Solar PV

Solar is the only domestic option already deployed at scale: Singapore reached 2 GWp in 2025, had 2.3 GWp by October 2026, is on track for 3 GWp by 2030 and has set a 4 GWp target for 2035, with more than 80% of capacity on rooftops [42], [46]. Average irradiation is about 1,580 kWh/m² a year [52], but limited land means solar can supply only about 10% of 2050 energy needs [46]. SERIS estimated Singapore's total technical PV potential at about

8.6 GWp by 2050 and projects specific yields of 1.32 to 1.45 MWh/kWp/yr in 2030 [30]. Installed rooftop cost is about S$1,000 to 1,400/kWp, with operating cost of 1% to 1.45% of upfront cost per year [37].

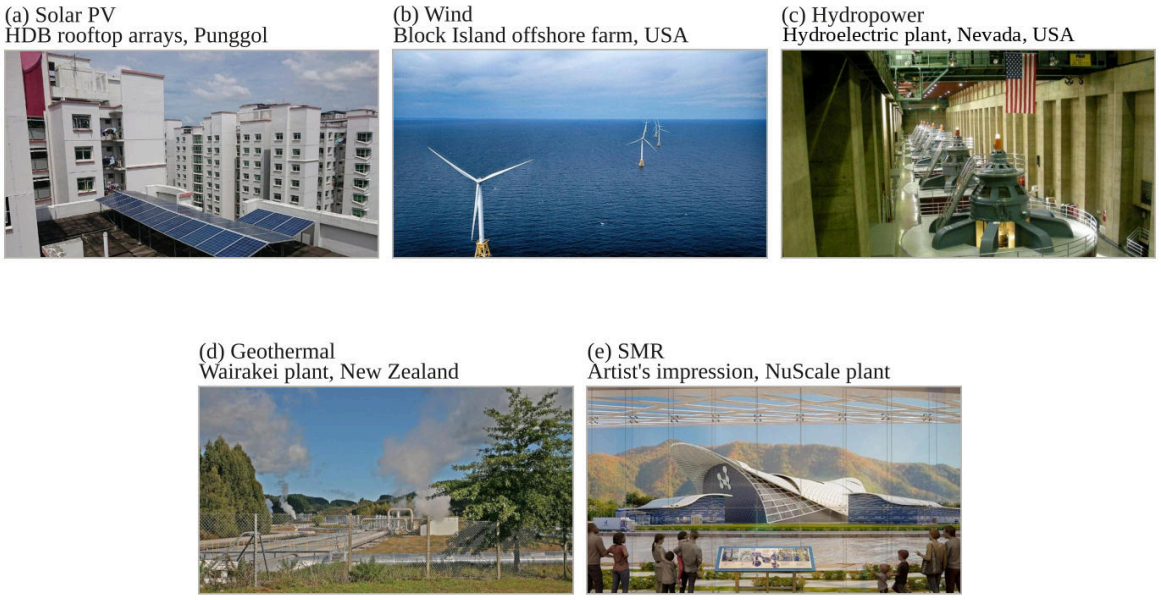


Figure 2.1. Domestic low-carbon electricity technologies assessed: (a) rooftop solar PV on HDB blocks (Deoma12); (b) Block Island offshore wind farm (Ionna22); (c) turbine deck of a hydroelectric power plant in Nevada (BuzzWeiser196); (d) Wairakei geothermal plant (W. Bulach); (e) artist's impression of a NuScale SMR plant (NuScale Power, via Oregon State University). Photographs used under CC BY-SA 4.0, except (e) under CC BY-SA 2.0.

Wind

Singapore's average wind speed of 2 to 3 m/s is below the roughly 4.5 m/s needed for viable turbines, and offshore sites are constrained by shipping traffic [53]. This makes wind energy not viable in Singapore.

Hydropower

Singapore has no river system with fast-flowing water for hydropower [52], making domestic hydropower not viable in Singapore.

Geothermal

Shallow geothermal resources of 60 to 80 °C are too cool for standard geothermal power plants [54], although recent surveys found temperatures up to 122 °C at 1.76 km deep, and about 230 °C may be reached at a depth of 5 km [55]. Enhanced geothermal systems (EGS) have high associated costs at US$140/MWh in the US today [56]. Singapore has stated that it is still evaluating the technical and economic feasibility of adopting geothermal energy and has not announced a deployment timeline.

Nuclear (SMR)

SMRs offer a consistent, low-carbon baseload power with a compact footprint. This energy production pattern matches the typical load of a data centre. SMRs are carried forward as Scenario B (Section 2.4).

Hydrogen and low-carbon ammonia

Hydrogen and ammonia are energy carriers whose emissions depend on how they are produced and shipped. Singapore's National Hydrogen Strategy sees hydrogen and its derivatives supplying up to half of power needs by 2050 [48]. Singapore has taken steps to ensure the domestic infrastructure is hydrogen ready, having introduced a law in 2024 that requires all new Combined Cycle Gas Turbines (gas-powered power plants) to be at least 30% hydrogen ready with the ability to be retrofitted to become 100% hydrogen ready [43]. Ammonia is a ready hydrogen carrier with an established international supply chain, can be stored and shipped over long distances and releases no CO_2 when burned [20], and the Jurong Island project gives it a local generation pathway, so it is carried forward as Scenario C.

Regional electricity imports and gas with carbon capture

Singapore has stated its aim to import 6 GW of low-carbon electricity from the region by 2035 (Section 1.5). An example is the ongoing Lao PDR-Thailand-Malaysia-Singapore Power Integration Project (LTMS-PIP) that started in 2022. The project allows Singapore to import up to 100 MW of hydropower from Lao PDR via Thailand and Malaysia. Carbon capture costs about 21% of plant output in Singapore's roadmap pilots, and Singapore has no suitable storage sites [57], so gas with CCS is screened out.

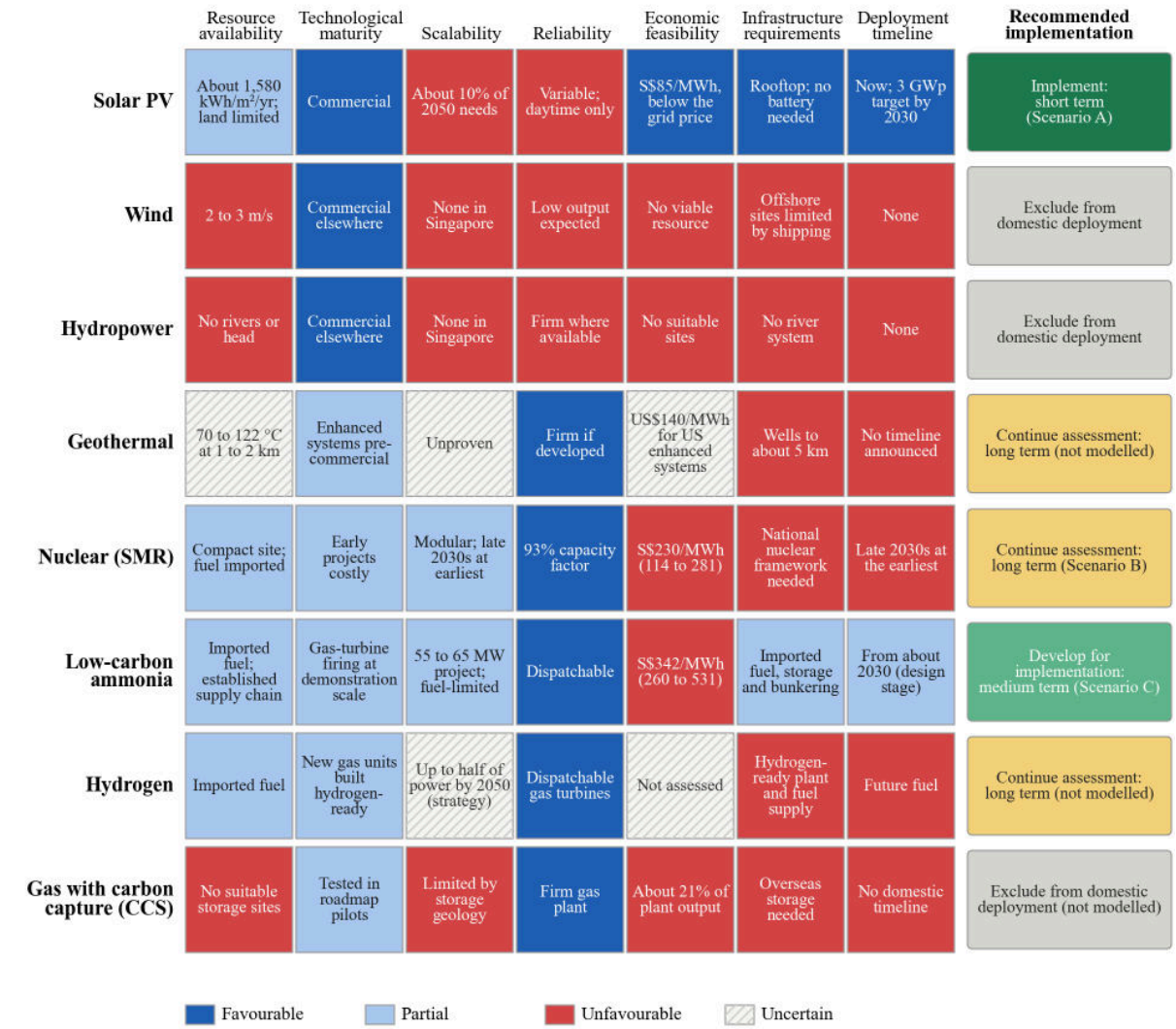


Figure 2.2. Feasibility screening matrix and recommended implementation. Dark blue cells are favourable, light blue partial and red unfavourable for deployment in Singapore, and hatched cells (uncertain) mean the evidence is insufficient to rate the criterion.

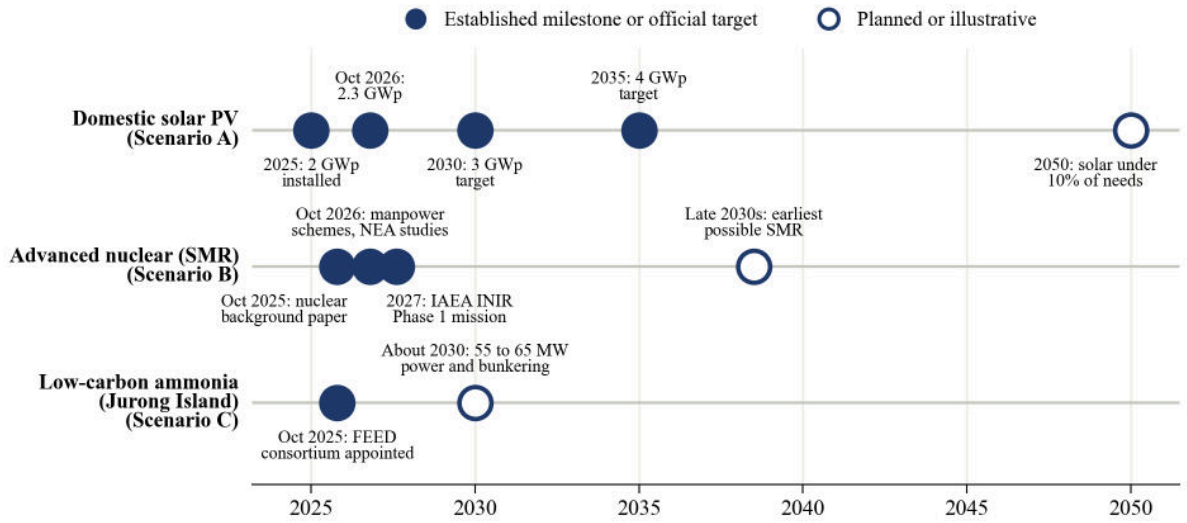


Figure 2.3. Development timeline of the supply options modelled in Scenarios A, B and C. Filled markers are milestones or official targets; hollow markers are planned or illustrative, and the late-2030s SMR date is this report's assumption [19], [32]. Sources: [42], [46], [20], [59], [18], [60], [61], [19].

Rooftop PV is recommended for short-term implementation, ammonia is developed for medium-term implementation as the dispatchable scenario, and SMRs remain under assessment as the long-term firm scenario (Figures 2.2 and 2.3).

2.3 Scenario A: Solar Photovoltaic Generation

The 50 MW Digital Loyang II data centre comprises five storeys with approximately 34,000 m² of gross floor area on a 12,800 m² land parcel [29], [62]. Assuming approximately equal floor plates across the five storeys, the building footprint and corresponding rooftop area are about 6,800 m², and the gross floor area equates to 680 m² per MW of IT capacity. Scaling these relationships to the 100 MW reference facility gives approximately 68,000 m² of gross floor area and 13,600 m² of rooftop area. Applying the SERIS rooftop utilisation factor of 60% and the lower-bound installed capacity density of 0.14 kWp/m² produces a reference solar PV installation of approximately 1.14 MWp, which at a specific yield of 1.385 MWh/kWp/yr generates 1,582 MWh annually [30], or 0.17% of facility demand.

The PV output peaks at 0.57 MW against an average load of 106 MW, so it is consumed entirely on site (Figure 2.4) and needs no battery. The whole roof at the highest density and yield gives 0.40% of demand, and doubling the area to include car parks gives 0.81%. At S$85/MWh (Section 2.8), rooftop PV generates below the grid price and belongs on every suitable roof.

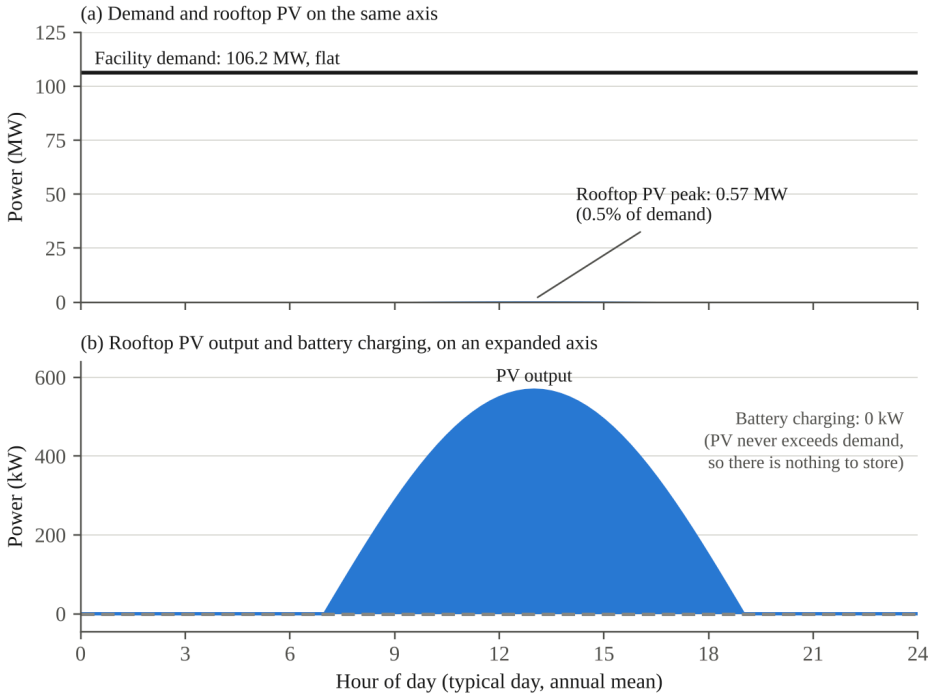


Figure 2.4. Typical-day solar PV supply profile for the reference data centre (Scenario A). A daytime half-sine profile is scaled to the annual yield of 1,582 MWh/yr; demand is flat at 106.25 MW, so battery charging is zero.

2.4 Scenario B: Advanced Nuclear Energy and Small Modular Reactors

SMRs are factory-built fission reactors of up to about 300 MWe with passive safety systems [27]. In a pressurised-water design, fission heat passes through a steam generator to a turbine-generator (Figure 2.5), and firm output at a 93% capacity factor [32] matches a flat data centre load more closely than any other low-carbon source. The technology is moving to construction: Ontario Power Generation received a licence for the first 300 MWe BWRX-300 at Darlington in 2025, placed the reactor basemat in April 2026 and targets grid connection by the end of 2030 [63].

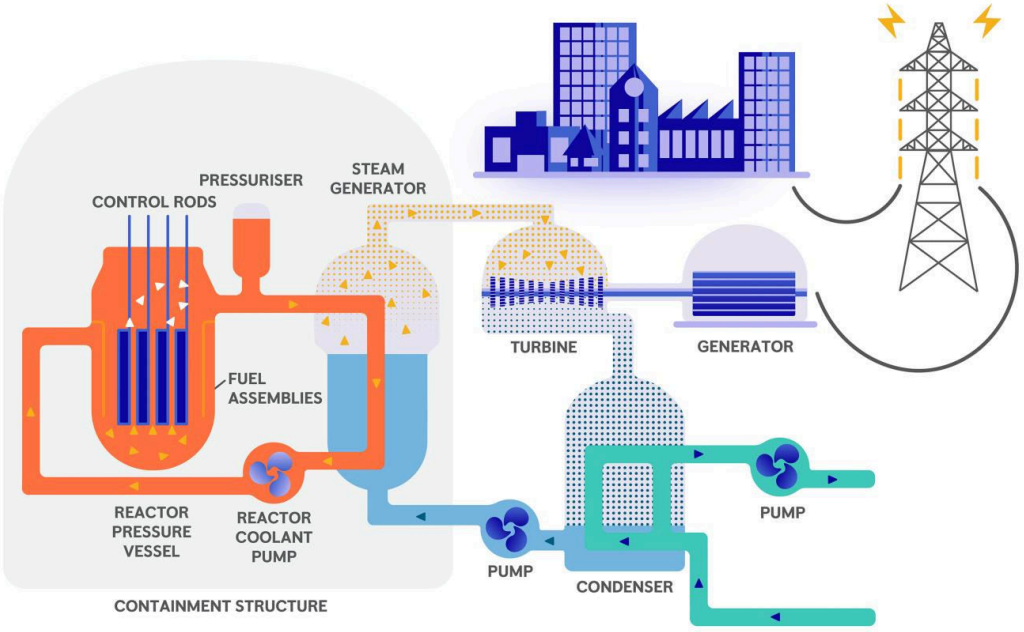


Figure 2.5. Heat-to-electricity process of a pressurised-water SMR: reactor heat passes through the steam generator to the turbine-generator, and the condenser returns feedwater. Source: Rolls-Royce SMR [33]; the 77 MWe reference module is an integral pressurised-water design using the same steam cycle.

Singapore is building the capability to decide on this option. EMA's 2025 background paper assesses nuclear energy against safety, reliability, affordability and environmental sustainability and records that the Singapore Nuclear Research and Safety Initiative became a full institute in July 2025 [18]; the Government has invested more than S$150 million in nuclear research [64]. In October 2026 METI and EMA launched three manpower schemes with partners including AtkinsRéalis, CNNC Overseas and X-energy [60], [65], and NEA commissioned nuclear safety, regulatory and environmental studies [61], preparing for the IAEA INIR Phase 1 mission in 2027 [19], which reviews the 19 infrastructure issues of the IAEA Milestones Approach (Figure 2.6) [66].

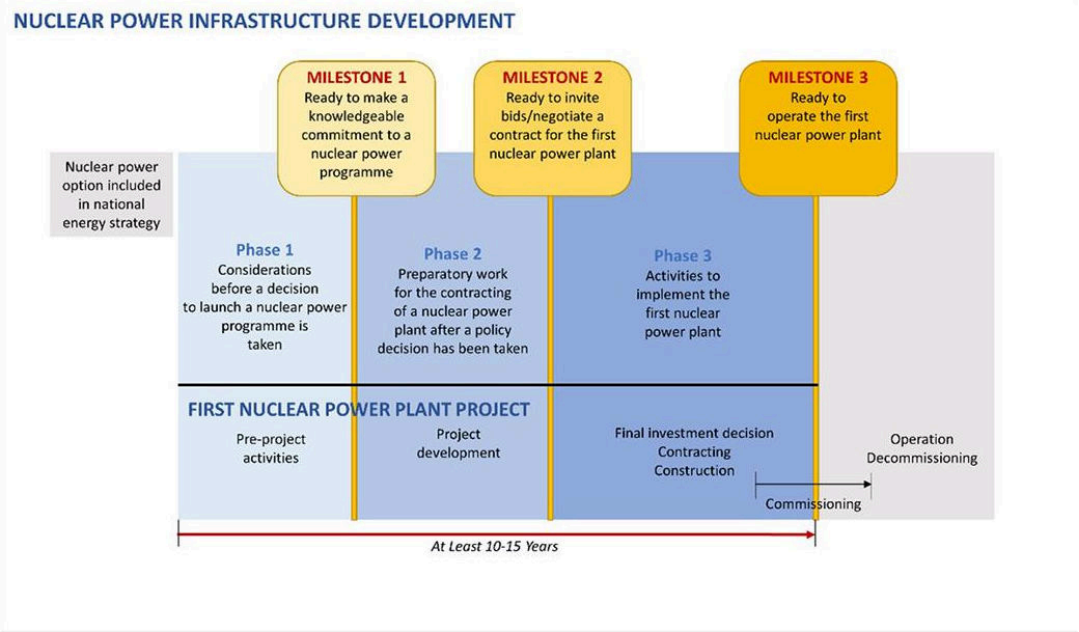


Figure 2.6. The IAEA Milestones Approach: three phases and three milestones from including nuclear power in the national energy strategy to operating the first plant, a process of at least 10 to 15 years. Source: IAEA [66].

EMA notes that the smaller size of SMRs could allow a smaller safety buffer zone where land is limited [27]. Costs fall with series construction: the first-of-a-kind NuScale project, six modules totalling 462 MWe, or 77 MWe each, was estimated at US$20,139/kW and US$89/MWh after subsidy [31] before its cancellation [67]; Lazard's 2025 range for new-build nuclear is US$141 to 220/MWh [38]; and OPG expects later Darlington units to cost C$4 to 5 billion each, against C$6.1 billion for the first [63].

The representative case adopts one 77 MWe module [31] at a 93% capacity factor [32]: $E_{SMR} = 77 \times 0.93 \times 8{,}760 = 627{,}304$ MWh/yr, or 67.4% of reference demand (Table 1.4). With construction times of 43 to 71 months [32] and the IAEA's 10 to 15 years from national consideration to first operation [66], an SMR can supply a Singapore data centre from the late 2030s.

2.5 Scenario C: Low-Carbon Ammonia-Fired Generation

Low-carbon ammonia provides dispatchable electricity from an imported carbon-free fuel. In October 2025 EMA and MPA appointed a Keppel-led consortium for front-end engineering design of a Jurong Island project to generate 55 to 65 MW from imported low- or zero-carbon ammonia by direct combustion in a combined-cycle gas turbine (GTCC), with bunkering of at least 0.1 Mt a year; the project awaits a final investment decision [20]. The reference plant is a 60 MWe direct-ammonia GTCC in the new-energy area that also hosts the data centre park [50].

Combustion and plant configuration

Ammonia burns without carbon: $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$ under ideal combustion. Its low laminar burning velocity, high minimum ignition energy and narrow flammability limits make flame stability the central design challenge, and fuel-bound nitrogen raises NO_x, so ammonia gas turbines use staged rich-quench-lean combustion, hydrogen enrichment and selective catalytic reduction (SCR) [58]. N_2O stays marginal: Ahmad et al. model 0.18 ppmv

in direct liquid-ammonia combustion [34], and Shin et al. find N_2O below 1 ppm with ammonia slip at ppb levels [39].

In a GTCC, the turbine exhaust raises steam in a heat-recovery steam generator (HRSG) for a bottoming steam turbine (Figure 2.7). Simulating an ammonia GTCC based on a 43 MW plant, Ahmad et al. obtained net efficiencies of 47.7% for direct liquid-ammonia injection, 49.1% for vaporised ammonia and about 54% with full cracking, with liquid injection giving the lowest NO_x [34]. This report adopts the direct liquid-ammonia case, matching the project's direct-combustion design, on a lower heating value of 18.8 MJ/kg. Annual fuel demand follows from plant output:

$$m_{NH3} = SFC_{NH3} \times E_{NH3} = 3.6 \times E_{NH3} / (\eta_{NH3} \times LHV_{NH3}) \quad (2.2)$$

At an 85% capacity factor, $E_{NH3} = 60 \times 0.85 \times 8{,}760 = 446{,}760$ MWh/yr, and Equation 2.2 gives $3.6/(0.477 \times 18.8) = 0.401$ tNH_3/MWh and 179,350 t of fuel a year, about 490 t a day; a seven-day tank [34] holds about 3,400 t.

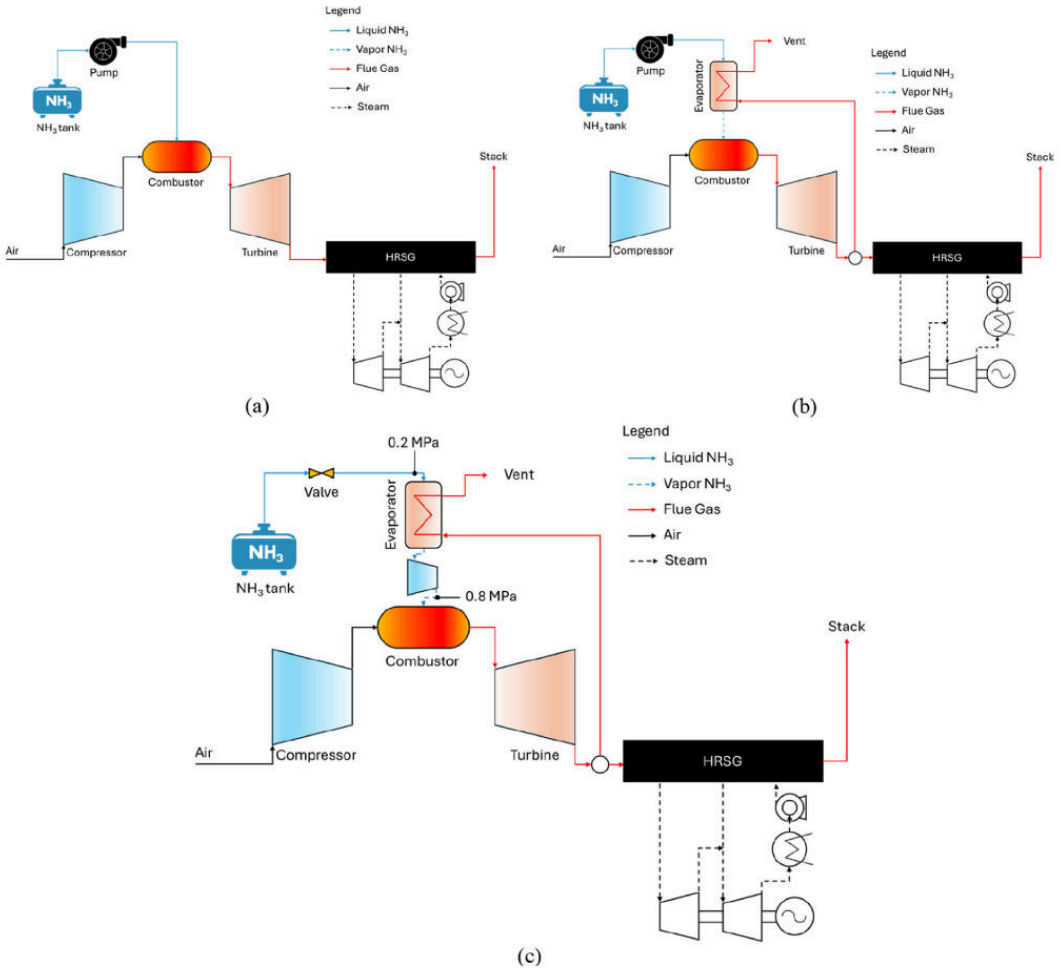


Figure 2.7. Ammonia combined-cycle configurations: (a) Case 1, direct liquid NH_3, adopted here; (b) Case 2a and (c) Case 2b, vaporised NH_3. Source: Ahmad et al. [34], Fig. 1, unaltered, CC BY-NC-ND 4.0.

Supply chain and life-cycle emissions

The climate value of ammonia is set upstream (Figure 2.8), and published intensities differ mainly in system boundary. Lee et al. assess imported ammonia on a cradle-to-gate boundary extended to shipping, counting construction and maintenance of newly built generating plant, electrolyser and air separation unit [36]. Their imported wind-based pathway reaches 1.03

tCO_2e/tNH_3: hydrogen production 0.80, synthesis and air separation 0.11, and shipping 0.12 over 12,553 km. Two inventory choices dominate: electrolyte production adds 0.46 tCO_2e/tNH_3, which Lee et al. identify as the main reason their result exceeds earlier studies, and a 70% Haber–Bosch conversion raises hydrogen demand 29% above stoichiometry [36]. With GREET 2025, Zhang et al. put the operational upstream footprint, without plant construction, at 0.03 tCO_2/tNH_3 for renewable and 0.66 for blue ammonia [35], and Shin et al. confirm that green ammonia has the lowest warming potential [39]. Table 2.2 converts each pathway with:

$$EF_{NH3} = SFC_{NH3} \times I_{NH3} + EF_{N2O} \quad (2.3)$$

Table 2.2. Life-cycle intensity of imported ammonia pathways and the resulting electricity emission factor

Pathway	Production and plant (tCO_2e/tNH_3)	Shipping (tCO_2e/tNH_3)	Delivered intensity, I_{NH3}	EF_{NH3} (tCO_2e/MWh)	Against the grid
Renewable ammonia, operational boundary	0.03 [35]	0.12 [36]	0.15	0.060	−85%
Renewable ammonia, life-cycle boundary (central)	**0.03 [35] + 0.126 [26], [36]**	**0.12 [36]**	**0.276**	**0.111**	**−72%**
Blue ammonia, operational boundary	0.66 [35]	0.12 [36]	0.78	0.313	−22%
Imported wind-based ammonia, full inventory	0.91 [36]	0.12 [36]	1.03	0.414	+3%
Imported blue ammonia, full inventory	2.28 [36]	0.12 [36]	2.40	0.964	+140%

SFC_{NH3} = 0.401 tNH_3/MWh; EF_{N2O} = 0.0003 tCO_2e/MWh; embodied electricity 0.011 tCO_2e/MWh × 11.5 MWh/tNH_3 (Appendix C.5). Grid 0.402 tCO_2/MWh; break-even at 1.0 tCO_2e/tNH_3.

The central factor puts ammonia on the life-cycle basis of rooftop PV and the SMR, whose IPCC factors include plant manufacture [26]. It adds to the GREET value the embodied emissions of the renewable electricity, at the IPCC onshore wind median of 0.011 tCO_2e/MWh [26] for Lee et al.'s electrolysis demand of 11.5 MWh per tonne [36], and Lee et al.'s shipping term: 0.03 + 0.126 + 0.12 = 0.276 tCO_2e/tNH_3, so EF_{NH3} = 0.111 tCO_2e/MWh, 72% below the grid. The shipping term is conservative for the shorter voyage from the Middle East to Singapore. The operational value (0.15) and Lee et al.'s full inventory (1.03) bracket the central case (Section 5.5).

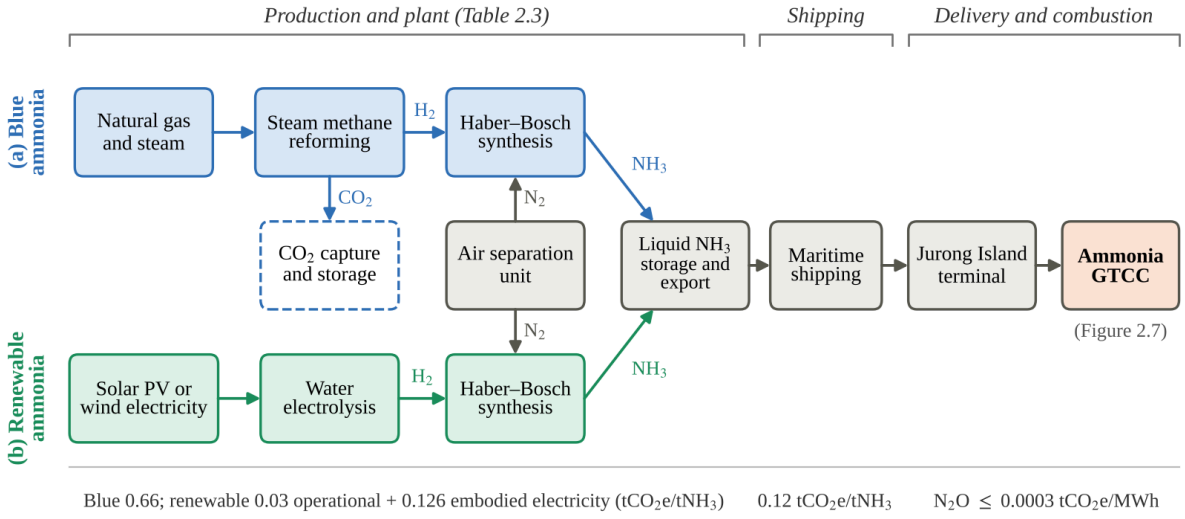


Figure 2.8. Imported ammonia supply chains of Table 2.2: (a) blue ammonia from steam methane reforming with carbon capture and storage; (b) renewable ammonia from electrolysis powered by solar or wind electricity; both are shipped as liquid ammonia to Jurong Island for the GTCC. Authors' schematic based on Lee et al. [36], Fig. 1, and Zhang et al. [35].

Generation cost and shared operation

Fuel dominates ammonia economics: Zhou and Duan found fuel at 68% to 70% of the levelised cost of ammonia combined cycles, against 29% for gas, and a 23.4% cost reduction from scaling 50 MW to 830 MW [68]. Applying Equation 2.4 to Ahmad et al.'s liquid-ammonia case [34] with the inputs of Table 1.4, capital and fixed costs contribute S$96/MWh and fuel at US$480/t contributes S$246/MWh (0.401 tNH_3/MWh × US$480/t × S$1.276 per US$), giving S$342/MWh (Appendix C.6); grid parity requires about US$203/t. The US$480/t base lies within Shin et al.'s current green ammonia range of US$400 to 850/t, which falls to US$275 to 450/t by 2040 [39].

A 60 MWe plant suits shared operation: it runs at design capacity factor, the data centre buys its allocated MWh at levelised cost, and other Jurong Island consumers take the balance. Alone at PUE 1.25, the plant supplies 48.0% of reference demand, close to the 50% green-energy share DC-CFA2 requires of new capacity, through an eligible ammonia pathway [14].

2.6 Reference Facility and Electricity Generation

Table 2.3 applies each technology to the reference facility at PUE 1.25, with the grid supplying the residual demand.

Table 2.3. Annual generation, share of demand and residual grid electricity of the supply scenarios at PUE 1.25

Scenario	Generation (MWh/yr)	Share of demand	Residual grid (MWh/yr)
A: Rooftop PV, 1.14 MWp	1,582	0.17%	929,168
B: SMR, 77 MWe	627,304	67.4%	303,446
C: Ammonia GTCC, 60 MWe	446,760	48.0%	483,990

Scenario	Generation (MWh/yr)	Share of demand	Residual grid (MWh/yr)
A + B + C portfolio	PV 1,582; SMR 627,304; ammonia 301,864	100%	0

In the combined portfolio the facility takes 301,864 MWh of ammonia electricity, 67.6% of plant output, and refuelling outages are covered by ammonia dispatch and the grid.

2.7 Comparative Emissions Assessment

Comparative emissions follow from Equation 1.2; for each standalone scenario, $CO_2e = E_i \times EF_i + (E_{DC} - E_i) \times EF_{grid}$, against a baseline of 374,162 tCO_2e/yr. Rooftop PV avoids 571 tCO_2e/yr, the ammonia plant 129,901 (34.7%) and one SMR module 244,648 (65.4%), and the A + B + C portfolio avoids 89.0% (Figure 2.9). Because every low-carbon factor is below 30% of the grid's, the share of demand supplied sets the emissions avoided, and for ammonia the delivered fuel intensity is decisive (Table 2.2).

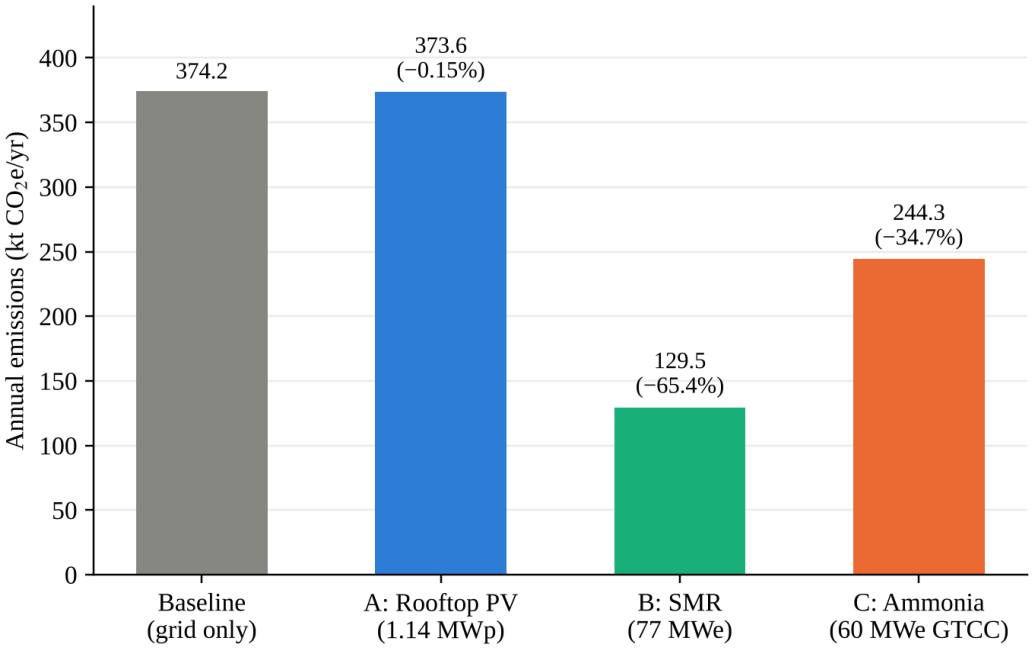


Figure 2.9. Comparative annual electricity-related emissions of the standalone supply scenarios at PUE 1.25, with the grid supplying the residual demand. Values are listed in Table 2.4.

2.8 Techno-Economic Assessment

Annual costs follow from Equation 1.3 with the generation costs of Table 1.4. PV, SMR and ammonia output are costed at their levelised cost of electricity:

$$LCOE = 1{,}000 \times [CAPEX \times CRF + FOM] / (H \times CF) + VOM + Fuel \qquad (2.4)$$

where CAPEX is capital cost (S$/kW), CRF the capital recovery factor (Equation 1.4), FOM fixed operating cost (S$/kW-yr), and VOM and Fuel variable costs (S$/MWh); the factor 1,000 converts S$/kWh to S$/MWh, since H × CF is in kWh per kW-year. CRF is 0.0858 (7%, 25 years) for PV and ammonia and 0.0750 (7%, 40 years) for the SMR.

Rooftop PV costs S$85/MWh, the SMR S$230/MWh and the ammonia plant S$342/MWh (Figure 2.10). The first-of-a-kind NuScale capital of US$20,139/kW is a cross-check only: its

capital recovery alone is S$237/MWh (Appendix C.3), so an unsubsidised first unit would exceed the central case.

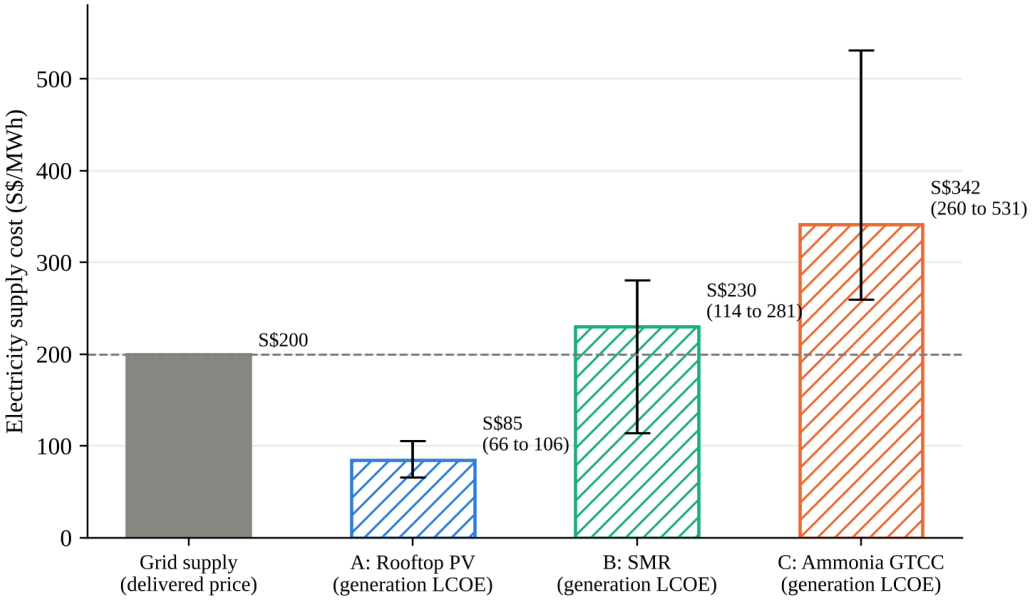


Figure 2.10. Electricity supply cost comparison. Hatched bars are levelised generation costs at the facility boundary (Equation 2.4); the solid bar is the delivered grid price, which includes network and balancing charges. Bars show central values and whiskers the ranges in Table 1.4.

Rooftop PV saves S$0.18 million a year at a net saving of S$319/tCO₂e. The SMR adds S$18.8 million at S$77/tCO₂e, from a net saving at the subsidised cost to S$208/tCO₂e at the top of Lazard's range. The ammonia plant adds S$63.4 million at S$488/tCO₂e, or S$206/tCO₂e at US$320/t, the same order as Zhang et al.'s US$242/tCO₂ for green ammonia in a gas combined cycle [35] (Figure 2.11).

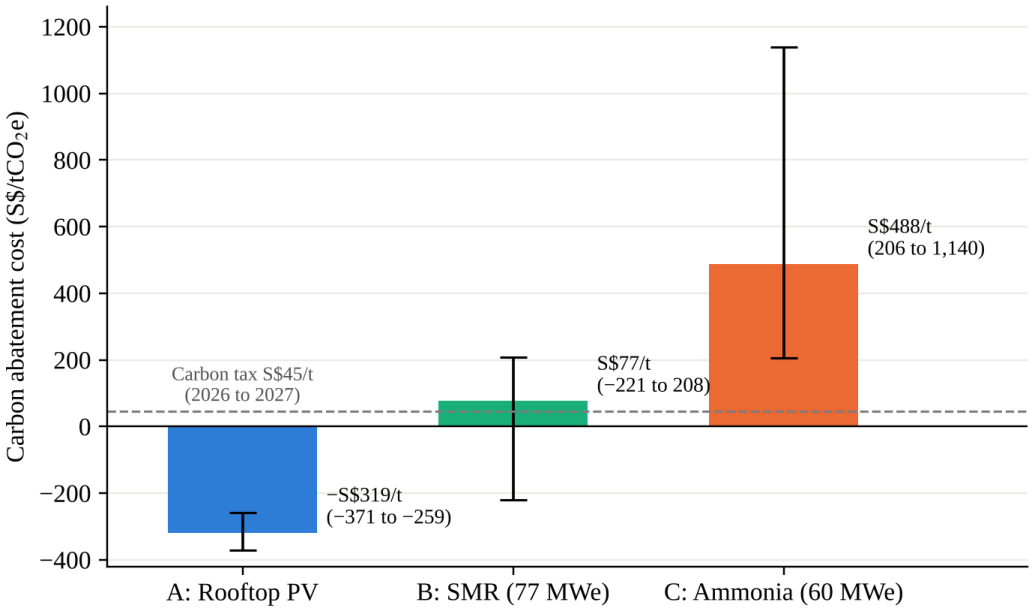


Figure 2.11. Carbon abatement cost by scenario. Bars show central values and whiskers the ranges in Table 2.4, derived from the cost ranges of Table 1.4; negative values are net savings. The dashed line is the 2026 to 2027 carbon tax [45].

2.9 Comparative Evaluation and Recommended Supply Strategy

The three technologies are complementary (Table 2.4). Rooftop PV is mature and cheaper than the grid but capped at 0.17% of demand. Ammonia is dispatchable this decade and, with renewable fuel, cuts the life-cycle emissions of almost half of demand by 72% at a cost set by the fuel price. SMRs deliver the largest and cheapest supply-side abatement with firm output, on a timeline set by a national decision.

Table 2.4. Annual electricity, emissions and cost results of the supply scenarios for the 100 MW reference data centre (PUE 1.25; S$M = S$ million)

Scenario	Low-carbon (GWh)	Emissions (tCO_2e)	Avoided (tCO_2e)	Cost (S$M)	ΔC (S$M)	AC (S$/$tCO_2e$)
Baseline (grid only)	0	374,162	0	186.15	0	-
A: Rooftop PV	1.6	373,590	571	185.97	−0.18	−319 (−371 to −259)
B: SMR, 77 MWe	627.3	129,513	244,648	204.97	+18.82	77 (−221 to 208)
C: Ammonia, 60 MWe	446.8	244,260	129,901	249.52	+63.37	488 (206 to 1,140)
A + B + C portfolio	930.8	41,171	332,990	247.60	+61.45	184

The recommended strategy installs rooftop PV now, contracts shared ammonia electricity from the Jurong Island plant as it enters service, and adds SMR output when Singapore's assessment permits, reducing the ammonia allocation to the residual demand (Figure 2.12), with the grid for balancing throughout. Together the pathways remove 89% of electricity-related emissions at S$184/$tCO_2e$.

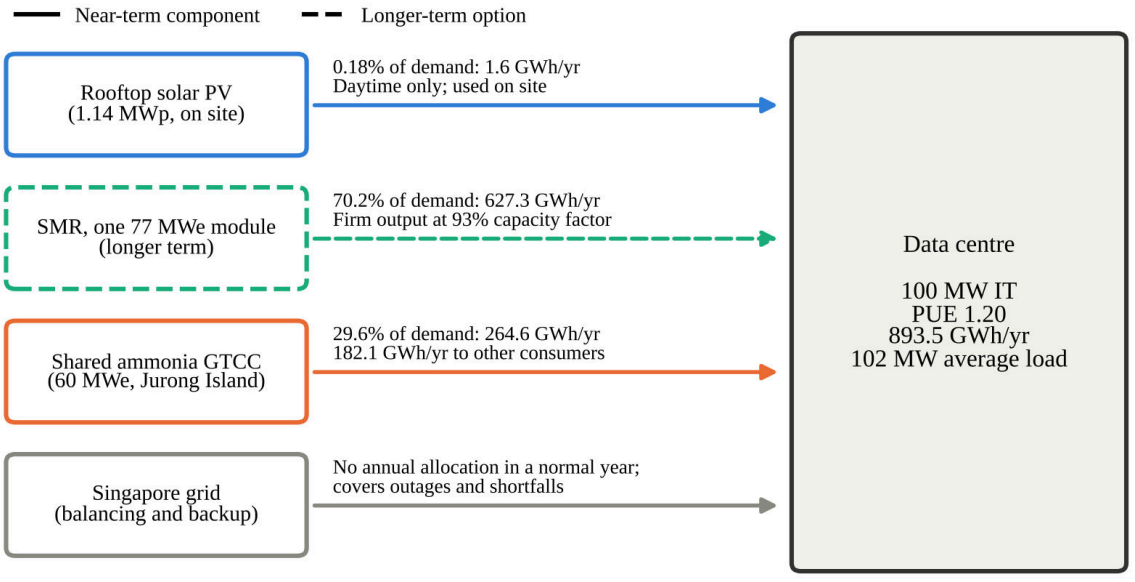


Figure 2.12. Proposed low-carbon supply arrangement for the reference data centre in the integrated long-term configuration (PUE 1.20). Percentages are shares of annual demand (Table C.1); the shared ammonia plant's remaining output serves other Jurong Island consumers.

Chapter 3: Data Centre Energy Efficiency and Optimisation

3.1 Introduction and Energy Efficiency Framework

Energy efficiency determines how much electricity a data centre requires before the question of supply arises. Demand-side measures, which reduce the electricity drawn for a given IT workload, are assessed on the same 100 MW reference data centre as the supply options of Chapter 2.

Supply and efficiency act on different terms of one physical identity (Figure 3.1). Low-carbon supply lowers the carbon intensity of each kilowatt-hour. Efficiency acts on the bracket: less IT energy per unit of useful work, and less facility overhead per unit of IT energy. Because domestic solar is land-capped, ammonia electricity carries fuel cost and SMRs await a national decision (Chapter 2), every kilowatt-hour removed from the bracket is one that low-carbon supply never has to provide.

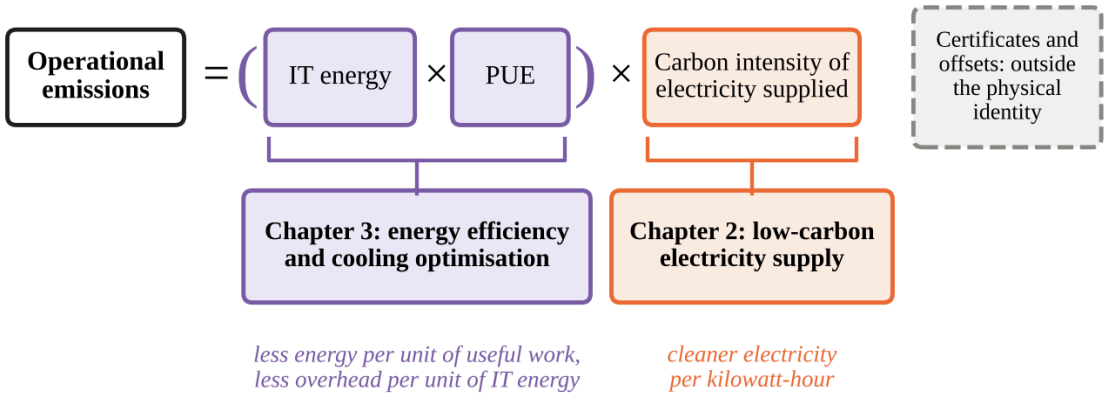


Figure 3.1. The physical emissions identity and the term each chapter acts on. Market instruments do not enter the physical equation.

Power usage effectiveness and useful work

Facility overhead is measured by power usage effectiveness (PUE), defined in ISO/IEC 30134-2 as the ratio of total facility energy to IT equipment energy over a year [69]:

$$\mathrm{PUE} = E_{\mathrm{facility}} / E_{\mathrm{IT}} = 1 + (E_{\mathrm{cooling}} + E_{\mathrm{power\ losses}} + E_{\mathrm{lighting,\ other}}) / E_{\mathrm{IT}} \qquad (3.1)$$

A PUE of 1.25 means that every kilowatt-hour delivered to servers requires a further 0.25 kWh for cooling, power conversion and other services. Figure 3.2 shows where that overhead arises: conversion losses in transformers, the uninterruptible power supply (UPS) and distribution, and the chillers and climate-control units that remove server heat. Cooling is the largest overhead: in the global installed base, IT equipment and cooling account for about 48% and 38% of data centre electricity [70], whereas the reference plant uses 0.175 of IT energy for cooling, about 14% of facility electricity. PUE is simple to measure and underpins Singapore's targets, but it rises when a server refresh cuts IT energy, contains no measure of useful work, understates the total saving of direct-to-chip liquid cooling (Section 3.2) and, as an annual average, hides the peak behaviour that sizes tropical plant. Absolute electricity, the cooling-system efficiency ratio (CSE = $P_{\mathrm{cool}}/P_{\mathrm{IT}}$) used by the NUS studies and PUE are therefore reported side by side (Section 3.5).

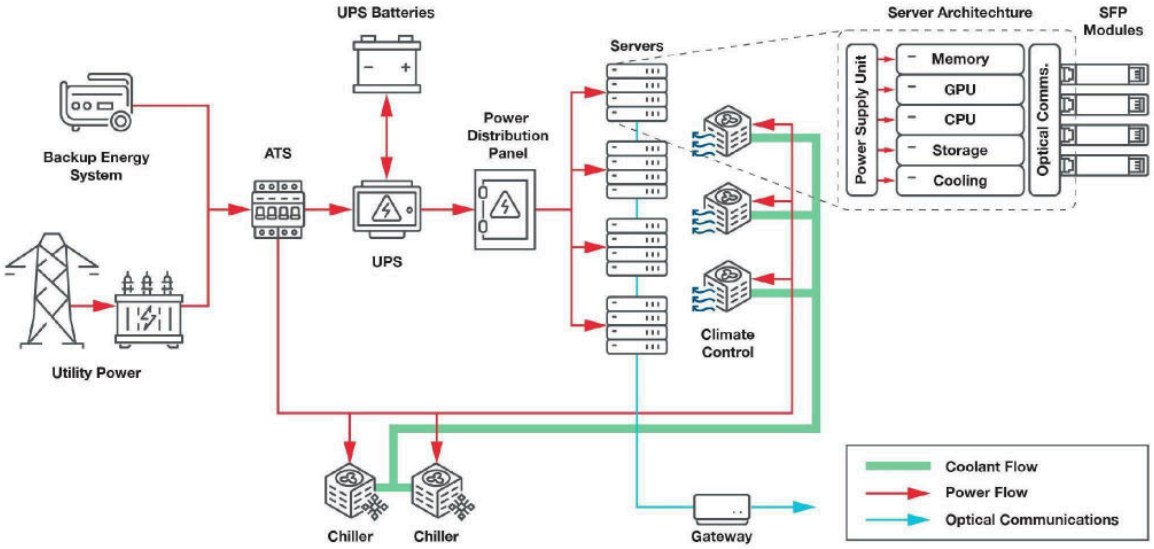


Figure 3.2. Power and coolant paths in a data centre, from utility supply and backup through the automatic transfer switch (ATS), UPS and distribution panel to the servers, with chillers and climate control removing heat. Source: Yokogawa Test & Measurement [71].

Following one watt through the data centre

Data centre energy performance can be assessed through three physical pathways, while policy provides the framework for their implementation (Table 3.1).

Table 3.1. Three physical energy pathways and the associated policy framework

Link	Metric	Question	Methods	International example	Singapore precondition
1. Compute demand	Utilisation	How much of the watt is useful compute?	Leaner software; carbon-aware scheduling; idle-server reclaim	Google shifts batch jobs to low-carbon hours [72]	A 94% gas grid is flat in carbon by hour; SS 715 consolidation still pays
2. Facility overhead	PUE, CSE	How much overhead does each watt drag?	Warmer halls; free cooling; liquid cooling; plant control	Air-side free cooling at sites in cool climates	No free cooling; SS 697 raises hall temperature; liquid cooling and control transfer
3. Waste heat	ERF	Where does the waste heat go?	Loop capture; heat pumps; an offtaker	Meta Odense recovers heat for about 6,900 homes [73]	No district heating demand
Policy framework	Compliance, disclosure	Does anything require links 1 to 3?	Statutory caps; mandatory reporting	Germany caps new sites at PUE 1.2 [74]; EU reporting above 500 kW [75]	The Digital Infrastructure Bill provides the legal hook

Singapore's climate removes the cheapest tool that cooler countries use, air-side free cooling, which needs outdoor air cooler than the hall supply temperature. Link 2 therefore offers Singapore the most room, because overhead can be cut at existing sites without new land or supply; the chilled-water plant and liquid cooling for new AI halls are the two principal opportunities.

3.2 Singapore-Specific Cooling Challenges and Opportunities

Why free cooling is unavailable

Air-side economisers need outdoor air below the hall supply temperature, but Singapore's outdoor air is hot and humid all year. Water-side economisers are also constrained by Singapore's warm, humid conditions. The STDCT cooling tower in Table 3.3 is rated at a design wet-bulb temperature of 30.2 °C, with condenser water entering at 40 °C and leaving at 34 °C; this is an equipment design point, not a typical annual wet-bulb temperature. At such conditions, the available cooling-water temperature is insufficient for the modelled chiller-bypass arrangement. Mechanical refrigeration therefore runs continuously, and cooling can account for up to 40% of a typical Singapore data centre's energy [12].

Where the room lies: the chilled-water plant

Figure 3.3 shows a typical primary-secondary chilled-water plant: chillers produce chilled water that secondary pumps deliver to computer-room air handlers, condenser-water pumps and cooling towers reject the heat, and a water-side economiser bypasses the chillers when conditions allow. In Singapore the economiser branch is effectively unused, so chillers, pumps and tower fans dominate facility overhead.

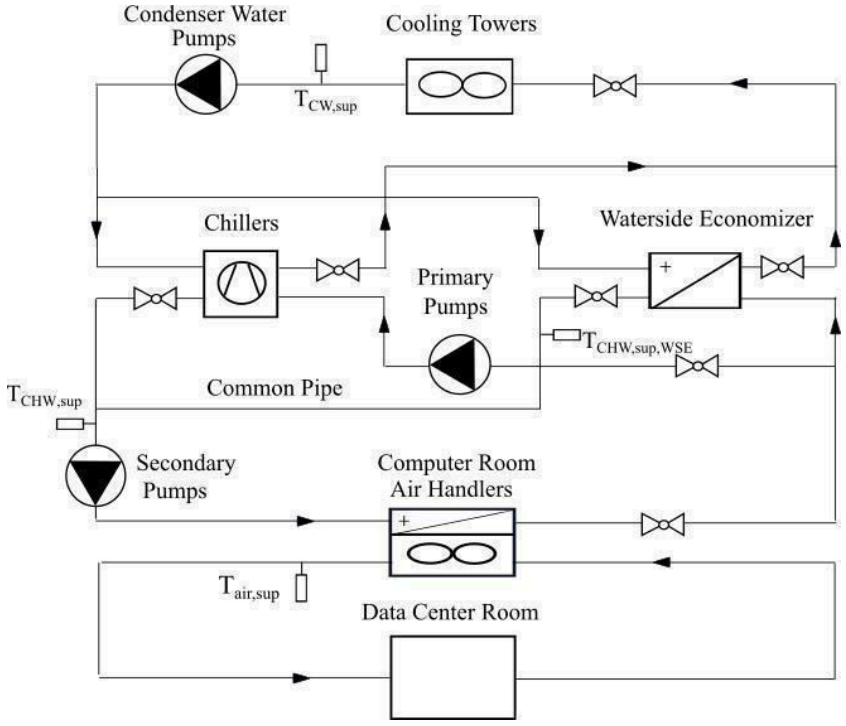


Figure 3.3. Chiller-cooled data centre plant with primary and secondary chilled-water pumps, condenser-water pumps, cooling towers, a water-side economiser and computer-room air handlers. Source: Modelica Buildings Library, Lawrence Berkeley National Laboratory [76].

Three physical relationships make this plant controllable. Chiller power falls as the chilled-water supply temperature rises and the condenser-water temperature falls, because both reduce the compressor lift. Pump and fan power fall with the cube of speed, so a 20% speed reduction cuts their power by almost half. A plant run on fixed, peak-sized setpoints therefore over-cools and over-pumps for most of the year, and resetting temperatures, speeds and chiller staging to the load recovers that energy once the plant is sub-metered and modelled.

Commercial evidence confirms the size of the prize. At Google, a machine-learning system trained on plant sensor data cut cooling energy by 40% on a live data centre [77], and model predictive control has cut cooling or chiller energy by 9% to 28% in live and real-plant tests (Section 3.4).

Liquid cooling for high-density AI halls

Global average rack power density rose from 2.4 kW in 2011 to 8.4 kW in 2020 [70], and AI racks can draw up to 50 times the power of a conventional rack. Singapore's SS 726:2026, the first liquid-cooling standard for tropical data centres, states that liquid cooling can cut data centre energy use by more than 30% against air cooling [16]. Direct-to-chip cold plates accept warm facility water, so their heat can be rejected through cooling towers with little chiller work even in the tropics. The STDCT itself targets a PUE below 1.2 for a combination of air and liquid cooling [78], which supports the PUE 1.20 target used for hybrid liquid cooling.

PUE also understates the benefit of liquid cooling. A study by NVIDIA and Vertiv of a high-density hybrid hall found that moving 75% of the load to liquid cooling cut facility power by 18.1% and total data centre power by 10.2%, yet PUE fell only from 1.38 to 1.34, because server fan power dropped by 80% and IT power by 7%, shrinking the denominator of the ratio as well as its numerator [79]. Hybrid air and liquid cooling is therefore the correct design for every new AI hall, with its benefit reported in total energy as well as PUE.

3.3 Physics-Informed Thermal Modelling of Data Centres

A thermal model that keeps physics as a hard constraint while letting its coefficients adapt predicted a Singapore data hall a week ahead with about one-fifth of the error of a fixed-coefficient model [80]. Accurate prediction is what allows predictive control to move setpoints towards their limits safely.

The dataset [80] covers six months of 30 s logs, 521,280 samples, with five months for training and one unseen month for testing.

The baseline Physically Consistent Neural Network (PCNN) treats the hall as a lumped thermal mass and predicts the next-step hall temperature as an energy balance plus a learned residual:

$$T_{DH}(k+1) = T_{DH}(k) - a[T_{DH}(k) - T_{SA}(k)]v_{SA}(k) + bP_{ITE}(k) + D_k \quad (3.2)$$

where T_{SA} and v_{SA} are supply-air temperature and flow, P_{ITE} is IT power and D_k is a learned term driven by outdoor conditions. The adaptive version (A-PCNN, Figure 3.4) lets the coefficients a and b vary with operating conditions but constrains them to be positive, so that, by construction, more IT power warms the hall and colder or faster supply air cools it, including outside the training data.

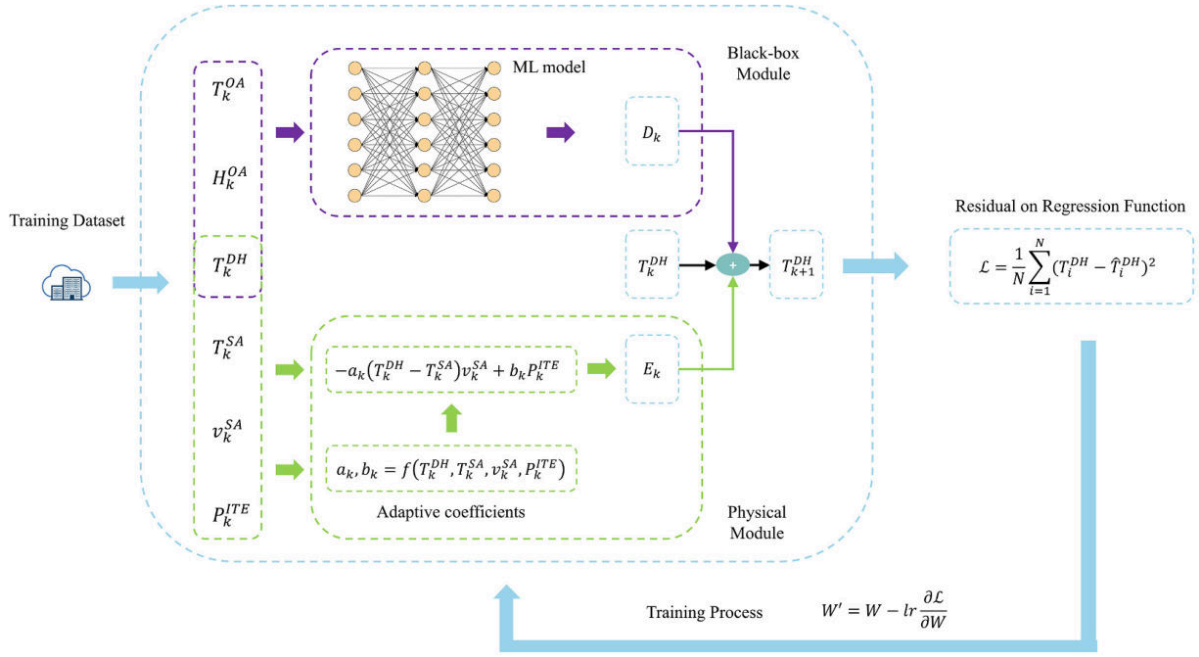


Figure 3.4. A-PCNN framework. Source: Chen, Chui and Lee [80], Fig. 2.

Table 3.2. Forecast error reduction of the adaptive models (five months of training, 30-day unseen test)

Comparison	MAE reduction	RMSE reduction
A-PC-MLP vs PC-MLP, 15-min horizon	17.3%	24.2%
A-PC-MLP vs PC-MLP, 7-day horizon	79.2%	79.6%
A-PC-MLP vs Transformer, 7-day horizon	38.1%	51.4%

The advantage grows with horizon (Table 3.2, Figure 3.5): at 7 days the adaptive model's error stays near 0.3 °C while fixed-coefficient models drift above 1.3 °C, the difference between a forecast that can schedule setpoints a day ahead and one that cannot. The adaptive models also tracked an unseen week whose return-air temperature fell below anything in the training set (Appendix A, Figure A.1) [80]. A forecast with physically signed coefficients is better suited than a fixed-coefficient model to supporting the controller of Section 3.4 outside its training range.

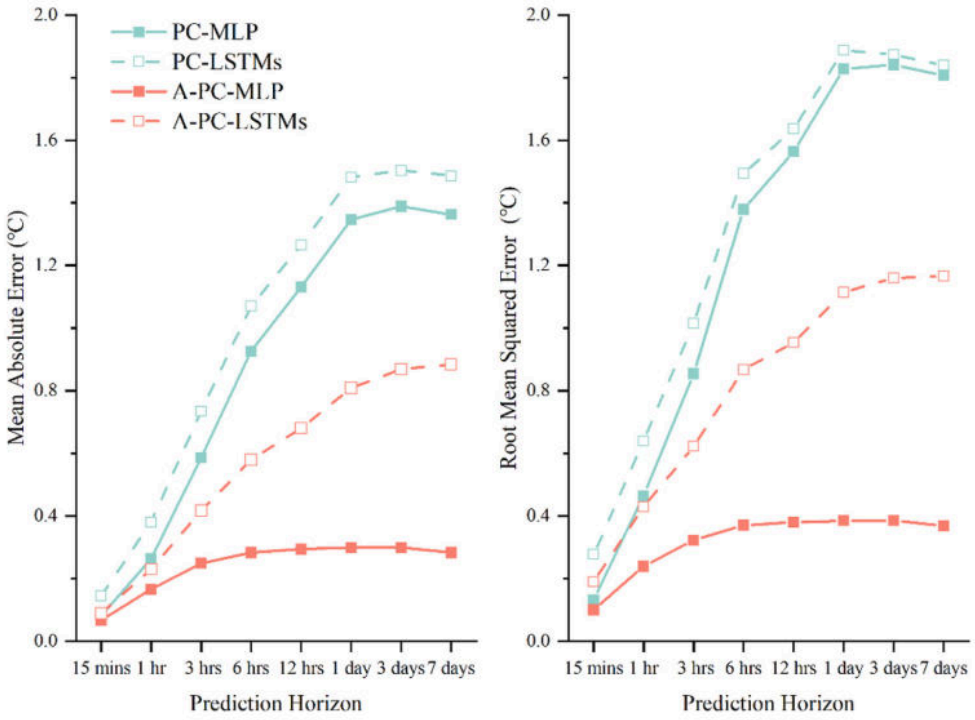


Figure 3.5. Forecast error versus horizon; red is the adaptive framework, blue the fixed-coefficient PCNN. Source: Chen, Chui and Lee [80], Fig. 6.

3.4 Physics-Informed Predictive Control for Cooling Optimisation

Embedding adaptive, physics-constrained component models in model predictive control (PIML-MPC) reduced the cooling-plant energy of the NUS Sustainable Tropical Data Centre Testbed (STDCT) by 12.6% over 30 days on seen load, raising mean supply-water temperature by 1.7 °C with no temperature violations [81]. Under an unseen, higher IT load the saving was 9.1%, with a worst reading 0.2 °C above the limit against violation rates of 25% to 57% for purely data-driven controllers (Table 3.6). The plant model is calibrated on the testbed's own meters and driven by measured IT load and weather, so the result is a calibrated simulation of that plant rather than a field measurement.

Testbed and calibrated plant model

STDCT opened in 2021 under a S$23 million programme funded by NRF and Meta and run with NTU [82]. Its data hall holds 22 racks of servers carrying live NUS IT load, cooled by a hybrid of air-handling units and two coolant distribution units with hot-aisle containment (Figure 3.6), served by the outdoor plant in Table 3.3.

Table 3.3. Plant specification of the STDCT cooling system

Plant element	Specification
Chiller	Trane water-cooled centrifugal, R134a, 571 kW nominal capacity, nominal COP 6.42
Cooling tower	Truwater counterflow, 756,000 kcal/h; 40 to 34 °C at 30.2 °C wet bulb; fan about 7.2 kW
Water flow	35 L/s total
Existing control	Fixed setpoints from operating experience, sized for peak load



Figure 3.6. The Sustainable Tropical Data Centre Testbed. Source: Chen, Chui and Lee [81], Fig. 7.

The authors modelled the chilled-water pump, chiller, condenser-water pump and cooling tower in OpenModelica, using manufacturer curves for the chiller and standard tower correlations, and calibrated the model against 14,613 metered chiller observations (MAE 0.79 kW, R^2 0.87). At a representative operating point every component was within 3.0% to 5.5% of metered power (Table 3.4; workflow in Appendix A, Figure A.2).

Table 3.4. Simulated and measured power at the representative operating point

Component	Simulated (kW)	Measured (kW)	Difference
Cooling tower fan	0.85	0.82	3.66%
Condenser-water pump	5.46	5.30	3.02%
Chiller	12.88	13.63	5.50%
Chilled-water pump	2.79	2.89	3.46%

Controller design

Each plant component has its own physics-shaped predictor; the cooling-tower model, for example, is constrained so that it cannot predict water colder than the wet-bulb temperature. Every 15 minutes the controller chooses the chiller supply-water temperature and the speeds of the chilled-water pump, condenser-water pump and tower fan to minimise total plant power within a 17 to 19 °C supply-water band, with a penalty on deviation from 18 °C (Figure 3.7). Commissioning needs about one month of sub-metered 15-minute plant data.

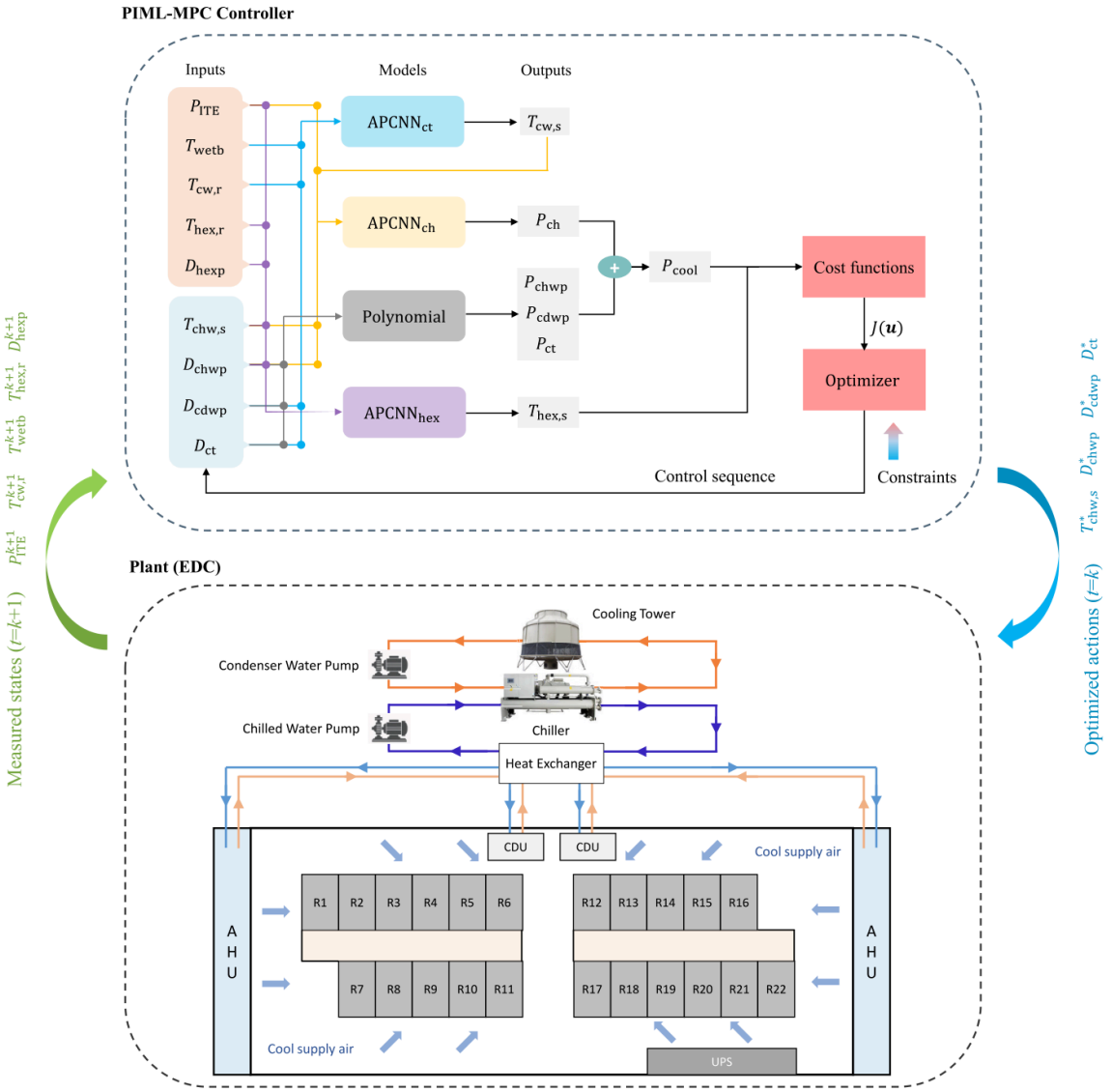


Figure 3.7. PIML-MPC structure. Source: Chen, Chui and Lee [81], Fig. 5.

Results against fixed setpoints

Table 3.5. Results over 30 days against the fixed-setpoint baseline

Metric (30 days)	Seen load: baseline	Seen load: PIML-MPC	Unseen high load: baseline	Unseen high load: PIML-MPC
Cooling energy (kWh)	15,352.1	13,422.0 (−12.6%)	19,863.0	18,047.2 (−9.1%)
CSE = P_{cool}/P_{IT}	0.137	0.119 (−13.1%)	0.135	0.122 (−9.6%)
Mean supply-water temperature (°C)	16.8	18.5	17.3	18.6
Maximum supply-water temperature (°C)	17.1	18.7	18.2	19.0
Temperature violation rate (above 19 °C)	0.0%	0.0%	0.0%	1.0%

The baseline over-cooled, holding supply water below 17 °C about 91.8% of the time. PIML-MPC ran at 18 to 19 °C and saved energy every day, 11.3% to 14.2% per day on seen

load and 4.6% to 12.9% under unseen high load (Figure 3.8). The saving comes from different components depending on load (Figure 3.9). On seen load the auxiliaries dominate: the chilled-water pump alone saves 929 kWh, 61.8% of its use, because the controller slows it to match the load. Under unseen high load the chiller becomes the largest absolute saver at 811 kWh.

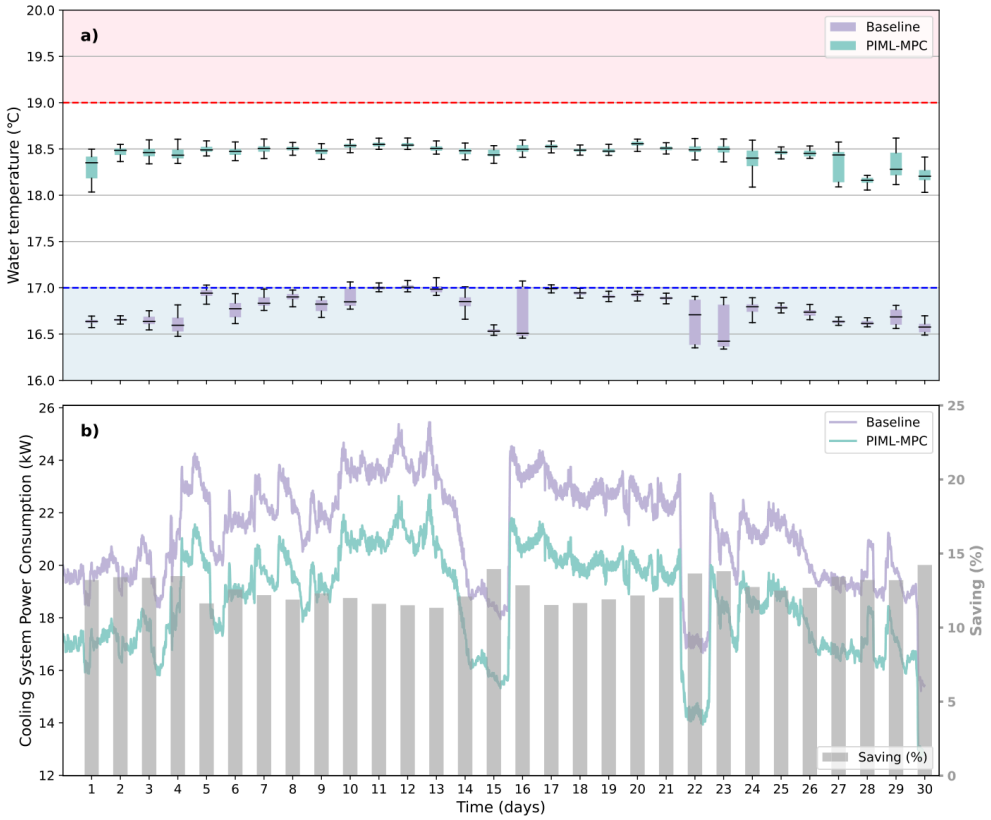


Figure 3.8. (a) Supply-water temperature with 17 and 19 °C limits; (b) cooling power and daily saving. Source: Chen, Chui and Lee [81], Fig. 10.

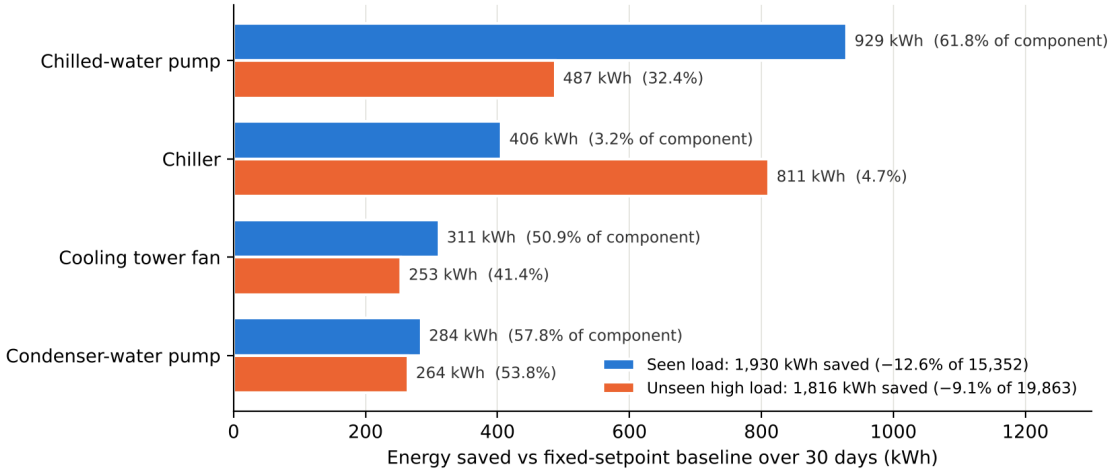


Figure 3.9. Energy saved by component over 30 days. Authors' plot of data from Chen, Chui and Lee [81], Tables 3 and 4.

Robustness against purely data-driven controllers

Physics constraints matter most when conditions leave the training range. Under unseen IT load, which peaked well above anything in training (Appendix A, Figure A.3), every purely data-driven controller breached the temperature limit a quarter or more of the time (Table 3.6, Figure 3.10).

Table 3.6. Controller comparison (mean of 10 seeds)

Controller	Seen: cooling kWh	Seen: TVR	Seen: CSE	Unseen: cooling kWh	Unseen: TVR	Unseen: CSE
PIML-MPC	13,422	0.0%	0.119	18,047	1.0%	0.122
ANN-MPC	13,309	10.7%	0.118	19,046	24.9%	0.128
NARX-MPC	13,435	34.3%	0.119	19,751	40.5%	0.132
ARX-MPC	14,319	68.6%	0.127	22,498	57.0%	0.150

ANN-MPC used marginally less energy on familiar load only because it broke the limit; on unseen load it used 5.5% more energy than PIML-MPC and violated the band a quarter of the time.

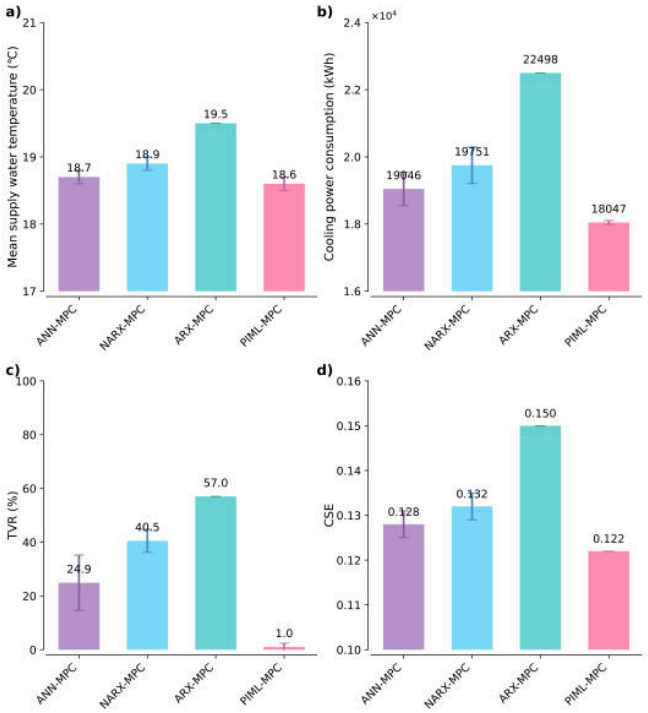


Figure 3.10. Four controllers under unseen load: (a) mean supply-water temperature, (b) cooling energy, (c) temperature violation rate, (d) CSE. Source: Chen, Chui and Lee [81], Fig. 16.

What the findings mean for a commercial plant

The findings support the application of physics-informed predictive control to commercial chilled-water plants in Singapore. The NUS PIML-MPC study achieved cooling-energy reductions of 12.6% under seen operating conditions and 9.1% under unseen higher IT loads, using a plant simulation calibrated on measured testbed data [81]. These results are consistent with the commercial evidence for predictive cooling optimisation, from about 9% on a live Google server floor [83] and 9.5% across the chiller plants of a Singapore

telecommunications operator [84] to 14% to 28% of chiller power during low-load hours under real-plant MPC [85]. The saving uses the existing variable-speed plant, and the worst excursion of 0.2 °C suits colocation service-level guarantees. Implementation involves plant sub-metering, model calibration, supervisory control integration and verification of temperature constraints and chiller staging against a measured baseline, with the existing control system retained as an operational fallback.

3.5 Energy, Carbon and Economic Assessment of Efficiency Measures

From cooling-system efficiency to PUE

The NUS studies report CSE, the cooling-plant part of PUE − 1:

$$PUE = 1 + P_{cool}/P_{IT} + P_{other}/P_{IT} = 1 + CSE + P_{other}/P_{IT} \tag{3.3}$$

With IT electricity and non-cooling overhead held constant (Section 1.3), a reduction in cooling electricity lowers PUE by the same amount: 0.137 to 0.119 on seen load gives ΔPUE = 0.018, and the unseen-load case gives 0.013. For a fixed IT energy E_{IT}, the annual saving of any measure that lowers PUE from PUE_0 to PUE_1 is

$$\Delta E = E_{IT}(PUE_0 - PUE_1) \tag{3.4}$$

This conversion reflects the distinction between facility-level and subsystem metrics reviewed by Isazadeh et al. [70]: PUE describes the facility as a whole but cannot locate where energy is lost, whereas a subsystem indicator such as CSE isolates the cooling plant. A CSE saving carries through to PUE only because the other terms are held fixed.

The electricity saving then converts into the three outcomes of the model in sequence. Facility demand falls from $E_{DC,0}$ to $E_{DC,0} - \Delta E$. Each MWh saved is a MWh not drawn from the grid, so the CO_2 avoided is $\Delta E \times EF_{grid}$ and the electricity expenditure saved is $\Delta E \times p_{grid}$ (Equations 1.2 and 1.3). Cooling-plant percentages must not be read as facility-wide savings: a 9% to 13% cut in plant energy lowers PUE by 0.013 to 0.018, which is only 1.0% to 1.4% of facility electricity at PUE 1.25, because cooling-plant energy in the testbed is about 0.14 of IT energy.

Optimise the temperature, do not maximise it

Raising water or hall temperature lowers cooling power monotonically, but warmer servers draw more power through CPU leakage and faster fans. Total power is therefore U-shaped in supply-water temperature (Figure 3.11), which is why the controller penalises deviation from 18 °C rather than pushing to the 19 °C limit. The SS 697 figure of 2% to 5% of cooling energy per 1 °C is a facility-side number, so this assessment carries 2%/°C as the central value and 5%/°C as the upper bound.

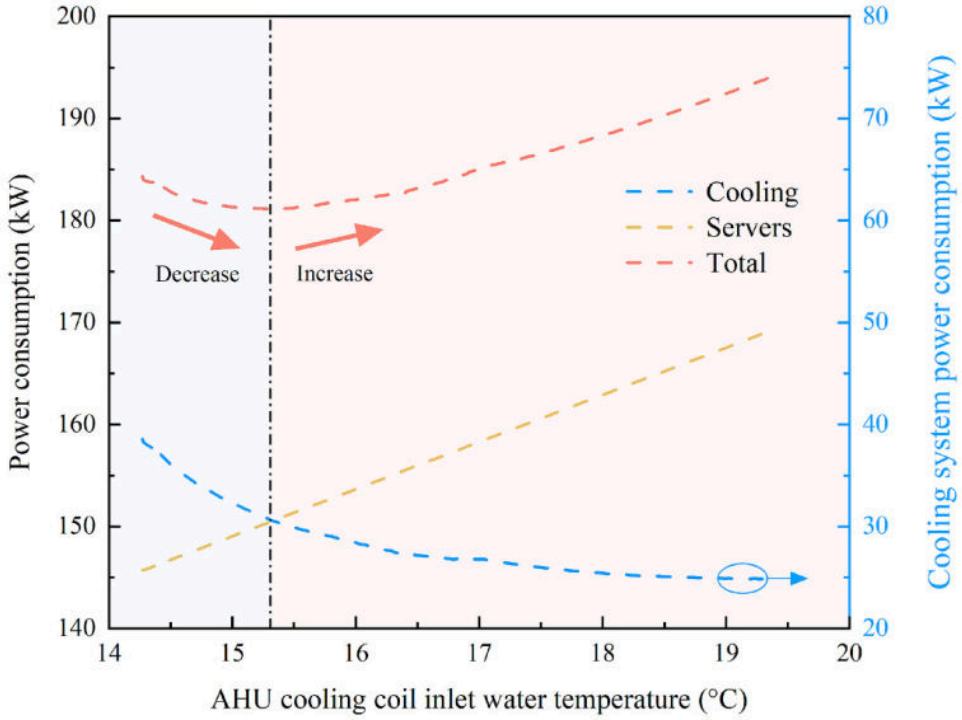


Figure 3.11. Total power is U-shaped in water temperature. Source: Chen, Chui and Lee [81], Fig. 3.

Reference case and the combined saving

Plant control (measure 1) and the SS 697 hall temperature rise (measure 2) both raise chilled-water temperature, so their savings overlap and cannot simply be added. PIML-MPC already raises supply water by 1.7 °C, and at STDCT that rise produced a chiller saving of about 2.6% of cooling energy, about 1.5% per °C, consistent with the low end of SS 697. The combined saving therefore applies measure 2 only to the remaining 2.3 °C of the 4 °C rise, and only to the cooling energy left after measure 1:

$$\Delta E_{1+2} = \Delta E_1 + s \times (4 - 1.7) \times (E_{cool,0} - \Delta E_1) \tag{3.5}$$

where s is the SS 697 saving per °C and $E_{cool,0} = 0.175\ E_{IT}$ is the baseline cooling energy (Table 3.7).

Table 3.7. Assumptions for the 100 MW reference data centre assessment

Parameter	Value
IT capacity and load factor	100 MW at 85% (Table 1.3)
Annual IT energy, E_{IT}	744.6 GWh (100 MW × 0.85 × 8,760 h)
Baseline PUE_0; facility electricity	1.25; 930.8 GWh (DC-CFA2 benchmark at 100% IT load [14])
Grid emission factor; electricity price	0.402 tCO_2/MWh [3]; S$200/MWh (Table 1.3)
ΔCSE from plant control	0.013 to 0.018, midpoint 0.0155 (Table 3.5 [81])
Cooling energy, $E_{cool,0}$	$0.175 \times E_{IT}$ = 130.3 GWh (design assumption: 70% of the 0.25 overhead; the remaining 0.075 covers UPS and distribution losses, lighting and other loads)
Hall temperature change	22 to 26 °C (SS 697 roadmap [12])
Cooling saving per °C	2% central, 5% upper (SS 697 [12])

Parameter	Value
Hybrid liquid-cooling target PUE	1.20 (design target; STDCT target below 1.2 for air and liquid cooling [78]; SS 726 [16])

Table 3.8. Annual savings for the 100 MW reference data centre (baseline PUE 1.25, 930.8 GWh/yr, 374 kt CO_2/yr, S$186 million/yr)

Measure and evidence basis	PUE after	Electricity saved (GWh/yr)	Cost saved (S$ million/yr)	CO_2 avoided (kt/yr)
1. Plant control (PIML-MPC): supply-water reset, pump and fan speed optimisation; calibrated STDCT model [81], commercial range 9.5% to 28% [84], [85]	1.232 to 1.237	9.7 to 13.4	1.9 to 2.7	3.9 to 5.4
2. Hall temperature 22 to 26 °C, alone; published standard SS 697 [12]	1.236 (1.215)	10.4 (26.1)	2.1 (5.2)	4.2 (10.5)
1 + 2 combined, overlap removed (Equation 3.5); combined design value	1.227 (1.214 to 1.230)	17.0 (15.2 to 26.8)	3.4 (3.0 to 5.4)	6.8 (6.1 to 10.8)
3. Hybrid air and direct-to-chip liquid cooling to PUE 1.20; design target [78]	1.20	37.2 from baseline; 20.2 beyond 1 + 2	7.4	15.0

Brackets give the SS 697 upper bound (5%/°C) or the seen-to-unseen load range. Combined cases: conservative 15.2 GWh (ΔCSE 0.013, 2%/°C), central 17.0 GWh (ΔCSE 0.0155, 2%/°C) and high 26.8 GWh (ΔCSE 0.018, 5%/°C).

Removing the overlap matters: simply adding measures 1 and 2 would give 20.1 to 39.5 GWh and over-count by 5 to 13 GWh (Table 3.8). The near-term package removes 17.0 GWh, 6.8 kt CO_2 and S$3.4 million a year, taking PUE to 1.227 (Figure 3.12). Hybrid liquid cooling then reaches the 1.20 design target, and its total saving exceeds the 20.2 GWh shown once server fan power is counted: if AI halls carried 30% of the IT load, the 7% IT power reduction measured by NVIDIA and Vertiv [79] would save a further 15.6 GWh a year, held in reserve as additional upside (Section 1.3), and its capital is assessed as a break-even value in Section 5.3.

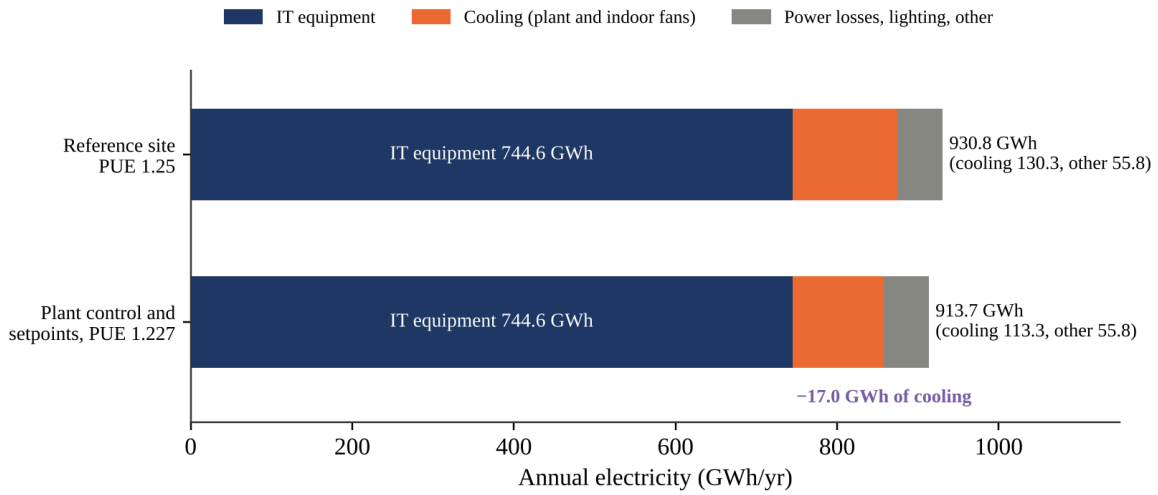


Figure 3.12. Annual electricity of the reference data centre by end use, before and after plant control and setpoint optimisation (central case).

The baseline matters as much as the measure: at the Uptime Institute's 2025 global average PUE of 1.54 [86], an existing site holds far more room than the new-build case, and Section 5.4 assesses such a site in full.

Implementation cost and payback

Implementation cost is site-specific and is not reported by either NUS study, so the S$1 to 5 million range of Section 1.3 is used. Measure 1 needs sub-metering of chillers, pumps and towers, a data historian and a supervisory controller on the existing building management system. Annualised at 7% over 10 years, S$1 to 5 million corresponds to S$11 to 73 per MWh saved, well below the S$200/MWh it displaces, and payback against the S$1.9 to 2.7 million annual saving is within 2.6 years (Table 3.9); the Singapore telco retrofit [84] indicates that real costs sit at the low end.

Table 3.9. Simple payback of plant control (measure 1) at the reference site against implementation cost

Implementation cost (S$ million)	Payback at S$2.7 M/yr saving (years)	Payback at S$1.9 M/yr saving (years)
0.5	0.2	0.3
1.0	0.4	0.5
2.0	0.7	1.0
3.0	1.1	1.5
5.0	1.9	2.6

The value grows with the electricity price: the combined package is worth S$2.6 million a year at S$0.15/kWh and S$5.1 million at S$0.30/kWh, and with USEP averaging S$196/MWh in early 2026 [44] the upper cases are the more realistic. Measure 3 needs coolant distribution units, manifolds and a warm-water loop, and is designed into new AI halls, where air cannot cool the rack densities at all.

3.6 National-Scale Assessment and Transferability

Table 3.10 scales measure 1, the measure with direct testbed and commercial evidence, to Singapore's 6.5 TWh of data centre electricity in 2024 at four adoption rates. Singapore's fleet-average PUE is not published, so the fleet is taken at PUE 1.5, close to the Uptime 2025 global average [86], giving 4.33 TWh of IT energy; at PUE 1.4 the savings are 7% higher.

Table 3.10. Singapore-scale saving from plant control (measure 1) by adoption rate (fleet PUE 1.5)

Share of fleet adopting	Electricity saved (GWh/yr)	CO_2 avoided (kt/yr)	Cost saved at S$0.20/kWh (S$ million/yr)
25%	14 to 20	5.7 to 7.8	2.8 to 3.9
50%	28 to 39	11.3 to 15.7	5.6 to 7.8
75%	42 to 59	17.0 to 23.5	8.4 to 11.7
100%	56 to 78	22.6 to 31.4	11.3 to 15.6

Ranges span the unseen-load (ΔCSE 0.013) and seen-load (ΔCSE 0.018) cases.

Full adoption saves 56 to 78 GWh a year, frees 6.4 to 8.9 MW of grid capacity and removes about 1% of the sector's 2.6 Mt CO_2, equal to 1.4% to 1.9% of the output of the national 3 GWp solar target (4,155 GWh a year), delivered on existing sites with no land. Adding the SS 697 roadmap and hybrid liquid cooling at new AI halls lifts the combined effect to about 2% to 4% of sector electricity.

Fleet transfer requires a measured baseline at each site, since sites already running reset logic will save less than 12.6%, and staging logic tested on multi-chiller plants. Efficiency under a capacity cap also frees headroom that new IT fills, as the roadmap's 300 MW unlocked through efficiency shows [87], so Chapter 4 pairs it with low-carbon supply.

3.7 Recommended Energy Efficiency Measures

Table 3.11 ranks the measures by expected saving at the reference site, cost, deployment effort and operational reliability, and separates existing sites from new high-density AI facilities.

Table 3.11. Prioritisation of efficiency measures for the 100 MW reference data centre

Measure	Expected saving (per year)	Applies to	Deployment	Cost	Reliability risk	Priority
Plant sub-metering and PIML-MPC control	9.7 to 13.4 GWh; 3.9 to 5.4 kt; S$1.9 to 2.7 M	Existing and new sites	About one month of data to commission; BMS integration	Low; payback under 2 years	Low: 0% violations on seen load	1
Hall and supply-water temperature optimisation (SS 697)	Adds 5.5 to 13.5 GWh to measure 1	Existing and new sites	Tenant contract update	Low capital; contract and commissioning cost	Low if run with measure 1's model	2
Hybrid air and liquid cooling (SS 726)	20 GWh beyond 1 + 2 at a new build; far more at AI rack densities	New builds and AI halls	Designed in at construction	Higher capital; required for AI racks	Medium: leak detection and fluid quality	3
IT efficiency and utilisation (SS 715)	Reduces E_{IT} itself; scales every other saving	All sites	Procurement and workload policy	Low to medium	Low	4
Measurement and verification of savings	Enables and proves measures 1 to 4	All sites	Monthly verified baseline	Low	None	Enabler

Operators should therefore deploy sub-metering and physics-informed control at existing sites first, raise temperatures to each site's measured optimum (Figure 3.11) under updated tenant contracts, design hybrid liquid cooling into every new AI hall under SS 726, and verify every claimed saving against a measured monthly baseline, which also gives IMDA a fleet-wide evidence base for the licence metrics now being drafted.

Efficiency determines how much low-carbon electricity must be procured. The near-term package removes 17.0 GWh a year at the reference site, the output of about 12 MWp of solar or roughly 9 ha of panels, and reduces the low-carbon electricity the facility must source by the same amount. Because every supply option is land-capped, fuel-priced or still under national assessment, each kilowatt-hour saved lowers both the CO_2 emitted and the volume

that supply must deliver, which is why efficiency is the first phase of the roadmap (Section 5.6).

Chapter 4: Strategic Challenges and Proposed Responses

4.1 From Current Strategy to Design Requirements

Singapore's approach to sustainable data centre growth is among the most comprehensive in the world, combining capacity allocation tied to efficiency and green energy, tropical and IT efficiency standards, a statutory licensing regime, low-carbon generation and import programmes, and research testbeds such as STDCT (Chapters 1 to 3). AI and cloud computing are nevertheless raising electricity demand faster than any single efficiency or supply measure can offset on its own.

Six gaps limit the current approach, and each translates into a design requirement for the integrated strategy (Table 4.1).

Table 4.1. Six gaps in Singapore's current approach and the engineering response to each

Gap	Challenge	Engineering response and where it is quantified
1. AI growth outpaces efficiency	Compute demand grows faster than energy per unit of compute falls	Cut energy intensity and electricity carbon intensity together, and plan low-carbon generation to the growth path (Section 4.2; Chapter 5)
2. Diminishing returns from PUE	New sites are already near PUE 1.2 to 1.3, and IT energy dominates	Target existing-site overhead and new-site IT efficiency; report useful work per kWh (Section 4.3; Chapter 3)
3. Intermittent renewables	Solar output does not follow a flat 24-hour load	Pair rooftop PV with firm, dispatchable ammonia and SMR output; verify hour by hour (Sections 4.4 and 4.5)
4. Fuel and supply security	Ammonia electricity depends on imported fuel, and SMR supply on a national decision	Certify and store fuel, contract with more than one supplier, phase generation and keep grid backup (Section 4.4)
5. Inconsistent measurement	No common threshold or method for a "low-carbon" claim	Report metered electricity, electricity-related emissions and an hourly carbon-free electricity score (Section 4.5)
6. Efficiency versus absolute consumption	Efficiency metrics improve while total demand rises	Track absolute electricity and emissions alongside intensity metrics (Section 4.2; Chapter 5)

4.2 Computing Demand Growth and Absolute Emissions

The International Energy Agency estimates that data centres used about 415 TWh of electricity in 2024, about 1.5% of world consumption, and projects about 945 TWh by 2030 in its Base Case, driven mainly by accelerated servers for AI [88], with Asia outside China, which includes Singapore, among the growing regions (Figure 4.1). In Singapore, data centres rose from about 7% of national electricity in 2020 to just over 10% in 2024 [2]. For a single facility, this growth means that CO_2 falls only if the electricity per unit of computing and the carbon per unit of electricity fall faster than demand rises.

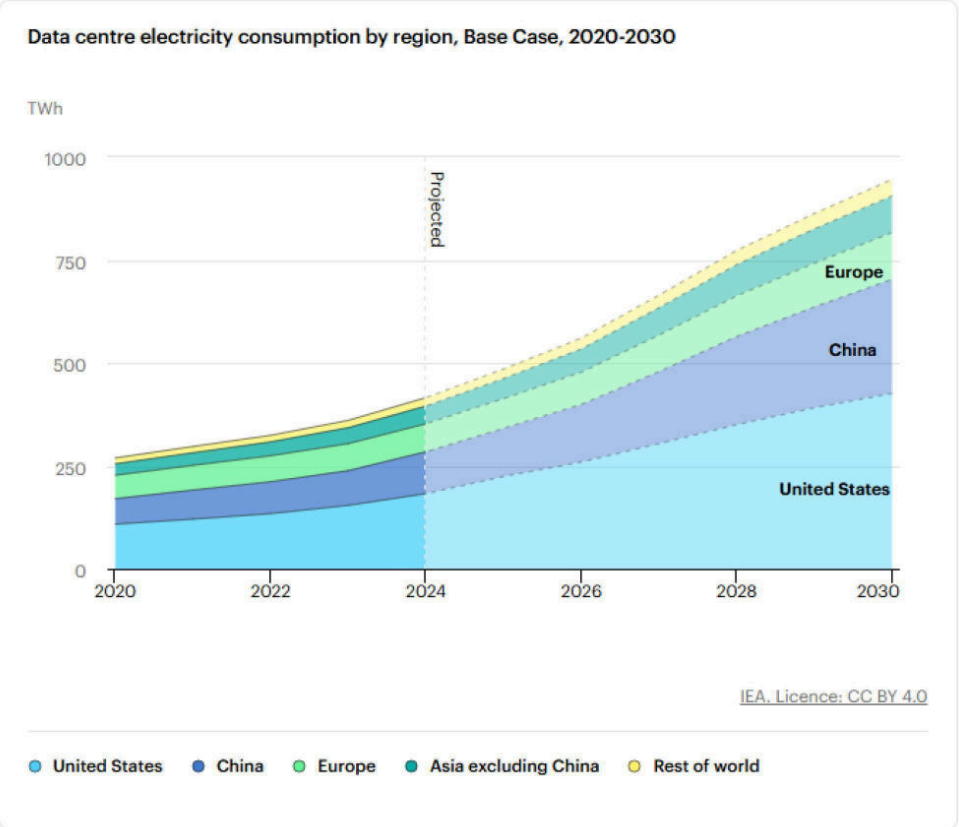


Figure 4.1. Data centre electricity consumption by region (United States, China, Europe, Asia excluding China, rest of world), Base Case, 2020 to 2030, in TWh. Source: IEA (2025), Energy and AI [88], licence CC BY 4.0.

Two mechanisms drive this growth. Exogenous demand for computing from AI, cloud migration and digitalisation rises whatever the efficiency of the facility, and the rebound effect means that cheaper computing stimulates additional demand. In Singapore the rebound is partly planned, since efficiency gains are explicitly used to unlock new capacity, as the Green Data Centre Roadmap's 300 MW shows [87]. Both mechanisms raise absolute consumption, so absolute emissions fall only when efficiency is paired with low-carbon supply.

The decomposition makes the requirement exact. Electricity-related emissions are the product of the computing workload W, the IT energy per unit of work, the facility overhead and the carbon intensity of supply:

$$CO_2 = W \times (E_{IT}/W) \times PUE \times EF \quad (4.1)$$

Emissions fall only if the product of the last three terms falls faster than W grows. In the worked example of Figure 4.2a, a 60% rise in computing demand combined with a 20% efficiency gain still raises electricity by 28% (100 × 1.6 × 0.8 = 128). Supplying half of that electricity from the low-carbon portfolio of Chapter 2 brings emissions down to 68% of today's level even as demand grows. Equally, if demand doubles while the emission factor halves, emissions stay flat, so the strategy moves both intensity terms at once.

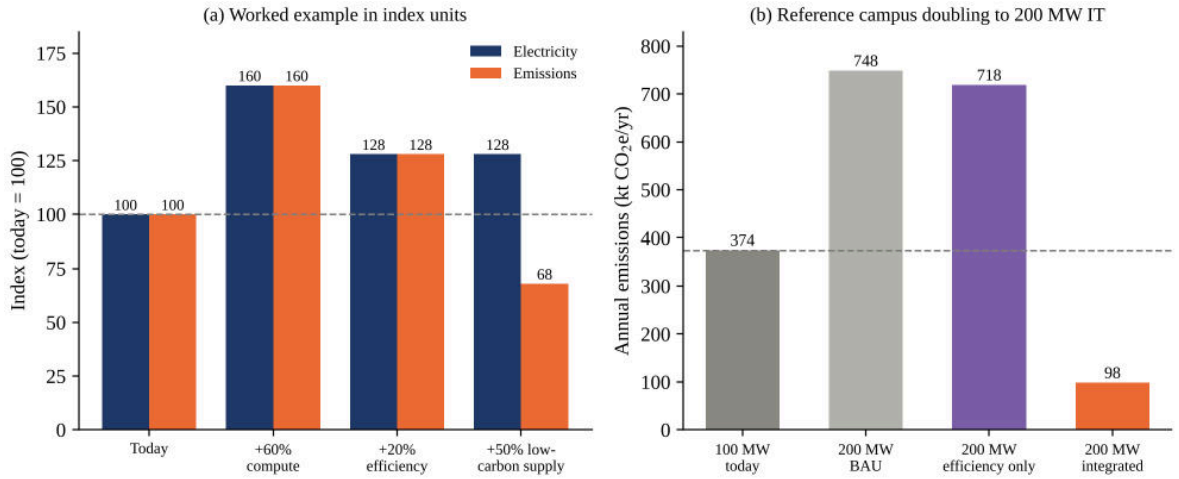


Figure 4.2. (a) Worked example in index units: computing growth, a 20% efficiency gain and a 50% low-carbon supply share. (b) Annual emissions of the reference campus doubled to 200 MW IT: business as usual (PUE 1.25, grid), efficiency only (PUE 1.20) and the integrated strategy (PUE 1.20, PV, two SMR modules, the full ammonia plant and grid for the remainder). Authors' calculation.

If the reference campus doubles to 200 MW of IT, business as usual would double emissions to 748 ktCO_2e a year and efficiency alone would hold them at 718 kt; the integrated strategy, with PUE 1.20, rooftop PV, two SMR modules and the full ammonia plant, cuts them to 98 kt, 276 kt below today's 100 MW baseline (Figure 4.2b). Nationally, the roadmap's further 300 MW [11] would add 1.12 MtCO_2 a year under business as usual (Table 1.6) but about 0.11 Mt at the portfolio's effective 0.041 tCO_2e/MWh, so growth is compatible with decarbonisation when generation is planned to the growth path.

4.3 Beyond PUE: Useful Computation per Kilowatt-Hour

Newly built large facilities in Singapore are designed for PUE 1.2 to 1.3, and DC-CFA2 requires 1.25 at full IT load [14]. The installed base worldwide is older and less efficient. Figure 4.3 shows that the average annual PUE reported to the Uptime Institute fell steeply from 2.50 in 2007 to 1.65 in 2014, but has stayed between 1.54 and 1.67 ever since, reaching 1.54 in 2025 [86]. Isazadeh et al. attribute the early fall to low-cost measures such as aisle containment and the plateau to later measures that are costlier and more complicated [70]. This global average covers facilities of all ages, but it shows that facility-level gains have flattened.

PUE improvements show diminishing returns. The largest share of facility energy that overhead removal can save is $1 - 1/\text{PUE}$ (Figure 4.4): 33% at PUE 1.5, 20% at 1.25 and 17% at 1.2. Moving from 2.0 to 1.5 removes far more energy than moving from 1.3 to 1.2. Existing sites at 1.4 to 1.5 therefore hold most of the remaining overhead, and plant control and setpoint optimisation should be deployed there first (Chapter 3), while new sites already near 1.25 hold their energy in the IT equipment and gain most from the IT side.

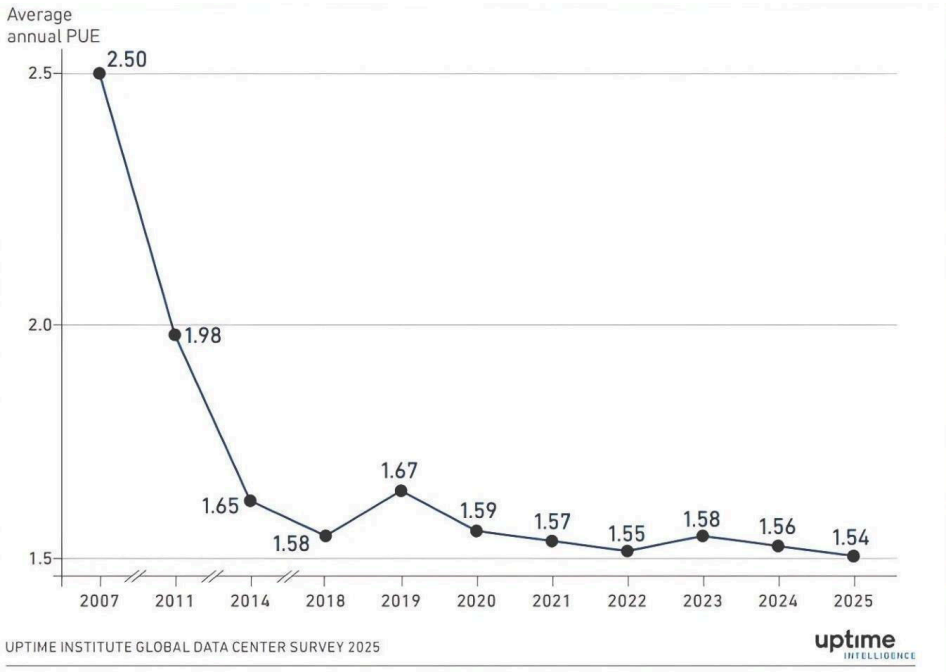


Figure 4.3. Average annual PUE reported by data centre operators, 2007 to 2025. Source: Uptime Institute Global Data Center Survey 2025 [86].

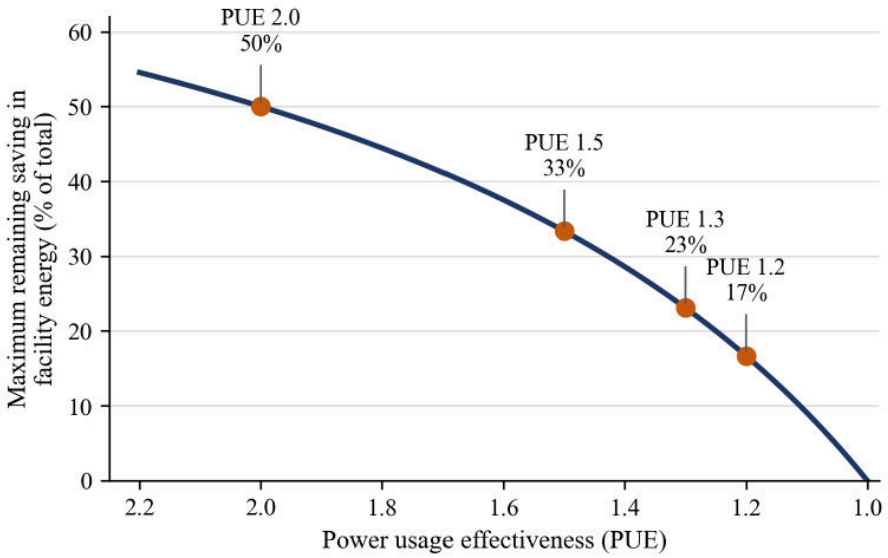


Figure 4.4. Maximum remaining saving in facility energy (1 − 1/PUE) at constant IT load, derived from Equation 3.1.

IT efficiency and facility efficiency are separate components of total consumption: IT measures reduce the electricity needed for a given computational workload, whereas facility measures reduce the additional energy for cooling, power distribution and supporting infrastructure. PUE relates facility energy to IT energy but does not measure computational productivity, and some measures can lower PUE without lowering energy use [70]. Equation 4.1 therefore keeps IT energy per unit of useful work and PUE as separate terms; the quantitative cooling scenarios change only the PUE term and hold IT energy constant, which avoids double counting, and IT-side improvements under SS 715 remain a complementary recommendation.

Free cooling, often listed among efficiency options, is unavailable in Singapore's hot and humid climate (Section 3.2), so once cooling overhead is minimised the dominant consumer is the IT equipment itself. SS 715 equipment and higher utilisation reduce the IT energy per unit of work [13], and direct-to-chip liquid cooling removes most server fan power, cutting IT power by about 7% in a high-density hybrid hall [79]. The leverage is large: a 10% reduction in IT energy at the reference facility, at PUE 1.227, saves 91.4 GWh and 36.7 kt CO_2 a year, more than five times the cooling-plant package of Chapter 3. The response to Gap 2 is therefore to keep PUE for facility overhead and add a useful-work metric, such as IT energy per unit of compute delivered, to licence reporting.

4.4 Firm Low-Carbon Electricity Supply and Energy Security

Rooftop PV supplies daytime generation, the ammonia GTCC dispatchable generation through Jurong Island's fuel network, and SMRs long-term firm generation; the grid, UPS and backup generators sustain availability however electricity is sourced, so the solar mismatch is a carbon-matching problem rather than a reliability problem.

On-site renewables supply only a small share of a Singapore facility's demand, so off-site generation procured through power purchase agreements (PPAs) carries most of the low-carbon volume (Figure 4.5); here, the shared ammonia plant and a future SMR.

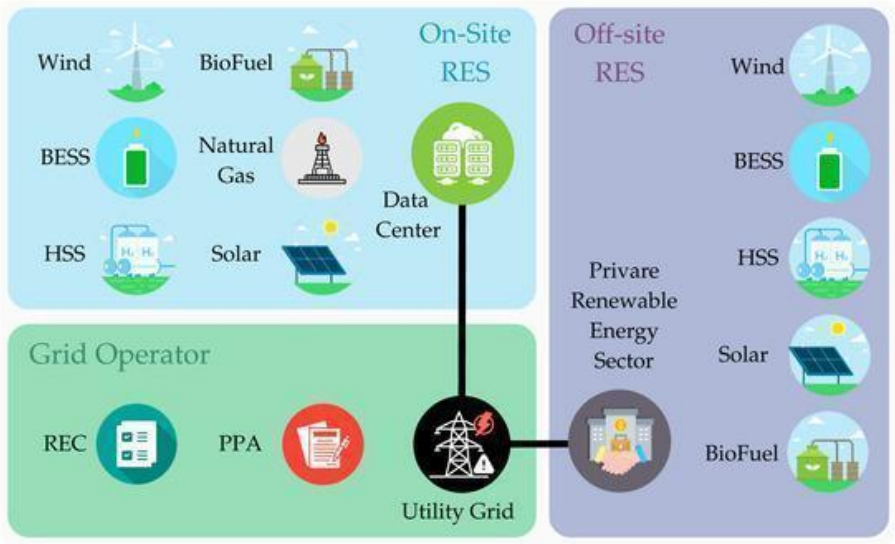


Figure 4.5. Strategies of renewable energy integration for data centres: on-site renewables owned by the data centre, and off-site renewables procured through PPAs and renewable energy certificates (RECs) via the grid operator. Source: Ziaei, Husev and Rabkowski [89], Fig. 11, CC BY.

Firm generation exchanges weather exposure for fuel and programme exposure. On 24 October 2026 Indonesia cut pipeline gas exports to Singapore by about 14% for 18 days, under 2% of daily gas demand [90], showing how imported fuels transmit shocks. An ammonia interruption leaves power supply intact, because storage and the grid back it up, and an 18-day loss of the long-term allocation would return 13.1 GWh to the grid and add 3.8 $ktCO_2e$, 1.1% of the year's avoided emissions. Table 4.2 sets out the response to each exposure.

Table 4.2. Supply risks of the integrated portfolio and the engineering and procurement response

Risk	Proposed response
Ammonia fuel supply	Certified fuel contracted from more than one supplier; 7 to 20 days of on-site storage [34], [39]; grid backup beyond that
Solar variability	Treat PV as daytime supply; no storage needed at 0.17% of demand (Section 2.3)
Generation availability	Cover SMR refuelling and GTCC maintenance with ammonia dispatch and the grid (Section 4.5)
Nuclear development	Contract SMR output only after a national decision informed by the 2027 INIR mission and NEA studies [19], [61]
Grid reliability	Keep a full grid connection; the reserve margin and growing storage cover outages [42], [49]
Carbon verification	Metered attributes for every allocated MWh; certified fuel at or below 0.276 tCO_2e/tNH_3 (Section 2.5)
Price exposure	Fuel-indexed prices with collars; each US$100/t moves cost by S$51/MWh (Section 2.5)

4.5 Measurement and Verification of Low-Carbon Performance

A credible low-carbon claim needs transparent indicators, clear boundaries and comparable accounting. Emissions must first be classified correctly under the GHG Protocol [21], with refrigerant leakage from owned chillers and backup generator fuel in Scope 1. For electricity the principal issue is timing: annual matching, in which a facility buys certificates equal to its yearly consumption, can report 100% renewable electricity while drawing fossil power at night. The GHG Protocol is revising its Scope 2 guidance around hourly matching and deliverability within the same grid region, with a joint GHG Protocol and ISO standard expected in the fourth quarter of 2028 [91]. The gap is large in Singapore: Google's asia-southeast1 cloud region ran on 5% carbon-free energy hour by hour in 2025, on a grid at 366 g CO_2/kWh [92].

Annual procurement is a practical basis for the quantity and cost of low-carbon supply (Section 1.3), but it does not show whether that supply coincides with consumption. Xu et al. [25] found that annually matched procurement had minimal impact on system-level CO_2 in the western United States, whereas hourly matching drove significant reductions and early uptake of clean firm generation and long-duration storage. Riepin and Brown [93] found consistent benefits across four European zones, with 90% to 95% hourly targets costing little more than annual matching, and He et al. [24] found that most of the cost of hourly matching with wind, solar and batteries lies in the last 5% of load. These costs come from the variability of wind and solar; firm SMR output and dispatchable ammonia generation follow a flat load and largely avoid them. Hourly matching is therefore adopted as the operational verification measure, quantified by the hourly carbon-free electricity score:

$$CFE_{hourly} = \Sigma_{h=1}^{8760} \min(E_{CFE,h}, E_{DC,h}) / \Sigma_{h=1}^{8760} E_{DC,h} \quad (4.2)$$

where $E_{CFE,h}$ is the eligible carbon-free electricity attributed to the facility in hour h and $E_{DC,h}$ is the metered facility consumption in that hour. Eligible supply must be generated or delivered in the same hour, from on-site generation or a source with a physical delivery path to Singapore's grid, and each MWh and its certificate are counted once only; supply above the hour's demand is not credited.

Equation 4.2 follows the hourly carbon-free electricity score used by Riepin and Brown [93], which measures how closely carbon-free procurement aligns in time with consumption. The typical-day model applies it to representative generation and demand profiles (Figure 4.6). Solar generation equal to 100% of annual demand scores 45%, because the midday surplus cannot cover the night. In the long-term configuration the SMR runs at 77 MW and the ammonia plant is dispatched for the remaining 25 MW, so every normal hour scores 100%; during SMR outages, 7% of the year, the ammonia plant supplies up to 60 MW, giving an illustrative annual score of 97.1% (Appendix C.10). Because the calculation uses assumed generation availability and demand profiles, it gives a consistent basis for comparing the temporal performance of supply portfolios; the operating score follows from metered hourly data and the chosen eligibility rules (Section 1.3).

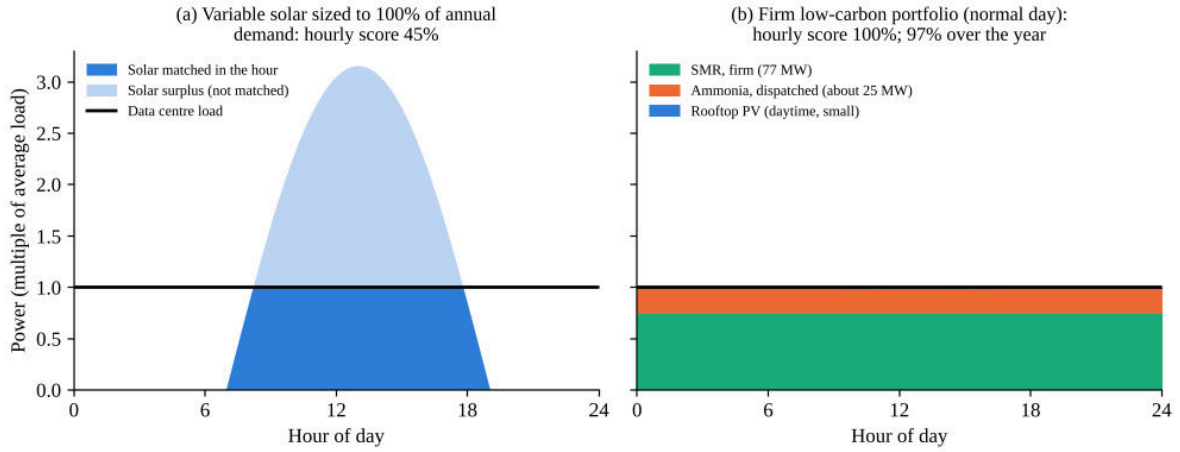


Figure 4.6. Hourly carbon-free electricity matching (Equation 4.2) for the reference load on a typical day: (a) solar generation equal to 100% of annual demand scores 45%; (b) the long-term portfolio of SMR, dispatched ammonia and rooftop PV scores 100% on a normal day and about 97% over the year. Authors' calculations, informed by Xu et al. [25] and Riepin and Brown [93].

The response to Gap 5 is a single performance dashboard, reported annually under the licence regime (Table 4.3). The operator records hourly facility consumption and eligible carbon-free delivery from rooftop PV and the allocated SMR and ammonia output, so the Equation 4.2 score measures the share of demand met by eligible carbon-free supply in the same hour, complementing the annual electricity, emissions and cost assessment of Section 1.4.

Table 4.3. Proposed low-carbon performance dashboard for the reference data centre

Indicator	Definition and unit	Reference case today	Integrated long term	Basis and source
Annual electricity	Total facility consumption (GWh/yr)	930.8	893.5	Equation 1.1; Table C.1
PUE	Facility energy / IT energy, annual (-)	1.25	1.20	ISO/IEC 30134-2 [69]; DC-CFA2 [14]; Table 3.8
Cooling energy ratio	Cooling energy / IT energy, annual (-)	0.175	0.125	CSE of Chen et al. [81]; Table 3.7
Useful-work intensity	IT energy per unit of compute delivered (kWh per work unit)	Report baseline	Report trend	SS 715 [13]; Isazadeh et al. [70]; Section 4.3
Low-carbon supply by source	Allocated rooftop PV, SMR and ammonia output (GWh/yr)	0	1.6, 627.3 and 264.6	Equation 5.2; Table C.1

Indicator	Definition and unit	Reference case today	Integrated long term	Basis and source
Grid electricity drawn	Annual grid allocation (GWh/yr)	930.8	0 (balancing only)	Equation 5.2; Table C.1
Electricity-related emissions	Equation 1.2 ($ktCO_2e/yr$)	374.2	37.0	GHG Protocol [21]; EMA [3]; Table 5.3
Ammonia fuel-chain intensity	Certified delivered intensity (tCO_2e/tNH_3)	Not applicable	0.276 (below 1.0)	Table 2.2 [35], [36]; Appendix C.5
Hourly carbon-free electricity score	Share of consumption matched by eligible low-carbon supply in the same hour (%)	0	97.1 (illustrative)	Equation 4.2; Riepin and Brown [93]; Google [92]
Water usage effectiveness (WUE)	Site water consumption / IT energy (m^3/MWh, equal to L/kWh)	2.1 (2025 sector median)	2.0 or lower (roadmap target)	ISO/IEC 30134-9 [94]; PUB [51]; IMDA [11]
Thermal reliability	Share of time supply-water temperature exceeds its limit (%)	Report	0 seen load; 1.0 unseen high load	Chen et al. [81]; Table 3.5

4.6 Synthesis

The six gaps define the design basis of Chapter 5: reduce energy and carbon intensity together so that absolute emissions fall as computing grows; target overhead at existing sites and IT efficiency at new AI sites; pair low-cost rooftop PV with firm ammonia and SMR generation secured by certified fuel, storage and grid backup; and report electricity, emissions and hourly matching side by side.

Chapter 5: Integrated Proposal, Implementation and Conclusions

5.1 The Proposed Integrated System

The integrated proposal combines the demand-side measures of Chapter 3 and the supply options of Chapter 2 into one engineering system for the 100 MW reference data centre, designed to meet the requirements identified in Chapter 4. It is assessed against the outputs of Section 1.4: the annual emissions emitted and avoided, the annual and incremental cost, and the abatement cost.

Development context: the Jurong Island low-carbon data centre park

The reference facility is placed in the low-carbon data centre park that JTC is developing on Jurong Island. Announced in October 2025, the park occupies about 20 ha within close to 300 ha set aside for new energies and can provide up to 700 MW of power capacity, making it Singapore's largest data centre park [50] (Figure 5.1). The new-energy area is also planned to host a hydrogen-ready natural gas power plant, an ammonia-to-power project of up to 65 MW, and shared energy storage and utilities [50]. In August 2026 DC-CFA2 provisionally awarded 50 MW each to Digital Realty, Equinix, Keppel Data Centres and ST Telemedia Global Data Centres for Jurong Island, each committed to powering more than half of its capacity from green energy and to using liquid cooling and fully energy-efficient IT equipment [95].

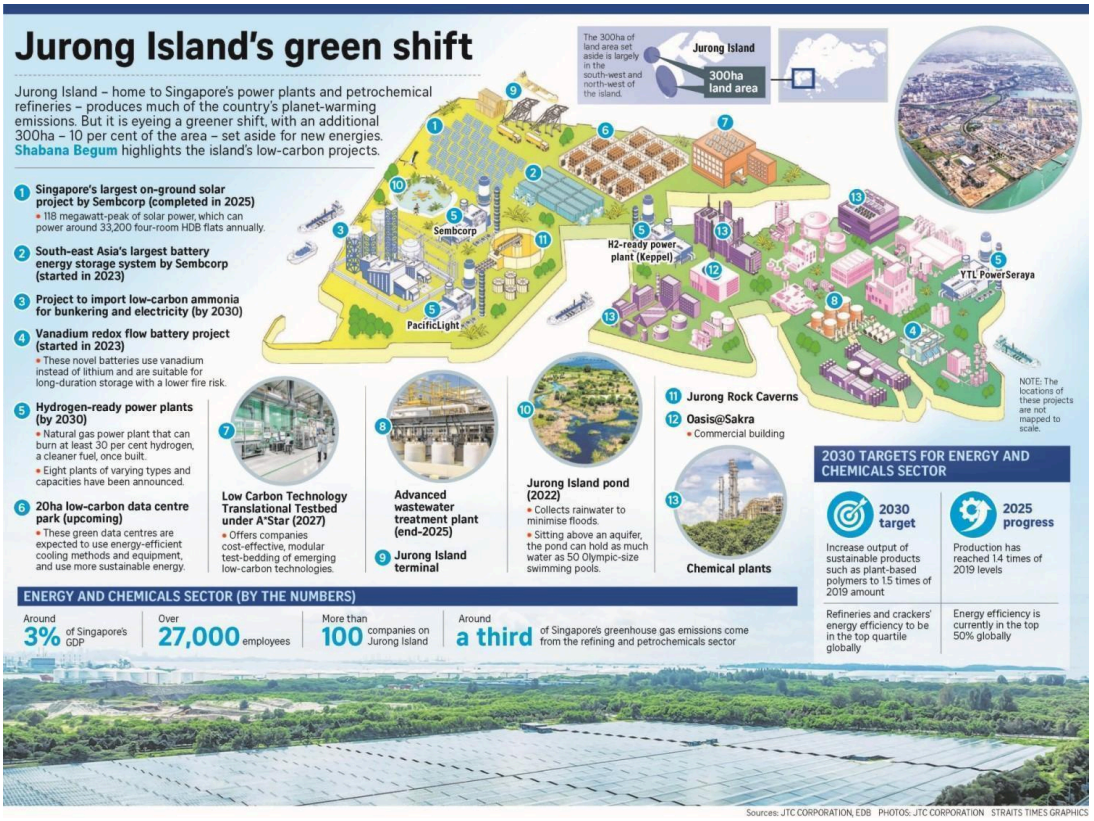


Figure 5.1. Jurong Island's low-carbon projects, including the planned 20 ha low-carbon data centre park (item 6) within the 300 ha set aside for new energies, largely in the south-west and north-west of the island (inset). Project locations are not mapped to scale. Source: The Straits Times graphics, using data from JTC Corporation and EDB; photographs by JTC Corporation [59].

The park suits the proposed strategy on three counts. Firstly, its power capacity accommodates the reference facility with room to grow: the facility needs about 125 MW at

full load (106 MW on average), about 18% of the park's 700 MW [50], and a campus doubled to 200 MW of IT (Section 4.2) would need 250 MW, or 36%. Each 50 MW DC-CFA2 allocation [95], taken as IT load like national capacity [1], corresponds to about 62.5 MW of facility power at PUE 1.25. Secondly, island-scale utilities and shared energy storage support the liquid cooling and backup requirements of the long-term configuration. Thirdly, the planned ammonia-to-power plant in the same new-energy area can supply the facility directly as shared generation, through an eligible DC-CFA2 green pathway [14].

Selection of measures

The industry view in Figure 5.2 frames the selection. Data analytics and AI-based control and virtual power purchase agreements combine high benefit with low implementation difficulty, immersion and direct-to-chip cooling offer high benefit at moderate difficulty, and 24/7 carbon-free load dispatching and advanced nuclear offer high benefit at high difficulty [96]. The quantified results of Chapters 2 and 3 confirm that ranking for Singapore: AI plant control pays back within two years, rooftop PV saves money on a small volume, and firm SMR and ammonia generation deliver most of the emissions reduction (Table 5.1).

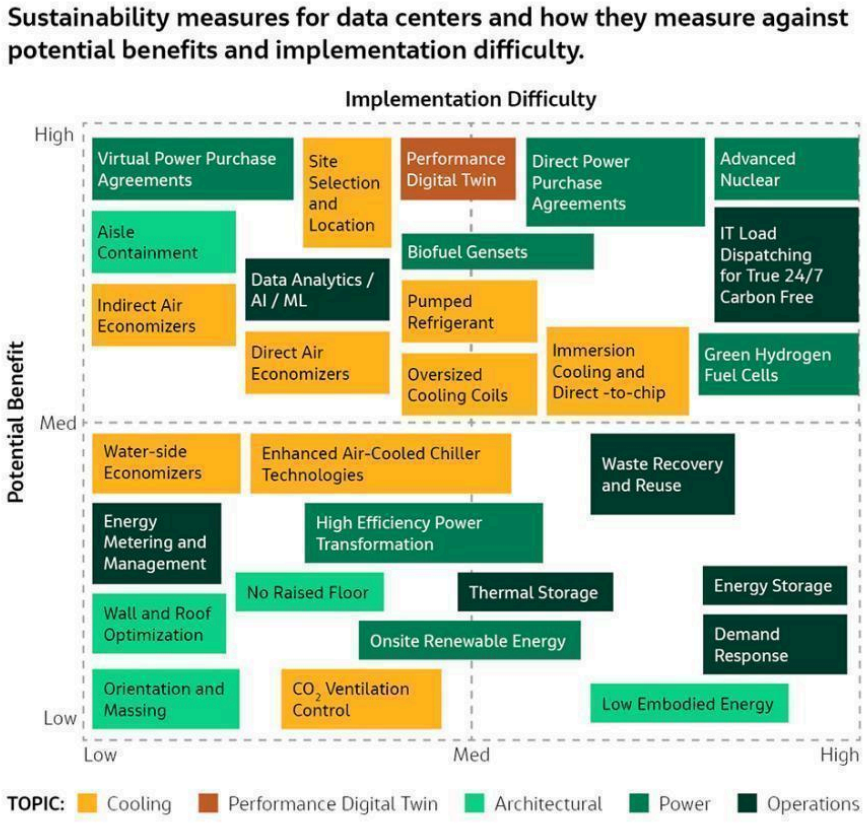


Figure 5.2. Data centre sustainability measures by potential benefit and implementation difficulty. Source: Jacobs, Driving sustainability in data centers, Figure 1 [96].

Table 5.1. Measures in the integrated system and their contribution

Measure (lever)	Contribution to emissions and cost, with evidence
Plant sub-metering and PIML-MPC control (demand)	9.7 to 13.4 GWh/yr; enables every other measure [81]; commercial plants 9% to 28% [83], [85]
SS 697 temperature optimisation (demand)	5.5 to 13.5 GWh/yr more, after removing overlap with plant control [12]

Measure (lever)	Contribution to emissions and cost, with evidence
Hybrid air and liquid cooling in AI halls (demand)	PUE 1.20 and 20.2 GWh/yr beyond plant control [78], [16]; also cuts server fan power [79]
Rooftop and facade PV (supply)	1.6 GWh/yr at S$85/MWh, below the grid price [30], [37]
Shared low-carbon ammonia GTCC (supply)	264.6 to 446.8 GWh/yr of dispatchable supply at 0.111 tCO_2e/MWh [34], [20]
SMR, 77 MWe (supply, longer term)	627.3 GWh/yr of firm supply at 0.012 tCO_2e/MWh [31], [32]
Singapore grid (balancing and backup)	Supplies residual load and covers outages [42], [49]
Measurement and verification (verification)	Makes every saving and allocation claim auditable [91]

Four design principles govern the system. Efficiency comes first, because each kilowatt-hour removed never has to be generated. Generation is allocated to verified post-efficiency demand, so every MWh is allocated once. Firm, dispatchable supply is preferred over annual certificates because it matches a flat load hour by hour (Section 4.5). Every result is verified with metered electricity, emissions and an hourly carbon-free electricity score (Equation 4.2). Figure 5.3 shows the three-layer architecture, in which a metering and control loop links plant data to the predictive controller, the dashboard and the ammonia allocation.

Table 5.2 defines six configurations, the authors' proposed engineering scenarios built from the evidence of Chapters 2 and 3, each adding one measure to the last. Cooling optimisation comes first because it cuts demand at a net saving (Section 3.5); rooftop PV follows as the only domestic low-carbon source deployable now (Section 2.3); ammonia adds dispatchable supply as the Jurong Island plant enters service (Section 2.5); and SMR electricity adds firm supply once a national decision permits (Section 2.4). The sequence follows the recommended implementation of Figure 2.2.

Table 5.2. Configurations assessed for the 100 MW reference data centre: the authors' proposed engineering scenarios

Configuration	PUE	Supply mix	Measures included
Baseline (grid only)	1.25	Grid	None
Efficiency only	1.227	Grid	Plant control and setpoints (Chapter 3)
Solar integration	1.227	PV + grid	Adds rooftop and facade PV
Ammonia integration	1.20	PV + full ammonia plant output + grid	Adds hybrid liquid cooling and the 60 MWe ammonia GTCC (446.8 GWh/yr)
Nuclear integration	1.20	PV + SMR + grid	Hybrid liquid cooling and one 77 MWe SMR module, without ammonia
Integrated long term	1.20	PV + SMR + shared ammonia; grid for balancing	Ammonia allocated to the residual demand (264.6 GWh/yr)

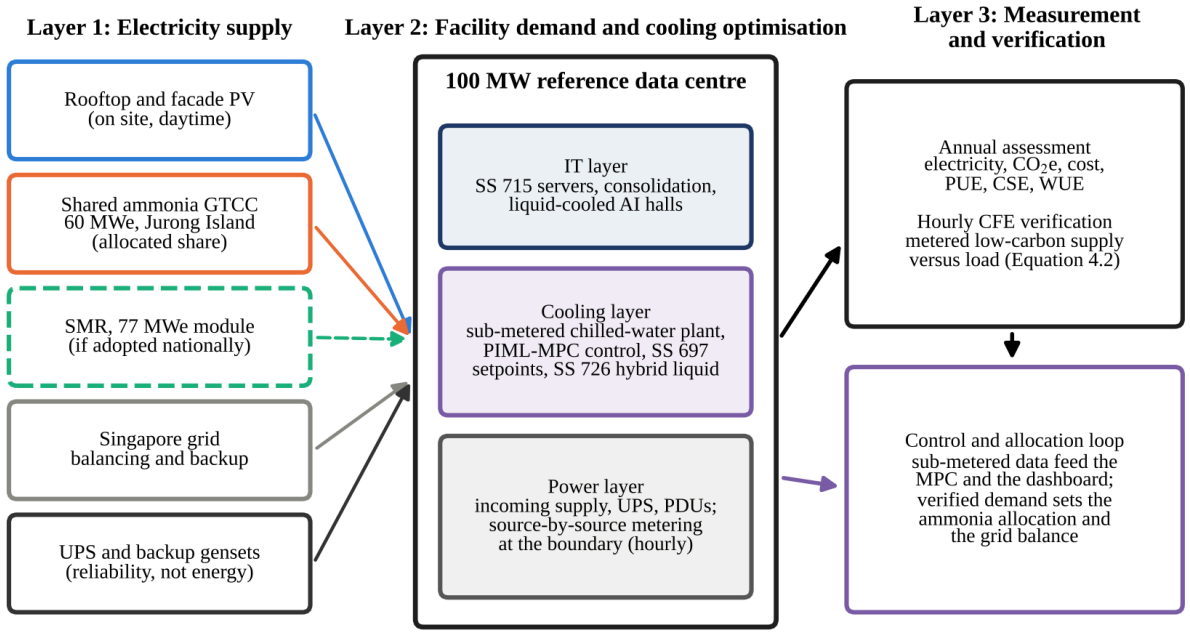


Figure 5.3. Integrated architecture for the reference data centre in three layers: electricity supply (left), facility demand and cooling optimisation (centre), and measurement and verification (right), separating annual assessment from hourly verification. The dashed box is the longer-term SMR option. Authors' design.

5.2 Application of the Integrated Model

Every configuration is calculated with Equations 1.1 to 1.7, the factors of Section 1.3 and the costs of Table 1.4. The annualised efficiency cost C_{eff} is reported separately (S$0.14 to 0.71 million a year for S$1 to 5 million at CRF = 0.1424), and hybrid liquid cooling through the capital its saving can carry (Section 5.3). With IT energy fixed, Equations 1.1, 1.2 and 1.5 give an exact decomposition of the emissions avoided into efficiency and supply terms:

$$\Delta CO_2 = E_{IT}(PUE_0 - PUE_1)EF_{grid} + E_{IT}PUE_1(EF_{grid} - EF_{eff}) \quad (5.1)$$

where $EF_{eff} = CO_{2,proposed} / E_{DC,proposed}$ is the effective emission factor of the proposal. The first term is the CO_2 removed by using less electricity, valued at the grid factor because each avoided MWh would have come from the grid; the second is the CO_2 removed by cleaning the electricity that is still used. The same decomposition applies to cost with prices in place of emission factors, and the two terms add exactly to the total, so no reduction is counted twice.

Generation is allocated after efficiency in order of carbon intensity, with ammonia filling the residual demand up to plant output and the grid taking any remainder:

$$E_{NH3,DC} = \min[E_{NH3}, E_{DC} - E_{PV} - E_{SMR}], \quad E_{grid} = \max[0, E_{DC} - E_{PV} - E_{SMR} - E_{NH3,DC}] \quad (5.2)$$

For the integrated long-term configuration, E_{DC} = 744,600 × 1.20 = 893,520 MWh/yr; after 1,582 MWh of PV and 627,304 MWh of SMR output, the facility takes $E_{NH3,DC}$ = 264,634 MWh of ammonia electricity, the plant's other 182,126 MWh serves other consumers, and the grid allocation is zero. Equations 1.2 and 1.3 give:

$$CO_2 = 1{,}582 \times 0.041 + 627{,}304 \times 0.012 + 264{,}634 \times 0.11124 = 37{,}030\ tCO_2e/yr \quad (5.3)$$

$$C = 1{,}582 \times 85 + 627{,}304 \times 230 + 264{,}634 \times 341.8 = S\$234.9\ million/yr \quad (5.4)$$

Against the baseline this avoids ΔCO_2 = 374,162 − 37,030 = 337,132 tCO_2e/yr (90.1%) for ΔC = +S$48.7 million/yr before C_{eff}, an abatement cost of S$145/$tCO_2e$ (Equations 1.5 to

1.7). Table 5.3 gives every configuration, and Appendix C.7 the six annual electricity balances.

5.3 Integrated Results for the Reference Data Centre

Table 5.3. Integrated electricity, emissions and cost results for the 100 MW reference data centre (annual values; S$M = S$ million; ΔC on an electricity-expenditure basis)

Configuration	E_{DC} (GWh)	Low-carbon (GWh)	Grid (GWh)	Emissions ($ktCO_2e$)	ΔCO_2e (kt)	C (S$M)	ΔC (S$M)	AC (S$/t)
Baseline	930.8	0	930.8	374.2	0	186.15	0	-
Efficiency only	913.7	0	913.7	367.3	6.8	182.75	−3.40	−497
Solar integration	913.7	1.6	912.2	366.8	7.4	182.57	−3.58	−484
Ammonia integration	893.5	448.3	445.2	228.7	145.4	241.89	+55.74	+383
Nuclear integration	893.5	628.9	264.6	114.0	260.2	197.34	+11.19	+43
Integrated long term	**893.5**	**893.5**	**0**	**37.0**	**337.1**	**234.88**	**+48.73**	**+145**

Efficiency and solar integration avoid 7.4 $ktCO_2e$ a year for a net saving of S$2.9 to 3.4 million (Table 5.4), and the efficiency package stays cost-neutral up to S$23.9 million of capital. Ammonia integration adds hybrid liquid cooling, following the roadmap sequence, supplies half of the 893.5 GWh demand and avoids 145 kt a year at S$383/$tCO_2e$, driven by fuel cost. Nuclear integration avoids 260 kt at S$43/$tCO_2e$, below today's S$45/t carbon tax [45]. The integrated long-term configuration combines both firm sources, removes annual grid allocation and avoids 337 kt a year, 90.1% of the baseline, at S$145/$tCO_2e$.

Hybrid liquid cooling saves 20.2 GWh and S$4.0 million a year beyond plant control, repaying up to S$36.8 million of capital (S$368/kW of IT) at 7% over 15 years, or S$28.4 million over 10 years (Appendix C.9). No hybrid-specific capital cost is published, and a vendor analysis found no capital premium for immersive liquid cooling at equal rack density [97]. Charging the full 15-year break-even raises the long-term abatement cost to S$157 to 159/$tCO_2e$ (Table 5.4).

Table 5.4. Electricity expenditure change and net annualised change of the integrated configurations (S$ million/yr)

Configuration	ΔC	C_{eff}	Liquid cooling, annualised	Net change	Net AC (S$/t)
Solar integration	−3.58	0.14 to 0.71	0	−3.44 to −2.87	−464 to −388
Ammonia integration, no liquid-cooling premium	+55.74	0.14 to 0.71	0	+55.88 to +56.45	+384 to +388
Ammonia integration, liquid cooling at break-even capital (S$36.8 million)	+55.74	0.14 to 0.71	4.05	+59.93 to +60.50	+412 to +416
Nuclear integration, no liquid-cooling premium	+11.19	0.14 to 0.71	0	+11.33 to +11.90	+44 to +46
Nuclear integration, liquid cooling at break-even capital (S$36.8 million)	+11.19	0.14 to 0.71	4.05	+15.38 to +15.95	+59 to +61
Integrated long term, no liquid-cooling premium	+48.73	0.14 to 0.71	0	+48.87 to +49.44	+145 to +147

Configuration	ΔC	C_{eff}	Liquid cooling, annualised	Net change	Net AC (S$/t)
Integrated long term, liquid cooling at break-even capital (S$36.8 million)	+48.73	0.14 to 0.71	4.05	+52.91 to +53.48	+157 to +159

Table 5.5. Decomposition of the annual emissions and electricity-expenditure changes into efficiency and supply terms (Equation 5.1)

Configuration	Efficiency term (kt)	Supply term (kt)	Total ΔCO_2e (kt)	Efficiency saving (S$M)	Supply cost change (S$M)	Total ΔC (S$M)
Ammonia integration	15.0	130.5	145.4	−7.45	+63.18	+55.74
Nuclear integration	15.0	245.2	260.2	−7.45	+18.64	+11.19
Integrated long term	15.0	322.2	337.1	−7.45	+56.17	+48.73

Low-carbon generation delivers 96% of the long-term reduction and efficiency all of the cost saving (Table 5.5): efficiency removes 37.2 GWh a year that would otherwise be generated, saving S$7.45 million, while firm generation turns a small efficiency gain into a deep cut (Figure 5.4).

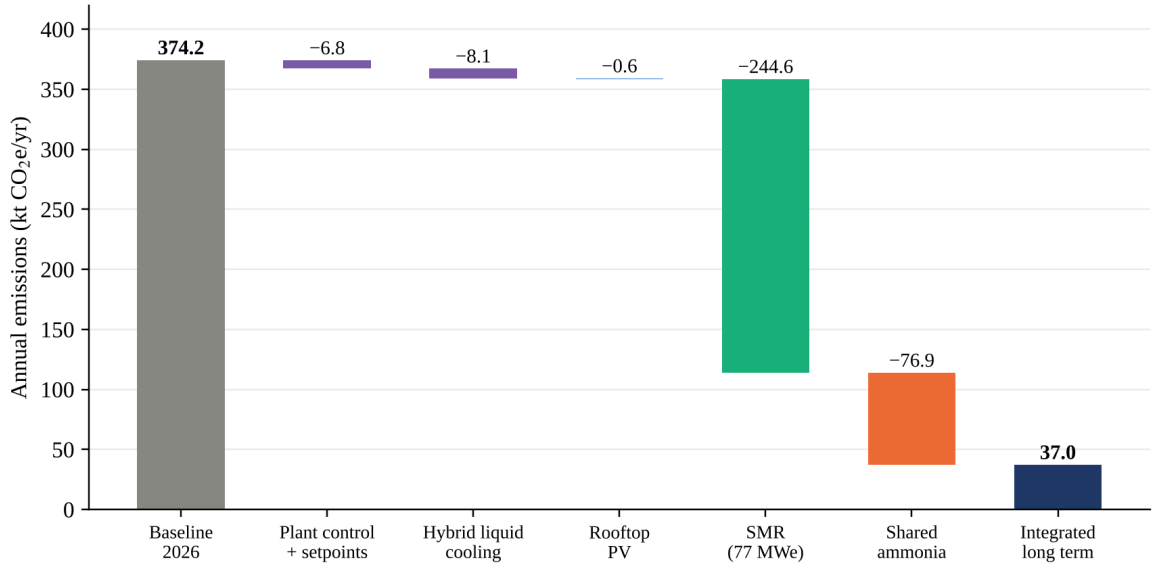


Figure 5.4. Annual emissions of the reference data centre from the 2026 baseline to the integrated long-term configuration, by measure (Tables 5.3 and 5.5). Authors' calculation.

5.4 Existing Data Centre Scenario

Most of Singapore's capacity is existing build: about 1.3 GW of the 1.6 GW of IT capacity predates the 280 MW allocated through competitive tenders since 2022 [2], and therefore the PUE 1.25 requirement. The same 100 MW of IT is therefore assessed at a baseline PUE of 1.50, close to the Uptime Institute's 2025 global average [86], with cooling at 70% of the 0.50 overhead, as in the new-build case. The site draws 1,116.9 GWh a year, emits 449.0 $ktCO_2e$ and spends S$223.4 million on electricity. Plant control and setpoint optimisation apply the same ΔCSE and SS 697 method as Chapter 3, and a plant retrofit then brings the site to PUE 1.30, the Green Data Centre Roadmap aspiration for all data centres [11], through containment, chiller and variable-speed upgrades (Table 5.6).

The PUE improvement from 1.50 to 1.30 is an efficiency target for an existing Singapore data centre undergoing cooling-system optimisation, and this level has been demonstrated locally: a KoolLogix phase-change cooling system, developed with A*STAR, was reported to have brought CDC's research and development data centre site in Singapore to a PUE of 1.29, against an Asia-Pacific average of 1.69 [98]. The 1.30 target is therefore consistent with efficiency already achieved under tropical conditions, and the retrofit scenario applies it to the reference facility with the same annual electricity and cost framework.

Table 5.6. Existing data centre at PUE 1.50 compared with the new-build reference case (break-even capex: implementation cost at which the annualised cost at 7% over 10 years equals the annual saving; annual values, S$M = S$ million)

Configuration	PUE	Saved (GWh)	Emissions (kt)	ΔCO_2e (kt)	ΔC (S$M)	AC (S$/t)	Break-even capex (S$M)
Existing baseline	1.50	0	449.0	0	0	-	-
Controls and setpoints	1.469	23.0	439.7	9.2	−4.60	−497	32
Retrofit to PUE 1.30	1.30	148.9	389.1	59.9	−29.78	−497	209
Retrofit + PV + ammonia	1.30	148.9	258.7	190.3	+33.40	+175	209
Retrofit + PV + SMR + ammonia	1.30	148.9	45.3	403.7	+36.95	+92	209
New build: controls and setpoints	1.227	17.0	367.3	6.8	−3.40	−497	24

The existing site offers the larger opportunity on every metric. The same software measures save 23.0 GWh a year, 35% more than at the new build, and the retrofit to PUE 1.30 removes 148.9 GWh, 59.9 ktCO_2e and S$29.8 million a year, staying cost-neutral up to S$209 million of capital (Appendix C.9). With the new build's generation, the site avoids 190 kt a year with PV and ammonia and 404 kt with the SMR added, at S$175 and S$92 per tonne, because the retrofit saving funds much of the generation premium. The strategy therefore applies first and most profitably to the installed base, which the Digital Infrastructure Bill's licence regime reaches [2].

5.5 Sensitivity Analysis

The delivered ammonia price and its life-cycle intensity dominate the long-term results (Figure 5.5). From US$320 to 850/t, generation cost runs from S$260 to 531/MWh and the long-term abatement cost from S$80 to 293/t$CO_2e$, with grid parity at about US$203/t. Renewable fuel avoids 351 kt a year on the operational boundary (0.15 tCO_2e/tNH_3) and 337 kt on the central life-cycle boundary (0.276); blue fuel (0.78) avoids 284 kt and Lee et al.'s full inventory (1.03) 257 kt, as those MWh carry about the grid's intensity. Certified renewable fuel and long-term price discovery are therefore the principal commercial tasks.

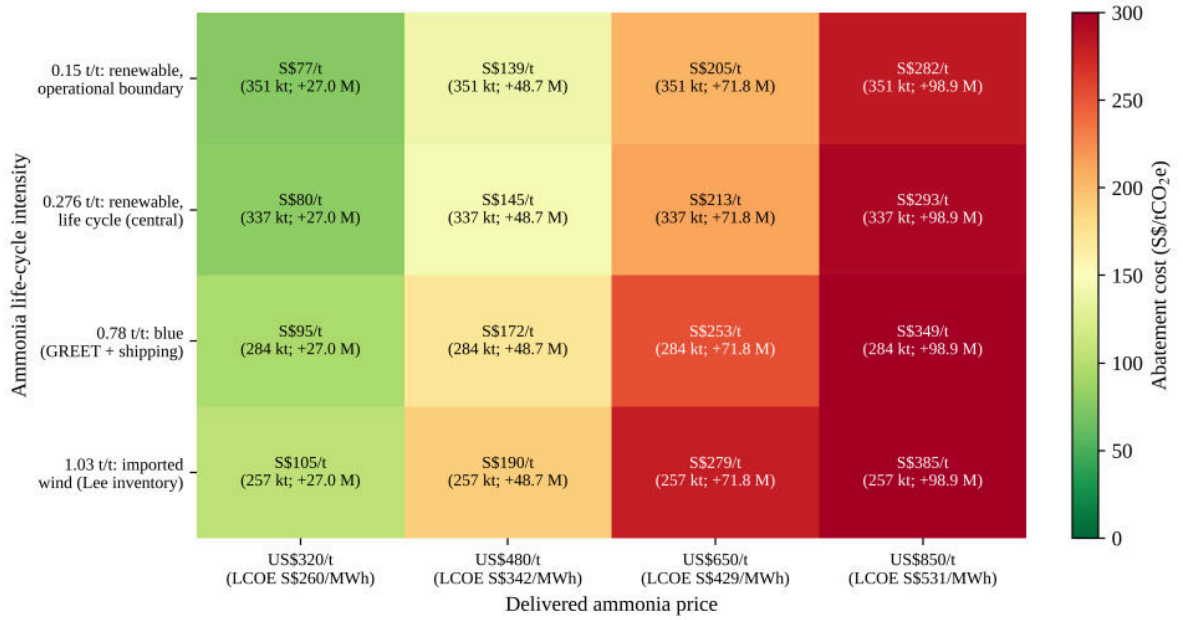


Figure 5.5. Long-term abatement cost, emissions avoided and annual cost change by delivered ammonia price and life-cycle intensity (Table 2.2), other inputs central. Authors' calculation.

Table 5.7. Sensitivity of the integrated results to the principal assumptions

Variable	Range tested	Effect on the result
Cooling saving	Conservative 15.2, central 17.0 and high 26.8 GWh/yr	Solar integration: 367.5, 366.8 and 362.8 kt; ΔC −3.2, −3.6 and −5.6 S$ million
Grid electricity price	S$150, 200, 250 and 300/MWh	Long-term ΔC +95.3, +48.7, +2.2 and −44.3 S$ million; AC 283, 145, 6 and −132 S$/t
Efficiency implementation cost	S$1 to 5 million capex	Solar integration net −S$3.4 to −2.9 million; break-even S$23.9 million
Liquid-cooling capital	S$0 to 36.8 million (7%, 15 years); S$28.4 million over 10 years	Long-term net AC S$145 to 159/t
SMR cost	S$114, 180, 230 and 281/MWh	Long-term ΔC −24.0, +17.4, +48.7 and +80.7 S$ million; AC −71, +51, +145 and +239 S$/t; nuclear integration cost-neutral at S$212/MWh
Ammonia fuel price	US$320, 480, 650 and 850/t	LCOE S$260, 342, 429 and 531/MWh; long-term AC 80, 145, 213 and 293 S$/t; grid parity at US$203/t
Ammonia life-cycle intensity	0.15, 0.276, 0.78, 1.03 and 2.40 tCO_2e/tNH_3	Long-term emissions 23.6, 37.0, 90.5, 117.1 and 262.6 kt; AC 139, 145, 172, 190 and 437 S$/t; grid parity at 1.0 tCO_2e/tNH_3
Ammonia plant utilisation	Capacity factor 50%, 70%, 85% and 93%	LCOE S$409, 362, 342 and 334/MWh; long-term AC 196, 161, 145 and 138 S$/t; the allocation needs at least 50.3%
IT load growth	100 MW to 200 MW IT	Integrated emissions 98 kt/yr with two SMR modules, 276 kt below today's baseline (Section 4.2)

The emissions result rests on the SMR and certified fuel, both specified by the operator, while the cost result depends mainly on the ammonia price and SMR cost (Table 5.7). Below about S$212/MWh the nuclear configuration costs less than the baseline, and at an all-in grid price

of S$250/MWh, which the 2026 rise in wholesale prices brings into view [44], the long-term configuration is close to cost-neutral.

5.6 Safety, Reliability and Implementation

Each measure is compatible with the reliability that colocation and cloud customers require, provided its principal hazard is engineered for (Table 5.8). The grid connection is sized for the full facility load, so an ammonia supply interruption or an SMR outage shifts supply back to the grid without interrupting the IT load, and the predictive controller operates within hard limits in the building management system and reverts automatically to fixed setpoints on any fault.

Table 5.8. Principal hazards of the integrated strategy and their engineering controls

Measure	Hazard	Engineering control
PIML-MPC plant control	Setpoints outside the thermal band; model drift; cyber intrusion	Physics constraints, hard limits in the BMS and automatic fallback to fixed setpoints; segregated control network with read-only optimiser paths
Warmer halls (SS 697)	Server faults and higher IT fan power	Operate at the measured total-power optimum; agree the ASHRAE envelope in tenant contracts [12]
Hybrid liquid cooling (SS 726)	Coolant leaks, corrosion, biofilm, floor loading	Leak detection and isolation, fluid-quality monitoring, compatible piping and structural checks [16]
Cooling towers	Legionella in warm recirculating water	Continuous biocide treatment, routine sampling and drift elimination
Chillers	Fugitive leakage of high-GWP refrigerant (Scope 1)	Leak detection, annual audits and low-GWP refrigerant at chiller renewal
UPS batteries	Thermal runaway and fire	Fire-rated separation, cell-level monitoring and suppression
Ammonia GTCC and fuel handling (Jurong Island)	Ammonia toxicity and corrosivity; leaks; flame instability; NO_x, N_2O and ammonia slip	Gas detection with automatic isolation, ventilation, secondary containment and separation distances; staged low-NO_x combustion with SCR and continuous emissions monitoring [58], [34]; emergency plans coordinated with neighbouring plants
SMR (long term)	Nuclear safety, security and waste	National regulatory framework under the IAEA Milestones Approach, prepared through NEA safety studies and the 2027 INIR mission [61], [19]; the operator contracts for output only

Ammonia is toxic and corrosive, so its import, storage and combustion require containment, detection, ventilation, emergency preparedness and NO_x, N_2O and slip control [58]. Nuclear integration follows Singapore's capability-building sequence to the 2027 INIR mission [18], [60], [61], [19], with reactor-specific safety assessment and licensing before any supply contract.

The proposal is delivered in five phases (Figure 5.6 and Table 5.9), each triggered by a measured result from the previous one and aligned with a national milestone: the licence regime, the 3 GWp solar target and the Jurong Island ammonia project around 2030, and Singapore's nuclear assessment after the 2027 INIR mission [2], [42], [19], [59].

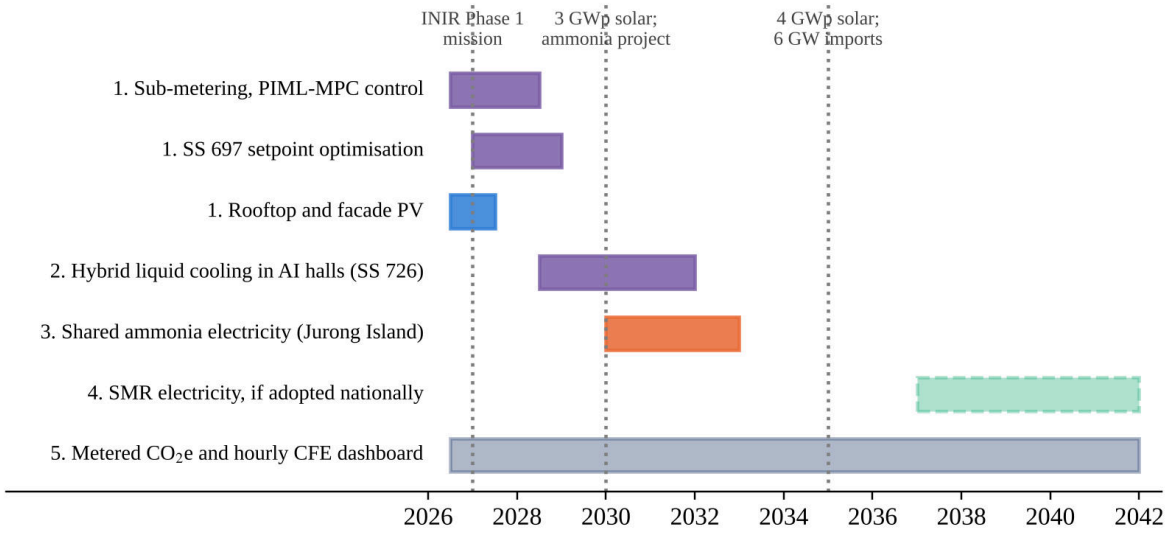


Figure 5.6. Implementation roadmap for the reference data centre, aligned with national milestones [2], [42], [19], [59]; the SMR phase depends on a national deployment decision. Authors' synthesis.

Table 5.9. Phased implementation plan with emissions and cost targets

Phase	Engineering and procurement actions	Target at end of phase
1. Cooling optimisation and rooftop solar (2026 to 2029)	Sub-meter the plant; install a data historian; field-trial and roll out PIML-MPC; raise temperatures to the measured optimum; install rooftop and facade PV	PUE 1.227; 17.0 GWh/yr saved and 7.4 $ktCO_2e$/yr avoided; net saving S$2.9 to 3.4 million/yr
2. Hybrid liquid cooling (2028 to 2032)	Hybrid air and direct-to-chip liquid cooling in all AI halls under SS 726	PUE 1.20; a further 20.2 GWh/yr and S$4.0 million/yr; capital up to S$36.8 million repaid
3. Ammonia electricity integration (from about 2030)	Contract certified ammonia electricity from the Jurong Island plant at a fuel-indexed, cost-reflective price; fuel at or below 0.276 tCO_2e/tNH_3	Up to 446.8 GWh/yr; 145 $ktCO_2e$/yr avoided (ammonia integration)
4. Potential SMR integration (late 2030s onwards)	Contract SMR output after a national decision; reduce the ammonia allocation to residual demand	627.3 GWh/yr of firm supply; 260 $ktCO_2e$/yr avoided with PV alone (nuclear integration)
5. Integrated operation and verification (throughout)	Operate the portfolio with grid balancing; report metered emissions and hourly CFE under revised GHG Protocol guidance [91]	37.0 $ktCO_2e$/yr remaining; 337 kt/yr avoided; about 97% hourly matching

Verification follows a measured baseline: before any control change, a month of sub-metered data establishes a regression of plant power on IT load and wet-bulb temperature, and savings are then reported monthly against it, each verified saving resizing the next ammonia allocation. The same meters record the hourly volumes for the dashboard of Table 4.3, so one data system supports both efficiency verification and supply claims.

5.7 Conclusions and Recommendations

Low-carbon data centres can be achieved in Singapore through an efficiency-first strategy in which predictive cooling control and hybrid liquid cooling reduce demand; rooftop solar, shared ammonia generation and future SMR electricity supply the reduced load; and metered, hourly reporting verifies every claim.

Applied to a 100 MW facility in the Jurong Island park, cooling optimisation and rooftop PV avoid 7.4 kt a year at a net saving of S$2.9 to 3.4 million, and hybrid liquid cooling reaches PUE 1.20, repaying up to S$36.8 million of capital. Shared ammonia generation with certified renewable fuel supplies up to half of the reduced demand at 0.111 tCO_2e/MWh, and a 77 MWe SMR supplies 70% at S$43 per tonne avoided. Together they cut annual emissions from 374 kt to 37.0 kt, a 90% reduction, at S$145 per tonne while matching about 97% of consumption hour by hour; an existing site starting from PUE 1.50 avoids 404 kt a year. Table 5.10 sets out the recommended actions.

Table 5.10. Recommendations by actor, type and implementation timeframe (short term 2026 to 2029; medium term 2030 to 2035; long term after 2035)

Actor	Recommendation	Type	Implementation timeframe
Operators	Sub-meter the cooling plant and deploy physics-informed predictive control against a measured monthly baseline, existing sites first	Commercial implementation	Short term (2026 to 2029): Phase 1
Operators	Raise hall and supply-water temperatures to the measured total-power optimum under SS 697, and install rooftop and facade PV	Commercial implementation	Short term (2026 to 2029): Phase 1
Operators	Design hybrid air and direct-to-chip liquid cooling into every new AI hall under SS 726, and buy SS 715 compliant IT	Commercial implementation	Short term (from 2028): all new capacity, including DC-CFA2 allocations (Phase 2)
IMDA and EMA	Write plant sub-metering and hourly reporting into licence conditions under the Digital Infrastructure Bill	Regulation	Short term: licence regulations following the Bill (Oct 2026) [15]
EMA, MPA and consortium	Complete front-end engineering design of the Jurong Island ammonia plant, including NO_x and N_2O control	Technology development	Short term: from the FEED award (Oct 2025) to final investment decision [20]
EMA and MPA	Certify delivered ammonia intensity, covering production and shipping, at or below 0.276 tCO_2e/tNH_3	Regulation	Short term: before the plant enters service (about 2030)
Operators	Contract shared Jurong Island ammonia electricity at a fuel-indexed price with certified renewable fuel	Commercial implementation	Medium term (2030 to 2035): from plant commissioning (Phase 3)
EMA and NEA	Continue nuclear capability building, NEA safety and regulatory studies, and the IAEA review	Technology development	Ongoing: INIR Phase 1 mission in 2027 [19], [61]
Operators	Contract SMR output after a national deployment decision, reducing the ammonia allocation to residual demand	Commercial implementation	Long term (after 2035): late 2030s at the earliest (Phase 4)
Operators	Report metered electricity, emissions and the hourly carbon-free electricity score on the Table 4.3 dashboard	Verification	Ongoing: annual reporting under the licence regime (Phase 5)

Singapore can sustain its role as a regional data centre hub while meeting its net-zero commitment. Sequenced correctly, efficiency pays for itself, rooftop solar saves money, ammonia supplies firm low-carbon electricity this decade and SMRs offer the deepest, lowest-cost long-term reduction. Operators should begin with efficiency across the existing fleet, where the opportunity is largest.

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Appendix A: Supplementary Figures from the NUS Studies

The following figures support Sections 3.3 and 3.4 and are reproduced from the original papers for this coursework.

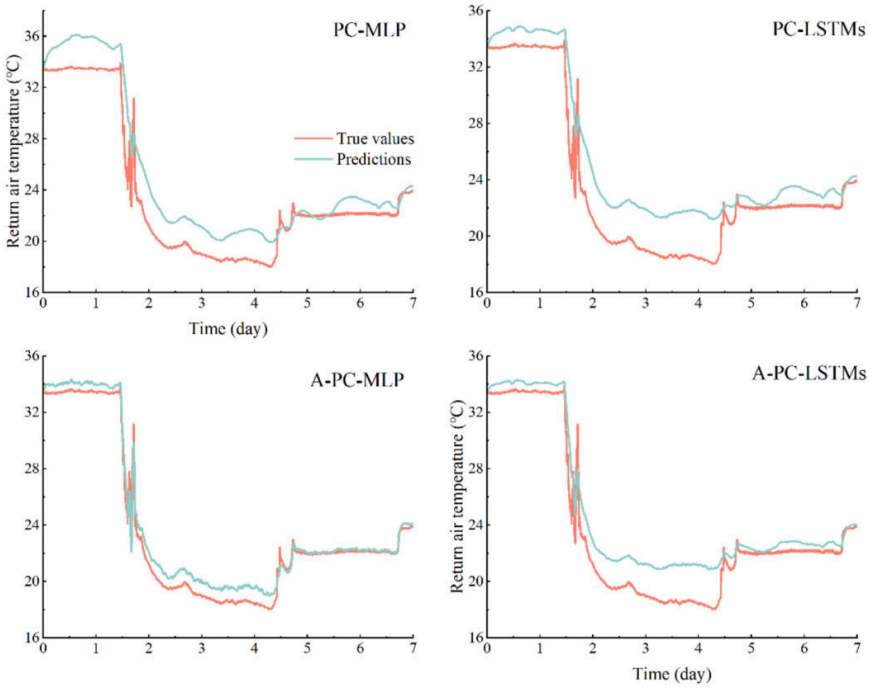


Figure A.1. Seven-day edge-case test of the adaptive thermal model; red is measured return-air temperature, blue the prediction. Source: Chen, Chui and Lee [80], Fig. 8.

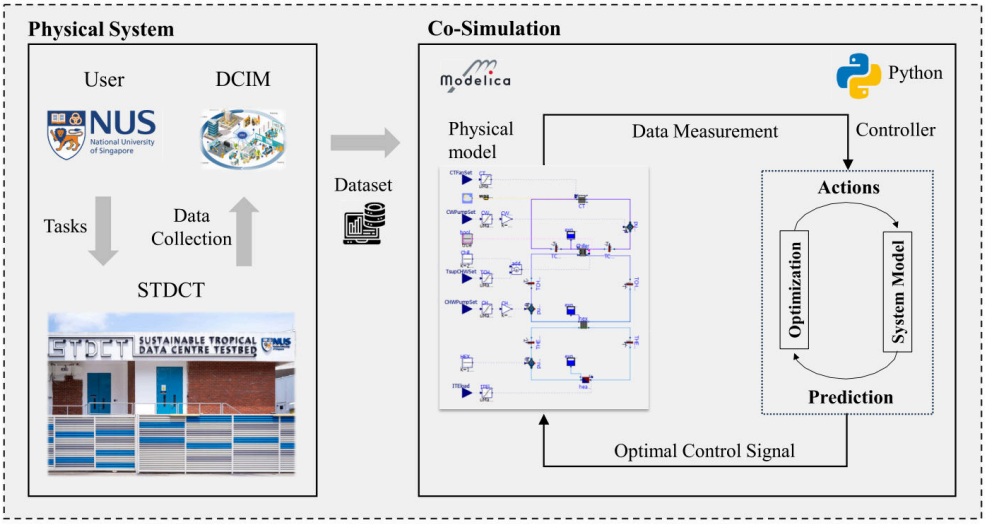


Figure A.2. Study workflow: STDCT DCIM data, calibrated Modelica plant model and model predictive control in Python. Source: Chen, Chui and Lee [81], Fig. 4.

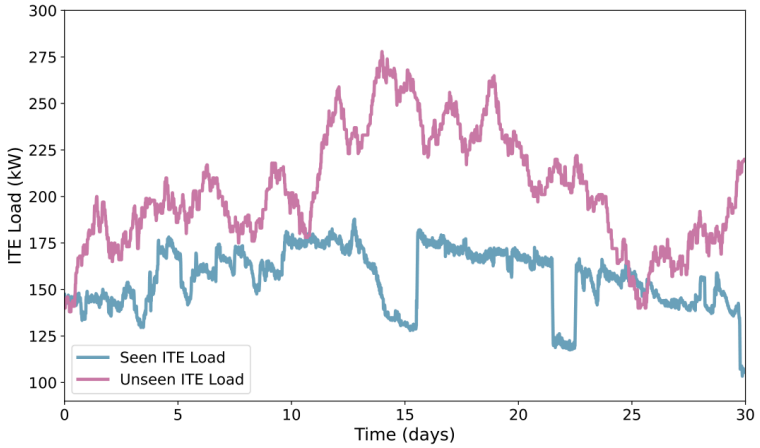


Figure A.3. Seen versus unseen IT load used to test the controllers. Source: Chen, Chui and Lee [81], Fig. 14.

Appendix B: Project Charter

Table B.1 records the charter agreed by the five-member group, built around one shared reference case so that the supply-side and demand-side chapters combine without double counting.

Table B.1. Project charter for the CN5303 Group 5 study

Item	Description
Project title	Low-Carbon Grid-Connected Data Centres in Singapore: An Integrated Efficiency and Low-Carbon Generation Strategy
Research question	How can a grid-connected data centre in Singapore achieve low operational emissions through energy efficiency improvements and low-carbon electricity generation from solar PV, ammonia and nuclear energy?
Objective	Propose and quantify an integrated, phased strategy for a 100 MW IT reference data centre, reporting annual emissions avoided, incremental annual cost and abatement cost
Scope (in)	Electricity-related emissions of a grid-connected facility, assessed with the comparative emission factors of Section 1.3; rooftop PV, shared low-carbon ammonia generation and SMR supply; cooling-plant and IT-side efficiency; emissions accounting; implementation roadmap
Scope (out)	Detailed Scope 3 life-cycle assessment; site-specific structural and plant design; electricity market reform beyond data centre supply
Common reference case	100 MW IT, 85% electrical load factor, PUE 1.25, 8,760 h/yr, grid emission factor 0.402 tCO_2/MWh, all-in electricity price S$200/MWh
Chapter 1 owner	research question, policy landscape, emissions boundaries, reference case and constraints
Chapter 2 owners	Wee Zhern Ray: renewable and low-carbon screening, supply modelling, Scenario A: Solar PV, Scenario B: Nuclear SMR, low-carbon supply pathways Scenario C: Low-carbon Ammonia, carbon assessment and supply strategy, techno-economics
Chapter 3 owner	energy efficiency framework, physics-informed thermal modelling and predictive cooling control
Chapter 4 owner	strategic challenges in Singapore's current approach and the engineering response to each
Chapter 5 owner	Wee Zhern Ray: integrated proposal, electricity and emissions modelling, techno-economic assessment, sensitivity analysis, implementation roadmap and conclusions

Appendix C: Supporting Calculations

The calculations below reproduce the principal derived values of the report from the stated inputs, so that each result can be checked independently.

C.1 Reference facility demand and cooling saving

E_{IT} = 100 × 0.85 × 8,760 = 744,600 MWh/yr and E_{DC} = 930,750 MWh/yr. With $E_{cool,0}$ = 0.175 E_{IT} = 130,305 MWh/yr, the central case of Equation 3.5 gives:

$$\Delta E_1 = 0.0155 \times 744{,}600 = 11{,}541 \text{ MWh/yr} \quad \text{(C.1)}$$

$$\Delta E_{1+2} = 11{,}541 + 0.02 \times 2.3 \times (130{,}305 - 11{,}541) = 17{,}004 \text{ MWh/yr} \quad \text{(C.2)}$$

so E_{DC} = 913,746 MWh/yr and PUE_1 = 1.2272. The conservative (ΔCSE 0.013, 2%/°C) and high (0.018, 5%/°C) cases give 15,229 and 26,847 MWh/yr. At the existing site (PUE 1.50, cooling 0.35 E_{IT}) the method gives 11,541 + 11,457 = 22,998 MWh/yr and PUE 1.4691. Hybrid liquid cooling at PUE 1.20 gives 893,520 MWh/yr, a further 20,226 MWh/yr.

C.2 Rooftop PV generation and levelised cost

P_{PV} = 13,600 × 0.60 × 0.14 = 1,142.4 kWp and E_{PV} = 1,142.4 × 1.385 = 1,582 MWh/yr (CF 0.158). With CRF = $0.07 \times 1.07^{25}/(1.07^{25} - 1)$ = 0.0858, S$1,200/kW and fixed costs of 1.225% (S$14.7/kW-yr):

$$LCOE_{PV} = 1{,}000 \times (1{,}200 \times 0.0858 + 14.7) / (8{,}760 \times 0.158) = \text{S\$85/MWh} \quad \text{(C.3)}$$

The low (S$1,000/kW, 1.0%, 1.45 MWh/kWp) and high (S$1,400/kW, 1.45%, 1.32 MWh/kWp) cases give S$66 and S$106/MWh.

C.3 SMR generation and capital recovery

E_{SMR} = 77 × 0.93 × 8,760 = 627,303.6 MWh/yr. NuScale's US$20,139/kW equals S$25,697/kW; at CRF = 0.0750 (7%, 40 years) and 8.147 MWh/kW-yr its capital recovery is 1,928/8.147 = S$237/MWh. The cost cases are 89, 180.5 and 220 US$/MWh × 1.276 = S$114, 230 and 281/MWh.

C.4 Ammonia electricity and fuel balance

E_{NH3} = 60 × 0.85 × 8,760 = 446,760 MWh/yr; SFC_{NH3} = 3.6/(0.477 × 18.8) = 0.40145 tNH_3/MWh [34], so m_{NH3} = 179,350 t/yr (491 t/day). In the long-term configuration the facility's 264,634 MWh uses 106,236 t and other consumers' 182,126 MWh 73,114 t, closing the fuel balance on the plant total.

C.5 Ammonia life-cycle emission factor

Lee et al.'s hydrogen demand (0.2294 t/tNH_3) at 4.5 kWh/Nm³ and 0.0899 kg/Nm³ needs 229.4/0.0899 × 4.5 = 11,486 kWh/tNH_3 [36], whose embodied emissions at the IPCC onshore wind median are 0.011 × 11.49 = 0.126 tCO_2e/tNH_3 [26]. With production (0.03 [35]) and shipping (0.12 [36]), I_{NH3} = 0.276 and the supply-chain term is 0.40145 × 0.2763 = 0.1109 tCO_2e/MWh. Combustion N_2O at 0.18 ppmv [34] in 138.17 t/h of flue gas (27.5 kg/kmol) at 41.87 MW is 0.00095 kg/MWh, at most 0.0003 tCO_2e/MWh at any warming potential up to 300:

$$EF_{NH3} = 0.1109 + 0.0003 = 0.1112 \text{ } tCO_2e/\text{MWh} \quad \text{(C.4)}$$

The other pathways give 0.060, 0.313, 0.414 and 0.964 tCO_2e/MWh for 0.15, 0.78, 1.03 and 2.40 tCO_2e/tNH_3, and parity with the grid occurs at (0.402 − 0.0003)/0.40145 = 1.00 tCO_2e/tNH_3.

C.6 Ammonia levelised cost

Ahmad et al.'s capital of US$179.23 million for 41.87 MW net [34] is US$4,281/kW = S$5,462/kW, with maintenance (2.5%) and insurance (2%) giving FOM = S$245.8/kW-yr. At CRF = 0.0858 and CF = 0.85 the fixed term is 1,000 × (5,462 × 0.0858 + 245.8)/(8,760 × 0.85) = S$96.0/MWh, and at US$480/t the fuel term is 0.40145 × 480 × 1.276 = S$245.9/MWh:

$$\mathrm{LCOE}_{\mathrm{NH3}} = 96.0 + 245.9 = \mathrm{S\$341.8/MWh} \tag{C.5}$$

Fuel is 72% of the total; each US$100/t adds 0.40145 × 100 × 1.276 = S$51.2/MWh, so US$320, 650 and 850/t give S$260, 429 and 531/MWh, and grid parity needs (200 − 96.0)/(0.40145 × 1.276) = US$203/t. At 70% and 50% capacity factor the fixed term is S$116.5 and S$163.1/MWh. Ahmad et al.'s own US$0.2871/kWh (10%, 25 years) equals S$366/MWh.

C.7 Annual electricity balances of the six configurations

Each balance satisfies $\Sigma E_i = E_{DC}$ (Equation 1.2), with ammonia allocated by Equation 5.2 (Table C.1).

Table C.1. Annual electricity balances of the six configurations (MWh/yr)

Configuration	E_{DC}	PV	SMR	Ammonia to facility	Ammonia to others	Grid
Baseline	930,750	0	0	0	0	930,750
Efficiency only	913,746	0	0	0	0	913,746
Solar integration	913,746	1,582	0	0	0	912,164
Ammonia integration	893,520	1,582	0	446,760	0	445,178
Nuclear integration	893,520	1,582	627,304	0	0	264,634
Integrated long term	893,520	1,582	627,304	264,634	182,126	0

C.8 Comparative emissions, expenditure and abatement cost

Ammonia integration (PUE 1.20): CO_2e = 1,582 × 0.041 + 446,760 × 0.11124 + 445,178 × 0.402 = 228,724 tCO_2e/yr and C = 1,582 × 85 + 446,760 × 341.8 + 445,178 × 200 = S$241.9 million/yr, so ΔCO_2e = 145,438 t, ΔC = +S$55.7 million and AC = S$383/t. Nuclear integration gives 113,975 t and S$197.34 million (264,634 MWh of grid at 0.402 and S$200/MWh), so ΔCO_2e = 260,186 t and AC = S$43/t.

C.9 Capital break-even thresholds

With CRF = 0.1424 (7%, 10 years) or 0.1098 (7%, 15 years), the capital an annual saving S repays is K = S/CRF:

$$K_{LC} = (20{,}226 \times 200)/0.1098 = \mathrm{S\$36.8\ million} \tag{C.6}$$

This is S$368/kW of IT capacity; over 10 years the threshold is 4.045/0.1424 = S$28.4 million. The new-build control package gives 3.40/0.1424 = S$23.9 million, existing-site controls (22,998 × 200)/0.1424 = S$32.3 million, and the existing-site retrofit 744,600 × (1.50 − 1.30) × 200/0.1424 = 29.78/0.1424 = S$209.2 million.

C.10 Net annualised results, hourly matching and supply interruption

The net annualised change adds C_{eff} (S$0.14 to 0.71 million) and any liquid-cooling capital at CRF = 0.1098 to ΔC:

$$\Delta C_{net} = \Delta C + C_{eff} + 0.1098\ K_{LC} \quad \text{(C.7)}$$

For the long-term configuration ΔC_{net} = S$48.87 to 49.44 million/yr, or S$52.91 to 53.48 million with K_{LC} = S$36.8 million; nuclear integration breaks even at 230 − 11.19/0.6273 = S$212/MWh. Hourly matching at 102.0 MW: for 8,147 h the SMR gives 77 MW and ammonia 25.0 MW, and for 613 h ammonia gives 60 MW, so CFE_{hourly} = (8,147 × 102.0 + 613 × 60)/893,520 = 97.1%. An 18-day ammonia loss returns 13,051 MWh to the grid and adds 13,051 × (0.402 − 0.1112) = 3,795 tCO_2e.

C.11 Life-cycle grid emission factor bound

At the IPCC gas combined-cycle median of 0.49 tCO_2e/MWh [26], the baseline is 456,068 tCO_2e/yr; the long-term configuration draws no annual grid electricity, so ΔCO_2e rises to 419,038 t (+24%) and AC falls to S$116/t. For ammonia integration:

$$CO_2 = 445{,}178 \times 0.49 + 446{,}760 \times 0.11124 + 1{,}582 \times 0.041 = 267{,}900\ tCO_2e/yr \quad \text{(C.8)}$$

so ΔCO_2e = 188,168 t against 145,438 t at 0.402.